Society for Computer Technology & Research's (SCTR's)

Pune Institute of Computer Technology (PICT), Pune

An Autonomous Institute affiliated to the Savitribai Phule Pune University (SPPU)
Approved by AICTE & Government of Maharashtra,
Accredited by NAAC (A+) & NBA [All eligible UG Programs]



Syllabus

T.Y. B. Tech. Information Technology (IT)

With effect from (A.Y. 2026-27)
National Education Policy (NEP) 2020 Compliant
For Approval by the Board of Studies (BoS IT) and Academic Council

Abbreviations used:

Sr. No.	Broad Category of the course	Sub- Category of course	Category Code
I.	**Basic Science/ Engineering Science Course (BSC/ ESC)**	Basic Science Course (BSC)	BS
		Engineering Science Course (ESC)	ES
II.	**Program Courses (PC)**	Program Core Course (PCC)	PC
		Program Elective Course (PEC)	PE
III.	**Multidisciplinary Courses (MC)**	Multidisciplinary Minor (MDM)	MD
		Open Elective (OE)	OE
IV.	**Skill Courses (SC)**	Vocational and Skill Enhancement Course (VSEC)	VS
V.	**Humanities Social Science and Management (HSSM)**	Ability Enhancement Course (AEC-01, AEC-02)	AE
		Entrepreneurship/Economics/ Management Courses (EEM)	EE
		Indian Knowledge System (IKS)	IK
		Value Education Course (VEC)	VE
VI.	**Experiential Learning Courses (ELC)**	Research Methodology (RM)	RM
		Community Engagement Project (CEP) / Field Project (FP)	CE
		Project (PRJ)	PR
		Internship/ On Job Training (IP/OJT)	IP
VII.	**Liberal Learning Courses (LLC)**	Co-curricular Activities (CCA)	CC
VIII.	**Honors Courses (HC)**	offered within the same branch depth in same branch (advanced specialization & research).	HC
IX.	**Minor Courses (MIC)**	offered by other branches **breadth across disciplines (interdisciplinary exposure, applied projects).**	MC
X.	**Double Minor Courses (DMC)**	Additional minor to be offered by other branches **breadth across disciplines (interdisciplinary exposure, applied projects).**	DM

Detailed Guidelines for Examination and Evaluation:

Link: Guidelines for examination

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Third Year B. Tech. Curriculum Structure (Semester - 5) with effect from the A.Y. 2026-27

Semester -5			Teaching Scheme (Hours/Week)				Credit Scheme				Examination / Evaluation Scheme and Marks						
Course Type	Course code	Name of the Course	L	P	T	Total	L	P	T	Total	Theory			Practical			Sem. Total
											CIE [20]	ISE [20]	ESE [60]	CIE TW	ESE P	ESE OR	
PCC	2635PC11	Theory of Computation (ToC)	3	-	-	3	3	-	-	3	20	20	60	-	-	-	100
PCC	2635PC12	Operating Systems (OS)	3	-	-	3	3	-	-	3	20	20	60	-	-	-	100
PCC	2635PC13	Artificial Intelligence and Machine Learning (AIML)	3	-	-	3	3	-	-	3	20	20	60	-	-	-	100
PCC	2635PC31	Operating System Laboratory (OSL)	-	2	-	2	-	1	-	1	-	-	-	25	25	-	50
PCC	2635PC32	Artificial Intelligence and Machine Learning Lab. (AIMLL)	-	2	-	2	-	1	-	1	-	-	-	25	25	-	50
PEC	2635PE1X	Program Elective-I (PEC-I)	2	-	-	2	2	-	-	2	20	20	60	-	-	-	100
PEC	2635PE3X	Program Elective-I Lab.	-	2	-	2	-	1	-	1	-	-	-	25	-	25	50
MDM	2635MD1X	MDM-3	2	-	-	2	2	-	-	2	20	20	60	-	-	-	100
MDM	2635MDXX	MDM-3 #	-	2	-	2	-	1	-	1	-	-	-	25	-	-	25
OE	2635OE5X	Open Elective-III *	-	-	2	2	-	-	2	2	-	-	50	-	-	-	50
AEC	2605AE31	Leadership and Management Skills (LMS)	-	2	-	2	-	1	-	1	-	-	-	25	-	-	25
Total			13	10	2	25	13	5	2	20	100	100	350	125	50	25	750

L: Lecture, P: Practical, T: Tutorial, CIE: Continuous Internal Evaluation, ISE: In-Semester Examination, ESE: End-Semester Examination, TW: Term work, OR: Oral, P: Practical examination.

#: Tutorial or laboratory as applicable. XX: Serial number of the courses under that particular category.

Program Elective Courses-I (PEC-1):

Domain Name	Course Code	Course Name
Network and Cyber Security	2635PE11 & 2635PE31	Information and Cyber Security (ICS) & Lab.
Intelligent Systems	2635PE12 & 2635PE32	Deep and Reinforcement Learning (DRL) & Lab.
Databases and Web Technologies	2635PE13 & 2635PE33	Data Warehouse & Data Mining (DDM) & Lab.
Advanced Cloud Computing	2635PE14 & 2635PE34	DevOps for Scalable Systems (DOSS) & Lab
Computer Graphics and Interactive Systems	2635PE15 & 2635PE35	Computer Graphics and Multimedia Systems (CGMS) & Lab.

Third Year B. Tech. Curriculum Structure (Semester -6) with effect from the A.Y. 2026-27

Semester -6			Teaching Scheme (Hours/Week)				Credit Scheme				Examination / Evaluation Scheme and Marks						
Course Type	Course code	Name of the Course	L	P	T	Total	L	P	T	Total	Theory			Practical			Sem. Total
											CIE [20]	ISE [20]	ESE [60]	CIE (TW)	ESE (P)	ESE (OR)	
PCC	2636PC11	Design and Analysis of Algorithms (DAA)	2	-	-	**2**	2	-	-	**2**	20	20	60	-	-	-	**100**
PCC	2636PC12	Cloud Computing and Virtualization (CCV)	2	-	-	**2**	2	-	-	**2**	20	20	60	-	-	-	**100**
PCC	2636PC31	IT Lab. - I (DAA & CCV)	-	4	-	**4**	-	2	-	**2**	-	-	-	25	25	-	**50**
PEC	2636PE1X	Program Elective-II	3	-	-	**3**	3	-	-	**3**	20	20	60	-	-	-	**100**
PEC	2636PE2X	Program Elective-III	3	-	-	**3**	3	-	-	**3**	20	20	60	-	-	-	**100**
PEC	2636PE3X	Program Elective-II Lab.	-	2	-	**2**	-	1	-	**1**	-	-	-	25	-	25	**50**
PEC	2636PE4X	Program Elective-III Lab.	-	2	-	**2**	-	1	-	**1**	-	-	-	25	-	25	**50**
MDM	2636MD1X	MDM-4	2	-	-	**2**	2	-	-	**2**	20	20	60	-	-	-	**100**
MDM	2636MDXX	MDM-4 #	-	2	-	**2**	-	1	-	**1**	-	-	-	25	-	-	**25**
OE	2636OE5X	Open Elective-IV *	-	-	2	**2**	-	-	2	**2**	-	-	50	-	-	-	**50**
VSEC	2636VE51	Seminar (Sem)	-	-	1	**1**	-	-	1	**1**	-	-	-	25	-	-	**25**
Total			**12**	**10**	**3**	**25**	**12**	**5**	**3**	**20**	**100**	**100**	**350**	**125**	**25**	**50**	**750**

#: Tutorial or laboratory as applicable. Choose one course from the MDM baskets. MDM: X is basket number, **Refer Annexure-I** for MDM details.

*: Open elective (OE) offered by online platform such as SWAYAM/NPTEL, **Refer Annexure-II** for details.

X: Serial number of the courses under that particular category.

Domain Name	Program Elective Courses-II (PEC-II)		Program Elective Courses-III (PEC-III)	
	Course Code	Course Name	Course Code	Course Name
Network and Cyber Security	2636PE11 & 2636PE31	Wireless and Adhoc Network Technologies (WANT) & Lab.	2636PE21 & 2636PE41	Blockchain Techniques (BT) & Lab.
Intelligent Systems	2636PE12 & 2636PE32	Computer Vision (CV) & Lab.	2636PE22 & 2636PE42	Scalable AI (SAI) & Lab.
Databases and Web Technologies	2636PE13 & 2636PE33	Modern Databases and Scalable Architectures (MDSA) & Lab.	2636PE23 & 2636PE43	Web Development Frameworks (WDF) & Lab.
Advanced Cloud Computing	2636PE14 & 2636PE34	Cloud-Native Application Development (CAD) & Lab.	2636PE24 & 2636PE44	Serverless Computing & Application Development (SCAD) & Lab.
Computer Graphics and Interactive Systems	2636PE15 & 2636PE35	Interface Design and Usability Engineering (IDUE) & Lab.	2636PE25 & 2636PE45	Virtual and Augmented Reality (VAR) & Lab.

Third Year B-Tech
(T. Y. B-Tech.)
Semester-5

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2635PC11]: Theory of Computation (TOC)

Semester	Credits	Teaching Scheme	Examination Scheme
5	03	L: 03 Hrs. / Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Set Theory, Relations and Formal Proof of Induction
- Knowledge of Tree and Stack

Course Objectives: The objective of this course is to provide students with

- To know the applicability of the model of computation to different problems.
- To understand in detail the relationship among formal languages, formal grammars and automata.
- To learn the design of Finite Automata, Pushdown Automata and Turing Machine for processing of formal languages.
- To learn the basics of a compiler to introduce the relationship between automata and the phases of a compiler.

Course Outcomes: After completing this course, students will be able to

CO1: Construct Finite Automata and write Regular Expressions for the Regular Languages.

CO2: Design and Simplify Context Free Grammar.

CO3: Design and Analyze Turing Machines for recognizing languages.

CO4: Construct Pushdown Automata for the Context Free Language.

COURSE CONTENTS

Module-I	Finite Automata and RE	12Hrs.	CO1

Introduction to Formal Languages: Alphabets, Strings, Languages; Finite Automata (FA): Deterministic Finite Automata (DFA); Non-deterministic Finite Automata (NFA); ε-NFA; Definitions and Constructions of FA; FA Conversions: NFA to DFA, ε-NFA to NFA to DFA, ε-NFA to DFA; Minimization of DFA. Finite automata with output (Moore and Mealy machines) and Inter conversion Regular Languages: Regular Expressions' operators, construction of RE, RE to FA, DFA to RE (Arden's Theorem); Closure properties and decision algorithms, Pumping Lemma for Regular Languages; use of RE in Lexical Analyzer

Module-II	Context Free Grammar	7 Hrs.	CO2

Context-Free Grammars (CFG): Definition, Derivations and Parse Trees; Ambiguity in CFGs; Chomsky hierarchy, Normal Forms (Chomsky Normal Form and Greibach Normal Form); Pumping Lemma for CFG

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Module-III	Turing Machine	9 Hrs.	CO3
Turing Machines: Formal Definition, Designing Turing Machines, Examples of simple Turing Machines, Step-by-step design and state diagram representation. Turing-recognizable (recursively enumerable) languages, Turing-decidable (recursive) languages. Variants. Halting Problem.			

Module-IV	Pushdown Automata	11 Hrs.	CO4
Definition, Model, Acceptance of CFL, Acceptance by Final State and Acceptance by Empty stack and its Equivalence, Equivalence of CFG and PDA. Non-deterministic PDA examples, Types of Parsers: Top-down & Bottom-up Parsers.			

Textbooks:

T1: J.E. Hopcroft et al., *Introduction to Automata Theory, Languages and Computation,* 3rd ed. Boston, MA, USA: Pearson (Addison-Wesley), ISBN-13: 978-8131720479, 2008.

T2: Daniel Cohen, *Introduction to Computer Theory*, 2nd ed. New York, NY, USA: Wiley, ISBN-13:978-8126513345,2007.

Reference Books:

R1: J. Martin, *Introduction to Languages and the Theory of Computation*, 4th ed. New York, NY, USA: McGraw-Hill, ISBN-13: 9780073191461,2010.

R2: M. Sipser, *Introduction to the Theory of Computation*, 3rd ed. Boston, MA, USA: Cengage Learning, ISBN-13: 9781133187813,2012.

R3: V. Kulkarni, *Theory of Computation*. 1st ed. New Delhi, India: Oxford University Press, ISBN-13: 9780198084587, 2008.

Relevant MOOCs Course (Course name and Weblink)

1. NPTEL- Theory of Computation, Prof. Raghunath Tewari, IIT Kanpur**,** https://nptel.ac.in/courses/106104148
2. NPTEL- Theory of Computation, Prof. Subrahmanyam Kalyanasundaram, IIT Hyderabad, https://nptel.ac.in/courses/106106242

Relevant Topics for Self-study:

1. https://www.geeksforgeeks.org/theory-of-computation/basic-theorems-in-toc-myhill-nerode-theorem/?utm_source=chatgpt.com
2. https://courses.grainger.illinois.edu/cs374/fa2015/slides/Undecidability-Rice.pdf?utm_source=chatgpt.com

Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2635PC12]: Operating Systems (OS)

Semester	Credits	Teaching Scheme	Examination Scheme
5	03	L: 03 Hrs. / Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Fundamentals of Computer Organization

Course Objectives: The objective of this course is to provide students with

- To introduce fundamental concepts of operating systems, covering process and thread management, interprocess communication, memory management, and I/O and file systems.
- To develop the ability to understand, apply, and evaluate operating system mechanisms for efficient resource management and system performance.

Course Outcomes: After completing this course, students will be able to

CO1: Discuss the core organization of computer systems and essential OS concepts involving processes, states, and threads.

CO2: Demonstrate the need and mechanisms of Interprocess Communication analyze concurrency challenges and apply synchronization tools to address classical problems and evaluate deadlock conditions with appropriate prevention, detection, avoidance, and recovery strategies.

CO3: Analyze memory management techniques, virtual memory with demand paging and address translation, and evaluate page replacement algorithms for efficient utilization.

CO4: Describe I/O techniques and disk scheduling, and evaluate file organization, directories, sharing, and storage management for efficient system performance.

COURSE CONTENTS

Module-I	Computer System Overview and Operating System Concepts	10 Hrs.	CO1

Computer system overview:
Basic organization of a computer (CPU, memory, I/O, system bus), Instruction Execution Cycle (Fetch–Decode–Execute, Datapath & control path), and Instruction Set Architecture (ISA – RISC vs CISC). I/O subsystems (Polling, Interrupts, DMA) and Memory Hierarchy (registers, cache, main memory, secondary storage, locality principle). Overview of Pipelined Architecture (pipeline stages, hazards).

Introduction to OS: Evolution, Types (Batch, Multiprogramming, Time-Sharing, Real-Time, Distributed, Mobile OS with Architecture); OS Services and System Calls; OS Structures – Monolithic, Layered, Microkernel.

Process: Concept of a Process, Process States, Process Description, Process Control (Process creation,

waiting for the process/processes, Loading programs into processes and Process Termination), Execution of the Operating System.

Threads: Processes and Threads, Concept of Multithreading, Types of Threads, Thread programming Using Pthreads.

Scheduling: Types of Scheduling, Scheduling Algorithms.

Module-II	**Interprocess Communication and Concurrency Control**	**10 Hrs.**	**CO2**

Interprocess Communication (IPC) – Pipes, Shared Memory, Message Passing. **Concurrency, Deadlocks & Synchronization** – Concurrency Issues: Critical Section Problem; Synchronization Mechanisms – Semaphores, Mutex, Monitors; Classical Synchronization Problems – Reader–Writer, Producer–Consumer, Dining Philosophers; Deadlocks – Prevention, Detection, Avoidance, Recovery.

Module-III	**Memory Management Techniques**	**10 Hrs.**	**CO3**

– Static and Dynamic Memory Allocation; Contiguous & Non-contiguous allocation; Paging & Segmentation; Virtual Memory – Concepts, Demand Paging, Thrashing, Address Translation; Page Replacement Algorithms – FIFO, LRU, Optimal.

Module-IV	**I/O Management and File Systems:**	**09 Hrs.**	**CO4**

I/O System (Hardware/Software Components, Programmed I/O, Interrupts, DMA) – Buffering, Caching & Spooling – Disk Scheduling Algorithms (FCFS, SSTF, SCAN, C-SCAN, LOOK).File Organization & Access (Pile, Sequential, Indexed, Hashed, B-Trees) – File Directories & Sharing (Structure, Naming, Access Rights, Simultaneous Access).Record Blocking – Secondary Storage Management (Allocation, Free Space, Volumes, Reliability).

Text Books:

T1: B. Stallings, *Operating Systems: Internals and Design Principles*, 9th ed. Boston, MA, USA: Pearson, 2018, ISBN-13: 978-9352866717

T2: A. S. Tanenbaum and H. Bos, *Modern Operating Systems*, 5th ed. London, UK: Pearson, 2015, ISBN-13: 978-9361595134

Reference Books:

R1: D. M. Dhamdhere, *Operating Systems: A Concept-Based Approach*, 3rd ed. New Delhi, India: McGraw Hill, 2017, ISBN: 978-1-25-906475-5

R2: A. S. Godbole and A. Kahate, *Operating Systems*, 3rd ed. New Delhi, India: McGraw Hill, 2016, ISBN: 978-0-07-070212-8

R3: M. J. Bach, *The Design of the UNIX Operating System*. Boston, MA, USA: Pearson, 1986, ISBN: 978-0-13-201799-2

R4: A. Silberschatz *et al*., *Operating System Concepts*, 10th ed. Hoboken, NJ, USA: Wiley, 2018, ISBN-13: 978-9357460569.

Relevant MOOCs Course (Course name and Weblink)

1. NPTEL: Operating Systems Fundamentals, IIT Kharagpur, Prof. Santanu Chattopadhyay
nptel.ac.in/courses/106105214

Relevant Topics for Self-study:

1. Cloud-native OS concepts and orchestration (Kubernetes)
2. Real-time operating systems (RTOS) fundamentals

Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2635PC13]: Artificial Intelligence and Machine Learning (AIML)

Semester	Credits	Teaching Scheme	Examination Scheme
5	03	L: 03 Hrs./ Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Basic Programming Skills
- Fundamentals of Data Structures
- Elementary Mathematics

Course Objectives: The objective of this course is to provide students with

- The fundamentals, scope, and applications of Artificial Intelligence and Machine Learning.
- To understand and apply various problem-solving and search strategies and solving AI-based problems efficiently**.**
- The knowledge and skills to understand, implement, and evaluate a variety of machine learning models for effective predictive modeling.
- To understand and apply distance-based clustering techniques, reinforcement learning concepts, and association rule mining methods, along with relevant performance measures, for solving diverse real-world data analysis and decision-making problems.

Course Outcomes: After completing this course, students will be able to

CO1: To describe the evolution and scope of AI & ML and differentiate between AI, ML, and related fields.

CO2: To formulate AI problems and implement appropriate search strategies to obtain efficient solutions.

CO3: To develop, optimize, and assess machine learning models to generate accurate and reliable predictive solutions.

CO4: To implement clustering, reinforcement learning and association rule mining techniques with appropriate performance evaluation to address real-world data analysis and decision-making tasks.

COURSE CONTENTS

Module-I	Introduction to Artificial Intelligence & Machine Learning	8 Hrs.	CO1

Introduction to Artificial Intelligence:

Foundations of Artificial Intelligence, History of Artificial Intelligence, State of the Art, Risks and Benefits of AI, Intelligent Agents, Agents and Environments, Good Behavior: Concept of Rationality, Nature of Environments, Structure of Agents.

Introduction to Machine Learning:

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Learning Tasks- Descriptive and Predictive Tasks, Types of Learning: Supervised Learning Unsupervised Learning, Semi-Supervised Learning, Reinforcement Learning.

Features: Types of Data (Qualitative and Quantitative), Scales of Measurement (Nominal, Ordinal, Interval, Ratio), Concept of Feature, Feature construction, Feature Selection and Transformation, Curse of Dimensionality.

Dataset Preparation: Training Vs. Testing Dataset, Dataset Validation Techniques – Hold-out, k-fold Cross validation, Leave-One-Out Cross-Validation (LOOCV).

Case Study: Data preprocessing for finance and social media.

Module-II	**AI Based Search & Optimization**	**8 Hrs.**	**CO2**

Problems Solving by Searching, Problem-Solving Agents, Example Problems, Search Algorithms, Uninformed Search Strategies, Informed (Heuristic) Search Strategies, Heuristic Functions, Search in Complex Environments, Local Search and Optimization Problems.

Case Study: AI for Game Design

Module-III	**Supervised Learning**	**13 Hrs.**	**CO3**

Regression: Introduction, Univariate Regression – Least-Square Method, Model Representation, Cost Functions: MSE, MAE, R-Square, Performance Evaluation, And Optimization of Simple Linear Regression with Gradient Descent. Types of regression-multivariate, polynomial.

Binary Classification: Linear Classification model, Performance Evaluation- Confusion Matrix, Accuracy, Precision, Recall, ROC Curves, F-Measure

Multi-class Classification: Model, Performance Evaluation Metrics Classification techniques -One vs One, One vs Rest.

Linear Models: Introduction, Linear Support Vector Machines (SVM) – Introduction, Soft Margin SVM, Introduction to various SVM Kernel to handle non-linear data – RBF, Gaussian, Polynomial, Sigmoid.

Tree Based Model: Decision Tree – Concepts and Terminologies, Impurity Measures -Gini Index, Information gain, Entropy, Tree Pruning -ID3/C4.5, Advantages and Limitations

Probabilistic Models: Naïve Bayes Classifier

Distance Based Models: K-Nearest Neighbor for Classification and distance-based methods

Case Study: Logistic regression for medical diagnosis

Module-IV	**Unsupervised and Rule Based Models**	**10 Hrs.**	**CO4**

Distance Based Models: K-means clustering, k-medoid algorithm-with example, Hierarchical Clustering, Performance Measures

Association Rule Mining: Introduction, Rule learning for subgroup discovery, Apriori Algorithm, Performance Measures – Support, Confidence, Lift.

Case Study: Reinforcement learning for autonomous driving

Text Books:

T1: Stuart Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach*, Third edition, Pearson, 2003, ISBN :10: 0136042597

T2: Deepak Khemani, *A First Course in Artificial Intelligence*, McGraw Hill Education (India), 2013, ISBN: 978-1-25-902998-1

Reference Books:

R1: Peter Flach, *Machine Learning: The Art and Science of Algorithms that Make Sense of Data,* Cambridge University Press, Edition 2012.

R2:	Ian H Witten et.al, *Data Mining, Practical Machine Learning Tools and Techniques*, Elsevier, 3rd Edition ,2011
R3:	Kevin P Murphy, *Machine Learning – A Probabilistic Perspective*, MIT Press, USA August 2012.

Relevant MOOCs Course (Course name and Weblink)

1. NPTEL- Introduction to Machine Learning, Prof. Balaraman Ravindran, IIT Madras
https://nptel.ac.in/courses/106106139
2. NPTEL-Deep Learning, Prof. Mitesh M. Khapra, IIT Madras
https://nptel.ac.in/courses/106106201
3. NPTEL-An introduction to Artificial Intelligence, Prof.Mausam, IIT Delhi
https://nptel.ac.in/courses/106102220

Relevant Topics for Self-study:

1. Ethical and Societal Implications of Artificial Intelligence
2. Feature Engineering in Machine Learning.

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)
[2635PC31]: Operating Systems Laboratory (OSL)

Semester	Credits	Teaching Scheme	Examination Scheme
5	01	P: 02Hrs. / Week	CIE (TW): 25 Marks ESE (PR): 25 Marks

Prerequisite: Students should have prior knowledge of

- Basic programming concepts in C/C++.
- Fundamentals of operating system concepts.

Course Objectives: The objective of this course is to provide students with

- Hands-on experience with Linux/Unix operating system environment, including shell scripting and system administration.
- Gain practical understanding of key operating system functions, including process management (creation, synchronization, inter-process communication), memory management, input/output (I/O), and disk management.
- Development of system-level programming skills using C and shell scripting.

Course Outcomes: After completing this course, students will be able to

CO1: Execute shell scripts for system administration tasks and automate routine operations in Linux Environment.

CO2: Implement process management concepts including process creation using fork(), exec() family functions, and handle orphan and zombie processes.

CO3: Develop and analyze various CPU scheduling algorithms (FCFS, SJF, Priority, Round Robin) and compare their performance metrics.

CO4: Implement inter-process communication mechanisms and synchronization techniques using semaphores, mutex locks, and shared memory.

CO5: Simulate and analyze memory management strategies including allocation algorithms and page Replacement Policies.

CO6: Implement and evaluate disk scheduling algorithms with seek time calculations for optimal disk I/O performance.

COURSE CONTENTS

Assign. No.	Description of Assignment	CO
1.	Title: **Familiarization with Linux/Unix environment:** **Basic commands, shell scripting, file permissions, and process utilities** **a)**Study and demonstrate usage of at least 20 basic Linux commands (ls, pwd, cd, cp, mv, cat, grep, find, chmod, chown, head, tail, diff, man, ps, top, kill, mkdir, rmdir, rm, touch etc.). Work with file/directory navigation and manipulation. **b**) Develop a mini student record management system using shell scripts that provides simple database operations (CRUD: Create, Read, Update, Delete) on a plain text (CSV) file. This project will help build skills in scripting, text parsing, and modular shell utility development.	CO1
2.	**Advanced Unix Process Management: fork(), Orphan & Zombie Handling, and exec()**	

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	Execution Replacement. **a)** Write a C program that creates a child process using fork(), where the parent prints even numbers and the child prints odd numbers independently. Display their PIDs and PPIDs to show process distinction. Demonstrate orphan and zombie processes by controlling the order of termination, and use wait() in the parent to properly clean up the child process **b)** Design two separate C programs—a parent and a child. The parent program creates a child process using fork() and, in the child, uses an exec() function (e.g., execv or execvp) to run the child program which calculates the factorial of a number passed as a command-line argument. The parent process waits for the child to complete and then prints a message confirming the child's successful execution.	CO2
3.	**Implementation and Analysis of CPU Scheduling Algorithms** Write a c program to implementation and analysis of CPU Scheduling Algorithms: FCFS, SJF, Priority, and Round Robin.	CO3
4.	**Implementation of IPC mechanisms.** Develop a simple chat application using **shared memory** for communication between two processes. Each process should be able to write messages to and read messages from a shared memory segment, implementing proper synchronization to ensure messages are exchanged without conflict.	CO4
5.	**Implementation of synchronization techniques using semaphores, and mutex locks and monitors**. **Producer–Consumer Problem:** Producer generates items and puts them in a buffer, while consumer removes them; synchronization (using semaphores/mutex) ensures no overflow or underflow. **Reader–Writer Problem:** Multiple readers can access data simultaneously, but writers need exclusive access; synchronization prevents conflicts and ensures data consistency.	CO4
6.	Simulation of memory allocation strategies – static, dynamic, contiguous (First Fit, Best Fit, Worst Fit) and non-contiguous allocation.	CO5
7.	Implement FIFO and LRU page replacement algorithms to simulate memory management. Input a page reference sequence, track page faults, and display memory pages after each reference. Provide the total count of page faults for both algorithms.	CO6
8.	Implementation of disk scheduling algorithms – FCFS, SSTF, SCAN, C-SCAN with seek time calculation.	CO6

Text Books:

T1: Sumitabha Das, *Unix Concepts and Applications*, 4th Edition, McGraw Hill, 2006, ISBN: 978-0-07-063546-3

T2: Neil Matthew and Richard Stones, *Beginning Linux Programming*, 4th Edition, Wiley, 2007, ISBN: 978-0-470-14762-7

Reference Books:

R1: W. Richard Stevens and Stephen A. Rago, *Advanced Programming in the UNIX Environment,* 3rd Edition, Addison-Wesley Professional, 2013, ISBN: 978-0-321-63773-4

R2: Andrew S. Tanenbaum and Herbert Bos*, Modern Operating System*, 4th Edition, Pearson, 2014, ISBN: 978-0-13-359162-0

R3: Abraham Silberschatze et.al, *Operating System Concepts*, 10th Edition, Wiley, 2018, ISBN: 978-1-119-45408-3cations, 4th Edition, McGraw Hill

Relevant MOOCs Course (Course name and Weblink):
1. NPTEL- Linux Programming & Scripting, Prof. Anand Iyer, IIT Madras nptel.ac.in/courses/117106113 2. NPTEL Operating Systems Fundamentals, Prof. Santanu Chattopadhyay, IIT Kharagpur nptel.ac.in/courses/106105214
Relevant Topics for Self-study:
1.Kernel compilation and customization in Linux. 2.Container orchestration using Kubernetes. 3.Real-Time Operating Systems (RTOS) and their applications.

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)
[2635PC32]: Artificial Intelligence and Machine Learning Laboratory (AIMLL)

Semester	Credits	Teaching Scheme	Examination Scheme
5	01	P: 02 Hrs. / Week	CIE (TW): 25 Marks ESE (PR): 25 Marks

Prerequisite: Students should have prior knowledge of

- Knowledge of Python programming and data structures
- Fundamentals of Probability, Statistics, and Linear Algebra
- Basic concepts of Artificial Intelligence and Machine Learning
- Data preprocessing and visualization techniques

Course Objectives: The objective of this course is to provide students with

- To develop the ability to design and implement AI-based search algorithms, such as A* and Greedy search, for solving game, graph, and optimization problems efficiently.
- To acquire skills in building and deploying basic chatbot applications for effective customer interaction using appropriate programming tools and techniques.
- To develop proficiency in preparing, analyzing, and preprocessing diverse real-world datasets, and applying appropriate machine learning techniques for predictive and descriptive analytics.
- To enhance the ability to evaluate machine learning models using relevant performance metrics and visualization techniques, enabling effective interpretation and decision-making in practical applications.

Course Outcomes: After completing this course, students will be able to

CO1: To implement and compare AI-based search algorithms, such as A* and Greedy search, to solve game, graph, and optimization problems effectively.

CO2: To design, develop, and deploy a basic chatbot application to address specific customer interaction scenarios.

CO3: To preprocess, analyze, and apply suitable machine learning models to real-world datasets for both predictive and descriptive analytics.

CO4: To assess machine learning model performance using appropriate metrics and visualizations to support data-driven decision-making.

COURSE CONTENTS

Assign. No.	Description of Assignment	CO
1.	Implement A star Algorithm for any game search problem.	CO1
2.	Implement Greedy search algorithm for any of the following application: I. Selection Sort II. Minimum Spanning Tree III. Single-Source Shortest Path Problem IV. Job Scheduling Problem V. Prim's Minimal Spanning Tree Algorithm VI. Kruskal's Minimal Spanning Tree Algorithm	CO1

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	VII. Dijkstra's Minimal Spanning Tree Algorithm	
3.	Develop an elementary chat boat for any suitable customer interaction application.	CO2
4.	Data preparation: Download heart dataset from following link. https://www.kaggle.com/zhaoyingzhu/heartcsv Perform following operation on given dataset. a) Find Shape of Data b) Find Missing Values c) Find data type of each column d) Finding out Zero's e) Find Mean age of patients f) Now extract only Age, Sex, ChestPain, RestBP, Chol. Randomly divide dataset in training (75%) and testing (25%).	CO3
5.	Download temperature data from below link. https://www.kaggle.com/venky73/temperatures- of-india?select=temperatures.csv This data consists of temperatures of INDIA averaging the temperatures of all place's month wise. Temperatures values are recorded in CELSIUS a. Apply **Linear & Logistic Regression** using suitable library function and predict the Month-wise temperature. b. Assess the performance of regression models using MSE, MAE and R-Square metrics c. Visualize simple regression model.	CO3
6.	Assignment on Classification technique Every year many students give the GRE exam to get admission in foreign Universities. The data set contains GRE Scores (out of 340), TOEFL Scores (out of 120), University Rating (out of 5), Statement of Purpose strength (out of 5), Letter of Recommendation strength (out of 5), Undergraduate GPA (out of 10), Research Experience (0=no, 1=yes), Admitted (0=no, 1=yes). Admitted is the target variable. Data Set Available on kaggle (The last column of the dataset needs to be changed to 0 or 1)Data Set : https://www.kaggle.com/mohansacharya/graduate-admissions The counselor of the firm is supposed check whether the student will get an admission or not based on his/her GRE score and Academic Score. So to help the counselor to take appropriate decisions build a machine learning model classifier using **Decision tree**& **Support Vector Machine** to predict whether a student will get admission or not. Apply Data pre-processing (Label Encoding, Data Transformation….) techniques if necessary. Perform data-preparation (Train-Test Split) C. Apply Machine Learning Algorithm D. Evaluate Model.	CO3
7.	Download the following customer dataset from below link: Data Set: https://www.kaggle.com/shwetabh123/mall-customers This dataset gives the data of Income and money spent by the customers visiting a shopping mall. The data set contains Customer ID, Gender, Age, Annual Income, Spending Score. Therefore, as a mall owner you need to find the group of people who are the profitable customers for the mall owner. Apply at least **two clustering algorithms** (based on Spending Score) to find the group of customers. a. Apply Data pre-processing (Label Encoding, Data Transformation….) techniques if necessary. b. Apply Machine Learning Algorithm c. Evaluate Model. d. Visualize clusters.	CO4
8.	Assignment on **Association Rule Learning**	CO4

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	Download Market Basket Optimization dataset from below link. Data Set: https://www.kaggle.com/hemanthkumar05/market-basket-optimization This dataset comprises the list of transactions of a retail company over the period of one week. It contains a total of 7501 transaction records where each record consists of the list of items sold in one transaction. Using this record of transactions and items in each transaction, find the association rules between items. There is no header in the dataset and the first row contains the first transaction, so mentioned header = None here while loading dataset. a. Follow following steps : b. Data Preprocessing c. Generate the list of transactions from the dataset d. Train Apriori algorithm on the dataset e. Visualize the list of rules F. Generated rules depend on the values of hyper parameters. By increasing the minimum confidence value and find the rules accordingly	

Text Books:

T1: Stuart Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach,* Third edition, Pearson, 2003, ISBN :10: 0136042597

T2: Ethem Alpaydin, *Introduction to Machine Learning*, PHI Learning Pvt. Ltd., 4th Edition, 2021, ISBN: 978-9356840225.

Reference Books:

R1: Christopher M. Bishop, *Pattern Recognition and Machine Learning*, Springer (Special Indian Edition), 2007, ISBN: 978-9380501413.

R2: Trevor Hastie et.al, *The Elements of Statistical Learning*, Springer (Indian Edition), 2nd Edition, 2017, ISBN: 978-8184894753.

R3: Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow*, Shroff Publishers (India), 3rd Edition, 2022, ISBN: 978-9355421456.

R4: Patrick Henry Winston, *Artificial Intelligence*, Addison-Wesley Publishing Company, ISBN: 0-201-53377-4

Relevant MOOCs Course (Course name and Weblink):

1.NPTEL: Machine Learning, Prof. Balaraman Ravindran, IIT Madras
https://nptel.ac.in/courses/106106139
2.NPTEL: Deep Learning, Prof. Mitesh M. Khapra, IIT Madras
https://nptel.ac.in/courses/106106201
3.NPTEL: The Joy of Computing using Python, Prof. Sudarshan Iyengar, IT Ropar
https://nptel.ac.in/courses/106106182

Relevant Topics for Self-study:

1. Exploring AI's potential benefits, risks, and impact on employment, privacy, bias, and decision-making in various industries.
2.Feature Engineering in Machine Learning

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Third Year B. Tech (T. Y B. Tech) AY (2026-27) Information Technology (IT) **Program Elective - I (PEC-I)**			
Semester	**Credits**	**Teaching Scheme**	**Examination Scheme**
5	**02**	**T: 2 Hr. / Week**	**CIE: 20 Marks** **ISE: 20 Marks** **ESE: 60 Marks**

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2635PE11]: Information & Cyber Security (ICS)

Semester	Credits	Teaching Scheme	Examination Scheme
5	2	L: 02 / Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Computer Network Technology

Course Objectives: The objective of this course is to provide students with

- Understand the foundational principles, models, and policies of information and cyber security.
- Analyse cryptographic techniques and digital security mechanisms to secure information systems.
- Apply cybersecurity fundamentals, architectures, and defence mechanisms to protect networks, applications, and endpoints.
- Evaluate emerging trends, tools, and governance frameworks in cybersecurity for practical and professional applications.

Course Outcomes: After completing this course, students will be able to

CO1: **Explain** (Remember/Understand) the fundamental concepts of information security, threats, vulnerabilities, and security models.

CO2: **Apply** (Apply) cryptographic techniques, authentication protocols, and key management systems to secure communication and data.

CO3: **Analyze** (Analyze) different types of cyber threats, security architectures, and IAM mechanisms for building secure systems.

CO4: **Evaluate & Design** (Evaluate/Create) advanced cybersecurity solutions using ethical hacking, penetration testing, and emerging technologies (cloud, IoT, AI, quantum).

COURSE CONTENTS

Module-I	Foundations of Information Security	06 Hrs.	CO1

Introduction to Information Security: Definitions, objectives, need, and importance, Evolution of information security & modern challenges, Threats, vulnerabilities, and attacks **Core Security Concepts**: CIA triad (Confidentiality, Integrity, Availability),Authentication, Authorization, Accountability (AAA), Non-repudiation, Privacy vs Security, Risk, exposure, and controls **Security Models & Policies:** Bell-LaPadula, Biba, Clark-Wilson models, IT Security policies: Organizational, IT, and data policies, Standards & regulations: ISO/IEC 27001, NIST, GDPR, HIPAA **Design Goals & Issues in Security:** Scalability, reliability, usability, Human and organizational factors, Cost-benefit analysis of security measures

Module-II	Cryptography and Digital Security	07 Hrs.	CO2

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Classical Cryptography: Substitution, transposition, polyalphabetic ciphers, Strengths and weaknesses
Modern Cryptographic Systems: Modes of operation (ECB, CBC, CFB, OFB, CTR), Symmetric key cryptography: DES, AES, Asymmetric key cryptography: RSA, Diffie–Hellman, ECC
Hashing, Authentication & Digital Security: Hash functions: SHA family, MD5 (limitations), Message Authentication Codes (MACs), Digital Signatures – concepts, process, applications
Key Management & Authentication Protocols: PKI, digital certificates, CAs, Kerberos protocol (Architecture & role of symmetric crypto), Enterprise authentication applications

Module-III	Cybersecurity Fundamentals	07 Hrs.	CO3

Introduction to Cybersecurity: InfoSec vs Cyber security, Cyber threats: Malware, Phishing, Ransomware, DoS/DDoS, MITM, Cyber kill chain & attack lifecycle
Cybersecurity Architecture & Defense: Defense-in-depth strategy, Perimeter security: Firewalls, IDS/IPS, Honeypots, Network security protocols: IPSec, SSL/TLS, HTTPS, SSH, VPNs
Endpoint & Application Security: Security for OS, databases, applications, Secure coding principles, Vulnerability scanning, patch management
Identity & Access Management (IAM): Authentication: Passwords, biometrics, MFA, Authorization: RBAC, ABAC, PBAC, Single Sign-On (SSO), Federated Identity Management

Module-IV	Advanced Cybersecurity and Emerging Trends	06 Hrs.	CO4

Ethical Hacking & Penetration Testing: Foot printing, scanning, enumeration, Exploitation & post-exploitation, Tools: Metasploit, Nmap, Wireshark
Cybersecurity in Emerging Technologies: Cloud security (SaaS, PaaS, IaaS challenges), IoT security (smart devices, IIoT threats), Mobile security (Android/iOS vulnerabilities)
Cybersecurity Governance & Legal Aspects: IT Act (India), international cyber laws & treaties, Digital forensics fundamentals
Future Directions in Cybersecurity: AI/ML in cybersecurity, Zero Trust architecture, Quantum cryptography, Cybersecurity workforce & career opportunities

Text Books:

T1: Michael E. Whitman & Herbert J. Mattord , *Principles of Information Security*, 7th Edition, 2021, Cengage Learning, ISBN: 9780357109310

T2: William Stallings & Lawrie Brown, *Computer Security: Principles and Practice*, 5th Edition, 2023, Pearson, ISBN: 9780137983790

Reference Books:

R1: Mark Stamp, *Information Security: Principles and Practice*, 3rd Edition, 2021, Wiley, ISBN: 9781119491796

R2: P.W. Singer & Allan Friedman ,*Cybersecurity and Cyberwar: What Everyone Needs to Know*, Updated Edition, 2022, Oxford University Press, ISBN: 9780190914666

R3: Jon Erickson ,*Hacking: The Art of Exploitation,* 2nd Edition, 2008, No Starch Press, ISBN: 9781593271442

R4: Behrouz A. Forouzan & Debdeep Mukhopadhyay, *Cryptography and Network Security*, 3rd Edition, 2019, McGraw Hill Education, ISBN: 9789332585225

Relevant MOOCs Course (Course name and Weblink)

1. NPTEL : Cryptography and Network Security - Prof. Sourav Mukhopadhyay ,IIT Kharagpur https://nptel.ac.in/courses/106105162
2. NPTEL : Foundations of Cryptography - Prof. Ashish Choudhury, IIIT Banglore https://nptel.ac.in/courses/106106221
3. NPTEL : Cyber Security and Privacy - Prof. Saji K Mathew – IIT Madras

https://nptel.ac.in/courses/106106248 4. NPTEL : Practical Cyber Security for Cyber Security Practitioners - Prof. Sandeep K. Shukla, IIT Kanpur https://nptel.ac.in/courses/106104467
Relevant Topics for Self-study:
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Third Year B. Tech. (T. Y. B. Tech) AY (2026-27)
Information Technology (IT)

[2635PE12]: Deep & Reinforcement Learning (DRL)

Semester	Credits	Teaching Scheme	Examination Scheme
5	02	L: 02 Hrs./ Week	ISE: 20 Marks CIE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Basic knowledge of Linear Algebra, Probability, Statistics
- Machine Learning and Programming in Python.

Course Objectives: The objective of this course is to provide students with

- To introduce the theoretical foundations, algorithms, methodologies, and application of neural networks and deep learning.
- To provide an understanding of Convolutional Neural Networks and their underlying principles.
- To understand advanced neural network architectures such as RNNs, LSTMs, Encoder-Decoder models, Autoencoders for data modeling.
- To introduce deep reinforcement learning and deep generative models, focusing on their frameworks, algorithms, architectures, and applications in decision-making and data generation tasks.

Course Outcomes: After completing this course, students will be able to

CO1: Understand the theoretical foundations, algorithms, and methodologies of neural network and deep learning.

CO2: Apply the concepts of Convolution Neural Networks and use of popular CNN architectures.

CO3: Analyze and implement RNNs, LSTMs, Encoder-Decoder models, and Autoencoders for solving problems in data modeling.

CO4: Apply reinforcement learning techniques such as Q-learning, Deep Q-Networks, and GAN architecture for solving sequential decision-making problems and generating synthetic data.

COURSE CONTENTS

Module-I	Introduction to Deep Learning	8 Hrs.	CO1

What is Deep Learning? Multilayer Perceptron, Feed-forward neural, Back Propagation, Gradient Descent, Vanishing Gradient Descent Problem,

Activation Functions: RELU, LRELU, ERELU, Optimization Algorithms

Hyperparameters: Layer size, Magnitude (momentum, learning rate), Regularization (dropout, drop connect, L1, L2)

Deep Reinforcement Learning: Introduction of Deep Reinforcement Learning, Markov Decision Process, Basic Framework of Reinforcement Learning, Dynamic Programming Algorithms for Reinforcement Learning, Q Learning and Deep Q-Networks, Deep Q recurrent networks

Module-II	Convolution Neural Network (CNN)	6 Hrs.	CO2

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Convolution Neural Network: Introduction to CNN, Convolution Operation, Parameter Sharing, Equivariant Representation, Pooling, Variants of the Basic Convolution Function, The basic Architecture of CNN, Popular CNN Architecture – AlexNet.

Module-III	Recurrent Neural Networks (RNN)	6 Hrs.	CO3

Recurrent Neural Networks: Types of Recurrent Neural Networks, Feed-Forward Neural Networks vs Recurrent Neural Networks, Long Short-Term Memory Networks (LSTM), Encoder Decoder Architectures, Recursive Neural Networks, Autoencoders

Module-IV	Deep Reinforcement Learning	6 Hrs.	CO4

Introduction to Deep Generative Model, Boltzmann Machine, Deep Belief Networks, Generative Adversarial Network (GAN), discriminator network, generator network, types of GAN, Applications of GAN networks

Case Study: Self-driving cars, Deep Learning for chatbots (customer support, ERP assistants), or GAN for detection of real or fake images.

Case Study: AI-based automated report generation in ERP and SAP systems.

Textbooks:

T1: Josh Patterson & Adam Gibson, *Deep Learning: A Practitioner's Approach*, O'Reilly Media, 2017 ,ISBN:978-1491914250

T2: Ian Goodfellow et.al. *Deep Learning*, MIT Press, 2017, Cambridge, Massachusetts, USA, ISBN-13: 978-0262035613

Reference Books:

R1: Richard S. Sutton and Andrew G. Barto, *Reinforcement Learning: An Introduction*, 2015

R2: N.D. Lewis, *Deep Learning Made Easy with R: A Gentle Introduction for Data Science*, January 2016

R3: Nikhil Buduma, *Fundamentals of Deep Learning Designing Next-Generation Machine Intelligence Algorithms,* O'Reilly,1st edition,

R4: Bengio and Yoshua, *Learning deep architectures for AI, Foundations and trends in Machine Learning,* 2009

Relevant MOOCs Course (Course name and Weblink)

1. NPTEL: Deep Learning, Prof. Sudarshan Iyengar, IIT Ropar
   https://nptel.ac.in/courses/106106184
2. NPTEL: Deep Learning, Prabir Kumar Biswas, IIT Kharagpur
   https://nptel.ac.in/courses/106105215

Relevant Topics for Self-study

1. Transfer Learning and Fine-Tuning Pre-trained Models
2. Attention Mechanisms and Transformers in Sequential Modeling
3. Ethical and Practical Challenges in GANs and Deep RL

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2635PE13]: Data Warehouse & Data Mining (DWDM)

Semester	Credits	Teaching Scheme	Examination Scheme
5	02	L: 02Hrs. /Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Basic knowledge of databases (SQL), programming (preferably Python/Java), data structures, and fundamental statistics/probability

Course Objectives: The objective of this course is to provide students with

- Understanding of the basic principles, concepts, and applications of data warehousing and data mining.
- An introduction to data mining as a vital phase in the knowledge discovery process .
- Familiarity with the conceptual, logical, and physical design of data warehouses along with OLAP applications and deployment.
- Knowledge of the fundamental concepts that form the foundation of data mining.

Course Outcomes: After completing this course, students will be able to

CO1: Design a data mart or data warehouse for any organization.
CO2: Extract knowledge using data mining techniques.
CO3: Apply the techniques of clustering, classification, association finding, feature selection and visualization to real world data.
CO4: Apply and compare classification techniques to improve predictive accuracy in real-world datasets.

COURSE CONTENTS

Module-I	DATA WAREHOUSING AND ON-LINE ANALYTICAL PROCESSING	8 Hrs.	CO1

What is data warehousing and data mining, A Multi-dimensional data model, Multi-dimensional Data Cubes, Star, Star Flakes, & Fact Constellation Schema, Concept Hierarchies, OLAP. Data Warehouse Architecture. Steps for design and construction of data warehouse, 3-tier data warehouse architecture, ROLAP, MOLAP, HOLAP, Data Pre-Processing, Overview, need for pre-processing Issues related to efficient data handling

Module-II	INTRODUCTION TO DATA MINING	6 Hrs.	CO2

Motivation, Importance, Definition of Data Mining, Kind of Data, Data Mining Functionalities, Kinds of Patterns, Classification of Data Mining Systems, Data Mining Task Primitives, Integration of A Data

Mining System with A Database or Data Warehouse System, Major Issues in Data Mining, Types of Data Sets and Attribute Values, Basic Statistical Descriptions of Data, Data Visualization, Measuring Data Similarity.

Module-III	MINING FREQUENT PATTERNS, ASSOCIATIONS AND CORRELATIONS	6 Hrs.	CO3

Basic Concepts, Efficient and Scalable Frequent Item set Mining Methods, Interestingness of Patterns, Pattern Evaluation Methods, Applications of frequent pattern and associations.

Frequent Pattern and Association Mining: A Road Map, Mining Various Kinds of Association Rules, Constraint-Based Frequent Pattern Mining, Extended Applications of Frequent Patterns.

Module-IV	CLASSIFICATION TECHNIQUES IN DATAMINING	6 Hrs.	CO4

Basic Concepts, Decision Tree Induction, Bayesian Classification Methods, Rule-Based Classification, Model Evaluation and Selection, Techniques to Improve Classification Accuracy: Ensemble Methods, Handling Different Kinds of Cases in Classification, Bayesian Belief Networks.

Text Books:

T1: J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed. Waltham, MA, USA: Elsevier (Morgan Kaufmann), 2012. ISBN: 978-0-12-381479-1.

T2: P. Ponniah, *Data Warehousing Fundamentals for IT Professionals*, 2nd ed. Hoboken, NJ, USA: Wiley, 2010. ISBN: 978-0-470-46207-2.

Reference Books:

R1: M. H. Dunham, *Data Mining: Introductory and Advanced Topics*, 2nd ed. New Delhi, India: Pearson Education, 2006. ISBN: 978-8131715209

R2: A. Sinha, *Data Warehousing*. India: Thomson Learning, 2007. ISBN: 978-8177587000.

R3: X. Wu and V. Kumar, *The Top Ten Algorithms in Data Mining*. Boca Raton, FL, USA: CRC Press, 2009. ISBN: 978-1420067365.

R4: A. Berson and S. J. Smith, *Data Warehousing, Data Mining, and OLAP*. New York, NY, USA: McGraw-Hill, 1997. ISBN: 978-0-07-005059-4.

Relevant MOOCs Course (Course name and Weblink)

1. NPTEL: Data Mining, Prof. Pabitra Mitra, IIT Kharagpur
   https://nptel.ac.in/courses/106105174
2. NPTEL: Business analytics and data mining Modelling using R, Prof. Gaurav Dixit, IIT Roorkee,
   https://nptel.ac.in/courses/110107092

Relevant Topics for Self-study:

1. Clustering Techniques in Data Mining
2. ETL (Extract, Transform, Load) Processes in Data Warehousing

Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2635PE14]: DevOps for Scalable Systems (DOSS)

Semester	Credits	Teaching Scheme	Examination Scheme
5	02	L: 02Hrs. /Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Basic Programming & Scripting Knowledge
- Foundations of Operating Systems & Networking
- Fundamentals of Software Development & Web Applications

Course Objectives: The objective of this course is to provide students with

- A strong foundation in DevOps principles and practices
- Infrastructure as Code (IaC) and cloud technologies
- To build, orchestrate, and monitor containerized applications at scale
- To integrate advanced DevOps practices for reliability, security, and scalability

Course Outcomes: After completing this course, students will be able to

CO1: Apply DevOps principles, version control, and CI/CD pipelines to streamline software development and deployment.

CO2: Implement Infrastructure as Code (IaC) practices to provision and manage scalable cloud infrastructure.

CO3: Deploy and manage containerized applications using Docker and Kubernetes with monitoring and performance optimization

CO4: Integrate DevSecOps, Chaos Engineering, and SRE practices to design secure, resilient, and highly scalable systems

COURSE CONTENTS

Module-I	DevOps Fundamentals for Scalable Systems	8 Hrs.	CO1

Introduction to DevOps and scalable systems: Core principles, collaboration, automation, continuous improvement. Relevance to building scalable architecture.

Version control and branching strategies for large projects: Advanced Git concepts and workflows, GitFlow and trunk-based development for distributed teams.

CI/CD pipelines for distributed systems: Continuous Integration & Deployment fundamentals, Orchestration of builds, tests, and deployments in large-scale systems.

Principles of scalable system design: Microservices architecture, load balancing, and fault tolerance, Patterns for scalability and reliability.

DevOps culture and collaboration in large teams: Fostering collaboration in distributed teams, Shared ownership, feedback loops, and communication strategies.

Module-II	Infrastructure as Code (IaC) and Cloud Platforms for Scalability	6 Hrs.	CO2

Introduction to IaC : Benefits of IaC in DevOps for managing infrastructure at scale.
Popular IaC tools and platforms: Terraform, AWS CloudFormation, Azure ARM, Ansible: features and comparisons.
Designing scalable cloud infrastructure: Best practices in deploying VMs, containers, networks, and storage.
Performance tuning and cost optimization.
Multi-cloud and hybrid cloud deployment with IaC : Strategies for managing workloads across multiple cloud providers.
Automation with IaC and CI/CD integration: Integrating IaC into CI/CD pipelines for automated infrastructure provisioning and configuration management.

Module-III	Containerization, Orchestration, and Monitoring	6 Hrs.	CO3

Containerization with Docker: Images, containers, networking, and volumes. Building scalable and portable applications.
Container orchestration with Kubernetes: Kubernetes architecture, deployments, services, and scaling.
Designing and deploying scalable applications with Kubernetes: Microservices on Kubernetes for high availability and fault tolerance.
Monitoring and logging for high-performance systems: Tools: Prometheus, Grafana, ELK Stack.
Centralized monitoring and observability practices.
Performance optimization and troubleshooting: Identifying bottlenecks and improving system performance.

Module-IV	DevSecOps and Advanced Practices	6 Hrs.	CO4

Integrating security into the DevOps lifecycle (DevSecOps) : Shifting security left and embedding it in CI/CD pipelines. Automated security testing.
Security best practices for cloud-native applications: Container security, secrets management, and access control.
Chaos Engineering and resilience testing: Simulating failures to strengthen distributed systems.
Site Reliability Engineering (SRE) principles: Service Level Objectives (SLOs), SLIs, error budgets, and reliability practices.
Case studies: Successful DevOps in large enterprises: Real-world insights on scaling DevOps in enterprise systems.

Text Books:

T1: G. Kim et al., *The DevOps Handbook: How to Create World-Class Agility, Reliability, & Security in Technology Organizations*, Portland, OR, USA: IT Revolution Press, 2021, ISBN: 978-1942788003.

T2: J. Arundel and J. Domingus, *Cloud Native DevOps with Kubernetes: Building, Deploying, and Scaling Modern Applications in the Cloud*, Sebastopol, CA, USA: O'Reilly Media, 2019, ISBN: 978-1492040767.

Reference Books:

R1: N. R. Murphy et al., *Site Reliability Engineering: How Google Runs Production Systems*, Sebastopol, CA, USA: O'Reilly Media, 2016, ISBN: 978-1491929124.

R2: M. Kleppmann, *Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems*, Sebastopol, CA, USA: O'Reilly Media, 2017, ISBN: 978-1449373320.

R3: J. Humble and D. Farley, *Continuous Delivery: Reliable Software Releases through Build, Test, and Deployment Automation*, Boston, MA, USA: Addison-Wesley Professional, 2010, ISBN: 978-0321601919.

R4:	N. Forsgren et al., *Accelerate: The Science of Lean Software and DevOps – Building and Scaling High Performing Technology Organizations*, Portland, OR, USA: IT Revolution Press, 2018, ISBN: 978-1942788331.978-1942788331

Relevant MOOCs Course (Course name and Weblink)

1. Introduction to DevOps – IBM (Coursera) : https://www.coursera.org/courses?query=devops
2. Scalable Deployment and DevOps for Web Development – Coursera (Board Infinity) : https://www.coursera.org/learn/scalable-deployment-and-devops-for-web-development
3. Cloud & DevOps: Continuous Transformation – MIT Professional Education : https://professionalprograms.mit.edu/online-program-cloud-devops/

Relevant Topics for Self-study:

1. Advanced Git workflows (rebasing, GitOps),
2. Cloud-native patterns (serverless computing, service meshes, disaster recovery),
3. Container orchestration enhancements (Helm charts, Kubernetes Operators, stateful workloads). Additionally, emphasis is placed on monitoring.
4. Observability with distributed tracing, chaos engineering, and autoscaling policies.
5. DevSecOps practices, covering vulnerability scanning, secrets management, and compliance frameworks,
6. AIOps, FinOps, and edge computing.

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2635PE15]: Computer Graphics & Multimedia Techniques (CGMT)

Semester	Credits	Teaching Scheme	Examination Scheme
5	02	L: 02 Hrs. / Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Basic Geometry, Trigonometry, Vectors and Matrices, Data Structures and Algorithm

Course Objectives: The objective of this course is to provide students with

- Understand the fundamentals of computer graphics: Explore the principles of graphics systems, including display devices, graphics hardware, and modern input-output technologies used in computer graphics.
- Master 2D and 3D transformations: Learn and apply various transformation techniques for both 2D and 3D graphics, including translation, scaling, rotation, and projection methods, to solve real-world problems.
- Develop animation and visual effects skills: Acquire knowledge of animation techniques and visual effects (VFX) principles, and understand their application in film production, gaming, and multimedia content creation.
- Explore multimedia technologies: Gain practical knowledge of multimedia elements, file formats, compression techniques, and tools for multimedia development, with an emphasis on integrating multimedia content into digital platforms.

Course Outcomes: After completing this course, students will be able to

CO1: Explain the concepts of raster and vector graphics, resolution, refresh rates, and the hardware/software components used in modern graphical systems.

CO2: Apply 2D and 3D transformations, such as translation, scaling, and rotation, and effectively implement projection techniques and clipping algorithms in graphical systems.

CO3: Create animated sequences using keyframe animation, procedural animation, and advanced techniques, along with integrating visual effects (VFX) in multimedia applications.

CO4: Demonstrate proficiency in employing multimedia tools for content creation, implementing file formatting and compression techniques, and integrating diverse multimedia elements into digital and web-based platforms effectively

COURSE CONTENTS

Module I	**Introduction to Computer Graphics and Modern Systems**	**08 Hrs**	**CO1**

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Overview of Computer Graphics : Definition and Applications, Difference between Raster and Vector Graphics, Refresh Rate, Frame Buffer, and Resolution Concepts, Graphics Hardware and Software Requirements.

Modern Input Output Devices : Display Devices (CRT, LCD, LED, OLED, Quantum Dot Displays, and MicroLED technology. etc.), Input Devices (Mouse, Keyboard, Light Pen, Touch Screen, Advanced touch interfaces, VR controllers, and motion capture devices, Graphics System Components: Frame Buffer, Display Controller, and pipeline overview.

Advanced Display Techniques : Working principles of HDR (High Dynamic Range) displays. Frame buffers for high-performance gaming. Introduction to Variable Refresh Rate (VRR) and resolution upscaling.

Basic Scan Conversion Algorithms. : Line Drawing: DDA Line drawing algorithm, Bresenham's Line drawing algorithm, Circle Drawing: Bresenham's circle drawing algorithm.

Case Study : Eclipse drawing algorithm

Module II	2D and 3D Transformations for Real-World Applications	06 Hrs	CO2

Basic 2D Transformations: Translation, Scaling, Rotation, Shearing, and Reflection, Homogeneous Coordinates and Matrix Representations, Composite Transformations and Transformation Matrices

3D Transformations: 3D Coordinate Systems and Axes, Translation, Scaling, Rotation in 3D, Projection Techniques: Parallel and Perspective Projections

Viewing Pipeline and Clipping: Window-to-Viewport Transformation, Clipping Algorithms: Cohen-Sutherland Line Clipping, Polygon Clipping (Sutherland-Hodgman Algorithm)

Case Study: Liang-Barsky Line Clipping Algorithm for clipping

Module III	Graphics, Animation & Visual Effects	06 Hrs	CO3

Illumination and Shading Models: Basic Illumination Models (Ambient, Diffuse, Specular Reflection), Shading Models: Gouraud Shading, Phong Shading, Color Models: RGB, CMYK, YUV

Rendering Techniques: Hidden Surface Removal Techniques, Ray Tracing and Rasterization Techniques, Anti-Aliasing Techniques

Animation: Introduction to Animation, Types of Animation: Traditional, 2D, 3D, Stop Motion
Animation Pipelines: Storyboarding, Keyframing, and Rendering. 12 Principles of Animation.
Techniques in Animation: Keyframe Animation and Tweening, Procedural Animation and Physics-based Simulation, Advanced Animation Techniques

Visual Effects (VFX): Introduction to Visual Effects, Role of VFX in Film and Gaming, Green Screen (Chroma Keying) and Matte Painting, Matchmoving and Camera Tracking
Compositing Techniques, Industry Tools for VFX

Case Study: Animation and VFX in a Film Production

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Module IV	Multimedia Technologies for the Digital Era	06 Hrs	CO4

Basics of Multimedia, Elements of Multimedia: Text, Graphics, Audio, Video, Animation, Interactivity. Applications of Multimedia in Engineering

Multimedia File Formats and Compression: BMP, JPEG, PNG, MP3, MP4, etc. Compression Techniques: Lossy Compression (MP3, H.264), Lossless Compression (PNG, FLAC), Role of Codecs in Multimedia Systems

Tools for Multimedia Development: Overview of Popular Tools: Inkscape, GIMP etc.

Video Editing: OpenShot, Shotcut etc. 3D Modeling: Blender. Basics of Multimedia Integration in Web (H5P), Audio recording (Audacity) and screen recording and live streaming Applications OBS Studio (Open Broadcaster Software)

Case Studies: Educational Multimedia Content or Marketing Campaigns using Multimedia

Text Books:

T1: D. D. Hearn and M. P. Baker, *Computer Graphics with OpenGL*, 4th ed. Boston, MA, USA: Pearson, 2014, ISBN: 978-0132309981.

T2: J. F. Hughes et al., *Computer Graphics: Principles and Practice*, 3rd ed. Boston, MA, USA: Addison-Wesley, 2013, ISBN: 978-0321399526.

Reference Books:

R1: S. Harrington, *Computer Graphics*, 2nd ed. New York, NY, USA: McGraw-Hill, 1987, ISBN: 0-07-100472-6.

R2: D.Salomon, *Data Compression: The Complete Reference*, 4th ed. New York, NY, USA: Springer, 2010, ISBN: 978-1441919635.

R3: M. Bailey and S. Cunningham, *Graphics Shaders: Theory and Practice*, 2nd ed. Boca Raton, FL, USA: A K Peters/CRC Press, 2012, ISBN: 978-1568817063.

Relevant MOOCs Course (Course name and Weblink)

1. NPTEL-Computer Graphics, Prof. Samit Bhattacharya, IIT Guwahati, https://nptel.ac.in/courses/106103224

Relevant Topics for Self-study:

1. Advanced Git workflows (rebasing, GitOps),
2. Cloud-native patterns (serverless computing, service meshes, disaster recovery),
3. Container orchestration enhancements (Helm charts, Kubernetes Operators, stateful workloads). Additionally, emphasis is placed on monitoring.
4. Observability with distributed tracing, chaos engineering, and autoscaling policies.

DevSecOps practices, covering vulnerability scanning, secrets management, and compliance frameworks,

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

Program Elective – I Laboratory (PECL-I)

Semester	Credits	Teaching Scheme	Examination Scheme
5	01	P: 02 Hr. / Week	CIE (TW): 25 Marks OR : 25 Marks

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)
[2635PE31]: Information & Cyber Security Laboratory (ICSL)

Semester	Credits	Teaching Scheme	Examination Scheme
5	01	P: 02 Hrs. / Week	CIE (TW): 25 Marks ESE (OR): 25 Marks

Prerequisite: Students should have prior knowledge of

- Computer Network Technology

Course Objectives: The objective of this course is to provide students with

- Introduction of fundamental concepts of classical and modern cryptography.
- Hands-on skills in implementing symmetric and asymmetric encryption algorithms.
- Designing a secure client–server communication using cryptographic methods.
- Understanding of digital signatures, authentication, and integrity mechanisms.
- Practical exposure to real-world security threats such as packet sniffing and MITM attacks.

Course Outcomes: After completing this course, students will be able to

CO1: Implement and analyze classical ciphers (Playfair, Vigenère) and modern ciphers (RSA, DES).

CO2: Design secure communication protocols using symmetric and asymmetric key cryptosystems.

CO3: Apply Diffie–Hellman key exchange for secure session key establishment.

CO4: Implement and verify RSA-based digital signature schemes for ensuring authenticity.

CO5: Demonstrate awareness of network vulnerabilities by simulating packet sniffing and MITM attacks.

COURSE CONTENTS

Assign. No.	Description of Assignment	CO
1.	**Implementation of Polyalphabetic Cipher (Playfair / Vigenère) in C/C++** 1. Generate the key matrix (for Playfair Cipher) or use a keyword (for Vigenère Cipher). 2. Implement encryption and decryption functionalities for given plaintext and ciphertext. 3. Handle edge cases such as repeated letters, special characters, and padding in the text. 4. Demonstrate the working of both encryption and decryption using sample input/output.	CO1
2.	**Secure Client–Server Communication using RSA Cryptosystem in Python** 1. Implement a TCP-based client and server running on different machines. 2. Generate RSA public and private keys for encryption and decryption.	CO2

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	3. Ensure that messages sent by the client are encrypted using the server's public key. 4. Ensure that the server decrypts the received messages using its private key. 5. Demonstrate successful and secure communication between the two entities.	
3.	**Client–Server Communication using DES with Diffie–Hellman Key Exchange in Python** 1. Establish a TCP connection between a client and server on different machines. 2. Implement Diffie–Hellman key exchange to securely share a secret session key. 3. Use the derived session key for message encryption and decryption with the DES algorithm. 4. Demonstrate secure end-to-end communication with encrypted messages. 5. Handle possible exceptions such as mismatched keys or corrupted messages.	CO3
4.	**Digital Signing and Verification using RSA in Python** 1. Implement a client and server that can communicate over a TCP connection. 2. Use RSA to generate key pairs for both signing and verification. 3. Implement a mechanism for the sender to digitally sign a message using their private key. 4. Implement verification at the receiver's end using the sender's public key. 5. Demonstrate message integrity and authenticity using the RSA digital signature scheme.	CO4
5.	**Packet Sniffing and MITM Attack Demonstration using Python** 1. Build a simple packet sniffer using Scapy (Python) or Jpcap (Java). 2. Capture and analyze network traffic, focusing on HTTP requests and responses. 3. Extract credentials (if present) from intercepted HTTP traffic. 4. Compare the captured traffic with HTTPS communication (where data is encrypted). 5. Demonstrate a basic Man-in-the-Middle (MITM) attack scenario and highlight security implications.	CO5

Text Books:

T1: William Stallings, *Cryptography and Network Security: Principles and Practice*, Pearson Education, 7th Edition, Global Edition, March 2017, ISBN: 978-0134444284**.**

T2: Behrouz A. Forouzan, *Cryptography and Network Security*, McGraw-Hill Education, 1st Edition, June 2007, ISBN: 978-0070660465**.**

Reference Books:

R1: Bruce Schneier, *Applied Cryptography: Protocols, Algorithms, and Source Code in C*, Wiley, 2nd Edition, October 1995, ISBN: 978-0471117094**.**

R2: Alfred J. Menezes, Paul C. van Oorschot, Scott A. Vanstone, *Handbook of Applied Cryptography*, CRC Press, 1st Edition, October 1996, ISBN: 978-0849385230**.**

R3: Christof Paar, Jan Pelzl, *Understanding Cryptography: A Textbook for Students and Practitioners*, Springer, 1st Edition, January 2010, ISBN: 978-3642041006**.**

R4: Douglas R. Stinson, Maura B. Paterson, *Cryptography: Theory and Practice*, CRC Press, 4th Edition, August 2018, ISBN: 978-1138197015.

Relevant MOOCs Course (Course name and Weblink):

1. NPTEL : Cryptography and Network Security - Prof. Sourav Mukhopadhyay, IIT Kharagpur

https://nptel.ac.in/courses/106105162

2. NPTEL : Foundations of Cryptography - Prof. Ashish Choudhury, IIIT Banglore
   https://nptel.ac.in/courses/106106221
3. NPTEL : Cyber Security and Privacy - Prof. Saji K Mathew – IIT Madras
   https://nptel.ac.in/courses/106106248
4. NPTEL : Practical Cyber Security for Cyber Security Practitioners - Prof. Sandeep K. Shukla, IIT Kanpur
   https://nptel.ac.in/courses/106104467

Relevant Topics for Self-study:

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2635PE32]: Deep & Reinforcement Learning Laboratory (DRLL)

Semester	Credits	Teaching Scheme	Examination Scheme
5	01	P: 02 Hrs. / Week	CIE (TW): 25 Marks ESE (OR): 25 Marks

Prerequisite: Students should have prior knowledge of

- Machine Learning
- Deep Learning

Course Objectives: The objective of this course is to provide students with

- Basics of neural networks and deep learning.
- Understanding of Convolutional and Recurrent Neural Networks.
- Applying Deep Reinforcement Learning models.
- Implementation of Deep Generative Models.
- Construction and application of reinforcement learning algorithms.

Course Outcomes: After completing this course, students will be able to

CO1: Implement fundamental deep learning models including DNNs, CNNs, and RNNs using industry-standard frameworks to solve classification tasks.

CO2: Analyze model performance and **diagnose** issues like overfitting by applying techniques such as regularization and visualization.

CO3: Design and **construct** reinforcement learning agents by formulating problems such as Markov Decision Processes and implementing value-based algorithms.

CO4: Develop and **evaluate** Deep Q-Networks (DQN) with key stability enhancements to solve continuous control tasks.

COURSE CONTENTS

Assign. No.	Description of Assignment	CO
1.	**Combating Overfitting in Image Classification** **Problem Statement:** Implement a Deep Neural Network (DNN) using Keras/TensorFlow to classify the **Fashion-MNIST** dataset. First, train a model that overfits the training data. Then, introduce **Dropout layers** and L2 regularization to mitigate overfitting. Compare the training and validation accuracy/loss curves of both models.**Dataset: Fashion-MNIST**	CO1
2.	**Feature Extraction using a Convolutional Neural Network (CNN)** **Problem Statement:** Build, train, and evaluate a Convolutional Neural Network (CNN) for image classification on the CIFAR-10 dataset. Your architecture must include at least two convolutional layers (with ReLU activation and max-pooling) and one fully connected layer. Visualize the feature maps (outputs) of the first convolutional layer for a few sample images to interpret what features the network learns. **Dataset: CIFAR-10**	CO2
3.	**Sentiment Analysis using Long Short-Term Memory (LSTM) Networks**	CO1

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	Problem Statement: Develop a sentiment analysis model that classifies movie reviews from the IMDB dataset as positive or negative. Use an Embedding layer followed by an LSTM layer. Limit the review length to the first 200 words and the vocabulary to the top 5000 words to manage training time. Report the final accuracy on the test set. **Dataset: IMDB Movie Reviews**	
4.	**Solving a Game using Tabular Q-Learning** **Problem Statement**: Implement the Q-Learning algorithm from scratch (using a Q-table) to solve the **FrozenLake-v1** environment from OpenAI Gym. The agent must learn to navigate from the start (S) to the goal (G) without falling into holes (H). Implement an ε-greedy policy for exploration. Plot the success rate over training episodes to show the learning progress. **Environment: FrozenLake-v1**	CO3
5.	**Mastering a Game with Deep Q-Networks (DQN)** **Problem Statement:** Implement a Deep Q-Network (DQN) agent to solve the CartPole-v1 environment. Use a neural network to approximate the Q-function. Utilize **experience replay** and a target network for stable training. Your agent must achieve an average reward of at least 195 over 100 consecutive episodes. Plot the total reward per episode to demonstrate the learning trajectory. **Environment:** CartPole-v1	CO4
6.	**Mini Project** **Problem Statement:** Design and implement a **Q-Learning agent** that learns to play Tic-Tac-Toe optimally against a random opponent. You must define the state representation (e.g., a flattened 3x3 board), action space (9 possible moves), and reward function (e.g., +1 for win, -1 for loss, 0 for draw). Train the agent and demonstrate that it never loses (it either wins or draws) in 100 games against the random opponent. **Environment:** A custom Tic-Tac-Toe environment. (Students can create a simple class for this or use a pre-built one from repositories like gym-tictactoe).	CO3

Textbooks:

T1: Ian Goodfellow et.al. *Deep Learning*, MIT Press, 2017, Cambridge, Massachusetts, USA, ISBN-13: 978-0262035613

T2: Charu C. Aggarwal, *Neural Networks and Deep Learning*, Springer, Cham, Switzerland, 2018, ISBN-13: 978-3319944630.

Reference Books:

R1: Richard S. Sutton and Andrew G. Barto, *Reinforcement Learning: An Introduction,* Second Edition MIT Press, Cambridge, MA, 2018, ISBN-13: 978-0262352703

R2: N. D. Lewis,*Deep Learning Made Easy with R: A Gentle Introduction for Data Science*, CreateSpace Independent Publishing Platform, Scotts Valley, CA, USA, 2016, ISBN-13: 978-1535398190.

R3: Nikhil Buduma, *Fundamentals of Deep Learning: Designing Next-Generation Machine Intelligence Algorithms*,O'Reilly Media, Sebastopol, CA, USA, 2017, ISBN-13: 978-1491925614.

Relevant MOOCs Course (Course name and Weblink):

1. NPTEL: Deep Learning, Prof. Sudarshan Iyengar, IIT Ropar
   https://nptel.ac.in/courses/106106184
2. NPTEL: Deep Learning, Prabir Kumar Biswas, IIT Kharagpur
   https://nptel.ac.in/courses/106105215

Relevant Topics for Self-study:

1. Regularization Techniques
2. How Convolutional Layers Learn

3. Solving the Vanishing Gradient Problem
4. The Exploration-Exploitation Dilemma
5. Value Function Approximation
6. MDP Design and Project Scoping

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)
[2635PE33]: Data Warehouse & Data Mining Laboratory (DWDML)

Semester	Credits	Teaching Scheme	Examination Scheme
5	01	P: 02 Hrs. / Week	CIE (TW): 25 Marks ESE (OR): 25 Marks

Prerequisite: Students should have prior knowledge of

- Basic knowledge of SQL, database concepts, and fundamentals of programming & statistics.

Course Objectives: The objective of this course is to provide students with

- Ability to build and query data warehouses using open-source tools.
- Implementation of data mining tasks with toolkits like WEKA.
- Understanding of datasets and applying preprocessing techniques.
- Demonstration of algorithms for association rule mining and classification.
- Applying data mining techniques with different parameters.

Course Outcomes: After completing this course, students will be able to

CO1: Execute PL/SQL commands and construct multidimensional data models using SQL and OLAP operations.

CO2: Apply preprocessing techniques in WEKA for attribute selection and handling missing values.

CO3: Implement decision tree algorithms and evaluate them using information gain measures.

CO4: Perform frequent pattern mining using FP-Growth and analyze discovered patterns.

COURSE CONTENTS

Assign. No.	Description of Assignment	CO
1.	Review and Perform various commands given in PL/SQL using MySQL.	CO1
2.	Perform multi-dimensional data model using SQL queries with Cube.	CO1
3.	Installation of WEKA Tool and Creating new Arff File using WEKA Tool.	CO1
4.	Pre-process a given dataset based on Attribute selection using WEKA.	CO2
5.	Pre-process a given dataset based on Handling Missing Values using WEKA.	CO2
6.	Data cube construction – OLAP operations Using WEKA.	CO3
7.	Generate a Decision Tree by using J48 algorithm.	CO3
8.	Calculating information gain measures.	CO3
9.	Implementation of FP- Growth Algorithm.	CO4

Text Books:

T1: Jiawei Han, MichelineKamber, Jian Pei , *Data Mining: Concepts and Techniques*, 3 rd edition, Elsevier, United States of America, 2012.

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T2: P. Ponniah, *Data Warehousing Fundamentals for IT Professionals*, 2nd ed. Hoboken, NJ, USA: Wiley, 2010.
T3: Alex Berson, Stephen J. Smith , *Data Warehousing, Data Mining, and OLAP*, McGraw-Hill, 1997.
Reference Books:
R1: Margaret H Dunham, *Data Mining Introductory and Advanced Topics*, 2nd edition, Pearson Education, New Delhi, India, 2006.
R2: Amitesh Sinha, *Data Warehousing*, Thomson Learning, India, 2007
R3: Xingdong Wu, Vipin Kumar, *The Top Ten Algorithms in Data Mining*, CRC Press, UK, 2009
Relevant MOOCs Course (Course name and Weblink):
1. NPTEL- Data Mining, Prof. Pabitra Mitra ,IIT Kheradpir, https://nptel.ac.in/courses/106105174 **2.** NPTEL- Introduction to Data Analytics, Prof.Nandan Sudarsanam, Prof. Balaraman Ravindran,IIT Madras**,** https://nptel.ac.in/courses/110106072
Relevant Topics for Self-study:
1. Clustering Techniques in WEKA 2. ETL (Extract, Transform, Load) Processes in Data Warehousing

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)
[2635PE34]: DevOps for Scalable Systems Laboratory (DOSSL)

Semester	Credits	Teaching Scheme	Examination Scheme
5	1	P: 2 Hrs. / Week	CIE (TW): 25 Marks ESE (PR): 25 Marks

Prerequisite: Students should have prior knowledge of

- Basic programming and familiarity with software development concepts.

Course Objectives: The objective of this course is to provide students with

- Introduction of version control, continuous integration, and build automation using Git and Jenkins.
- Practical knowledge of containerization and application orchestration with Docker and Kubernetes.
- Designing and implementing automated CI/CD pipelines with integrated testing.
- Monitoring and visualizing application performance using Prometheus and Grafana

Course Outcomes: After completing this course, students will be able to

CO1: Apply version control and build automation by creating Git repositories and configuring Jenkins jobs.

CO2: Implement containerization with Docker and deploy scalable applications on Kubernetes.

CO3: Design Jenkins pipelines with automated testing to ensure code reliability and quality.

CO4: Analyse and visualize application performance metrics using Prometheus and Grafana dashboards.

COURSE CONTENTS

Assign.	Description of Assignment	CO
1.	**Git Basics:** Create a GitHub repository for a sample application, commit code, and demonstrate basic Git operations (clone, branch, merge, push, pull).	CO1
2.	**Git Branching & Collaboration**: Implement branching strategies (feature branches, pull requests, and merging) to simulate collaborative development in a distributed team.	CO1
3.	**Jenkins Build:** Configure Jenkins to manually pull the sample application from GitHub and build it successfully.	CO1
4.	**Dockerization:** Containerize the given sample application using Docker so that it runs seamlessly on any machine without manual setup.	CO2
5.	**Kubernetes Deployment:** Deploy the Dockerized application on Kubernetes (using Minikube), configure scaling, and ensure it is accessible to users.	CO2
6.	**Jenkins Pipeline with Automated Testing**: Integrate pre-written test scripts into a Jenkins pipeline to ensure that the build fails automatically if any test cases fail.	CO3
7.	**Monitoring with Prometheus:** Set up Prometheus to collect and monitor application performance metrics in real time.	CO4
8.	**Visualization with Grafana:** Create Grafana dashboards to visualize and analyze the real-time metrics collected by Prometheus.	CO4

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Text Books:	
T1:	G. Kim et al., *The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Techno Organizations*. Portland, OR, USA: IT Revolution Press, 2021, ISBN: 978-1942788003.
T2:	J. Humble and D. Farley, *Continuous Delivery: Reliable Software Releases through Build, Test, and Deployment Automation*. Boston, MA, USA: Addison-Wesley Professional, 2010.
Reference Books:	
R1:	S. Sharma, *The DevOps Adoption Playbook: A Guide to Adopting DevOps in a Multi-Speed IT Enterprise*. Por USA: IT Revolution Press, 2017.
R2:	B. Beyer et al., *Site Reliability Engineering: How Google Runs Production Systems*. Sebastopol, CA, USA: O' Media, 2016.
R3:	J. Davis and R. Daniels, *Effective DevOps: Building a Culture of Collaboration, Affinity, and Tooling at Scale* Sebastopol, CA, USA: O'Reilly Media, 2016.
Relevant MOOCs Course (Course name and Weblink):	
1. Coursera -Introduction to DevOps: Practices and Tools, John Rofrano Introduction to DevOps | Coursera 2. Coursera-Continuous Integration and Continuous Delivery (CI/CD), John Rofrano Continuous Integration and Continuous Delivery (CI/CD) | Coursera	
Relevant Topics for Self-study:	
1. Cloud-native patterns (serverless computing, service meshes, disaster recovery) 2. Advanced Git workflows (rebasing, GitOps)	

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)
[2635PE35]: Computer Graphics and Multimedia Systems Laboratory (CGMSL)

Semester	Credits	Teaching Scheme	Examination Scheme
5	01	P: 02 Hrs. / Week	CIE (TW): 25 Marks ESE (OR): 25 Marks

Prerequisite: Students should have prior knowledge of

- Basics of trigonometry
- Knowledge of geometry (points, lines, circles, polygons)
- Vectors and Matrices, Data Structures and Algorithm

Course Objectives: The objective of this course is to provide students with

- The fundamentals of computer graphics algorithms, transformations, and clipping techniques.
- Skills in creating 2D/3D graphics, animations, and multimedia applications using open-source tools.
- Understanding of multimedia editing, authoring, and compositing for images, audio, video, and 3D content.
- Creativity, teamwork, and practical implementation of computer graphics concepts in real-world scenarios.

Course Outcomes: After completing this course, students will be able to

CO1: **Implement** basic graphics algorithms (line, circle, polygon drawing, clipping, transformations).

CO2: **Develop** simple 2D/3D animations and interactive graphics applications using OpenGL/WebGL/Blender.

CO3: **Apply** multimedia tools for editing, authoring, and compositing (image, video, audio, and VFX).

CO4: **Design and integrate** computer graphics and multimedia techniques in mini-projects with creativity and technical accuracy.

COURSE CONTENTS

Assign. No.	Description of Assignment	CO
1.	Line Drawing (DDA, Bresenham's Algorithm) Circle/Polygon Drawing	CO1
2.	Implement Cohen Sutherland line clipping method to clip the polygon with respect the viewport and window.	CO1
3.	Implementation of 2D Transformations (Translation, Scaling, Rotation, Reflection, Shearing)	CO1
4.	Keyframe & Tweening Animation, 2D/3D Animation using OpenGL/WebGL or Blender	CO2
5.	VFX Tool Practice (After Effects / Blender / Natron / Open-source alternatives)	CO3
6.	Multimedia Authoring & Editing Lab : Tools: Image Editing → GIMP, Inkscape Video Editing → OpenShot, Shotcut 3D Modeling → Blender	CO3

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	Audio Editing → Audacity Screen Recording/Streaming → OBS Studio	
7.	2D Animation Using Free Open Source Pencil2D. 3D Model Using Free Open Source Sweet Home 3D/Google Sketchup	CO2, CO4

Text Books:

T1: A. N. Sinha and A. D. Udai, *Computer Graphics*, 1st ed. New Delhi, India: Tata McGraw-Hill Education, 2008, ISBN-13: 978-0070634374.

T2: R. K. Maurya, *Computer Graphics Including Multimedia Systems*. New Delhi, India: Khanna Publishing House, 2018, ISBN-13: 978-9386173502.

Reference Books:

R1. J. D. Foley et al., *Computer Graphics: Principles and Practice*, 3rd ed. Boston, MA, USA: Addison-Wesley/Pearson, 2013, ISBN-13: 978-0321399526.

R2. P. Shirley et al., *Fundamentals of Computer Graphics*, 4th ed. Boca Raton, FL, USA: CRC Press (A K Peters), 2015, ISBN-13: 978-1482229394.

R3. R. Steinmetz and K. Nahrstedt, *Multimedia: Computing, Communications and Applications*, 1st ed. Boston, MA, USA: Pearson Education, 1995, ISBN-13: 978-0133244358.

1. NPTEL-Computer Graphics, Prof. Samit Bhattacharya ,IIT Guwahati, https://nptel.ac.in/courses/106103224
2. Computer Graphics Virtual Lab , IIT Hyderabad https://cse18-iiith.vlabs.ac.in/

Relevant Topics for Self-study:

1. Bezier Curves and B-Splines for modeling smooth shapes.
2. Texture Mapping and Shading Models in 3D graphics.
3. Image/Video Compression Techniques (JPEG, MPEG, H.264)

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Third Year B. Tech (T. Y B. Tech) AY (2026-27) Information Technology (IT) [2635MD1X]: Multidisciplinary Minor (MDM-3)			
Semester	**Credits**	**Teaching Scheme**	**Examination Scheme**
5	**02**	**T: 02 Hr. / Week**	**CIE: 20 Marks** **ISE: 20 Marks** **ESE: 60 Marks**
Refer Annexure-I			

Third Year B. Tech (T. Y B. Tech) AY (2026-27) Information Technology (IT) [2635MDXX]: Multidisciplinary Minor Laboratory (MDM-3#)			
Semester	**Credits**	**Teaching Scheme**	**Examination Scheme**
5	**01**	**P: 02 Hr. / Week**	**CIE (TW): 25 Marks** **ISE: NA** **ESE: NA**
Refer Annexure-I			

Third Year B. Tech (T. Y B. Tech) AY (2026-27) Information Technology (IT) [2635OE5X]: Open Elective-III (OE-III)			
Semester	**Credits**	**Teaching Scheme**	**Examination Scheme**
V	**02**	**T: 2 Hr. / Week**	**CIE: NA** **ISE: NA** **ESE: 50 Marks**
Refer Annexure-II			

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Third Year B-TECH (T. Y. B. Tech) AY (2026-27)
Information Technology (IT)

[2605AE31]: Leadership and Management Skills

Semester	Credits	Teaching Scheme	Examination Scheme
5	01	PR: 02 Hrs./ Week	TW: 25 Marks

Prerequisite:

- Course on Soft Skills (SS), and Professional Skills 1, Professional Skills 2

Course Objectives:

- Develop essential skills to influence and motivate others.
- Inculcate emotional and social intelligence and integrative thinking for effective leadership.
- Create and maintain an effective and motivated team to work for the society. Nurture a creative and entrepreneurial mindset.
- Help understand the personal values and apply ethical principles in professional and social contexts.

Course Outcomes:

CO1: Examine various leadership models and understand and/or assess their skills, strengths and abilities that affect their personal leadership style and can create a leadership vision.
CO2: Analyze and demonstrate a set of practical skills such as time management, self-management, handling conflicts, and team leadership.
CO3: Understand the basics of entrepreneurship and develop business plans.
CO4: Apply the design thinking approach for leadership.

COURSE CONTENTS

Expt. No	Title/Problem Statement	CO
1	**Leadership Skills:** Understanding Leadership and its Importance, Traits and Models of Leadership, Basic Leadership Skills	CO1
2	**Managerial Skills:** Basic Managerial Skills	CO2
3	**Managerial Skills:** Self-management Skills	CO2
4	**Entrepreneurship Skills:** Meaning of entrepreneurship, Classification and types of entrepreneurships, Traits and competencies of entrepreneur	CO3
5	**Creating Business Plan**: Problem identification and idea generation, Idea validation, Pitch making	CO3
6	**Innovative Leadership:** Innovative Leadership, Concept of emotional and social intelligence, Synthesis of human and artificial intelligence, why does culture matter for today's global leaders?	CO4
7	**Design Thinking:** What is design thinking? - Key elements of design thinking: - How to transform challenges into opportunities?	CO4

Text Books:

T1: Dr C. S. G. Krishnamacharyulu and Dr. Lalitha Ramakrishnan, *Leadership and Management Skills*, Himalaya Publishing House, Mumbai, India, 2022, ISBN-13: 978-9355960863.

Reference Books:

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R1: James C. Collins, *Good to Great: Why Some Companies Make the Leap... and Others Don't*, HarperCollins Publishers LLC, New York, NY, USA, 2001,ISBN-13: 978-0066620992.
R2: Stephen R. Covey, *The 7 Habits of Highly Effective People*, Simon & Schuster, New York, NY, USA, 2019, ISBN-13: 978-1982137274.
Any Special Guidelines **Students can avail additional resources to enhance soft skills further** **1.** NPTEL: Design Thinking - A Primer, Prof. Ashwin Mahalingam, Prof. Bala Ramadurai , IIT Madras https://nptel.ac.in/courses/110106124

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Third Year B-Tech
(T. Y. B-Tech.)
Semester-6

PICT IT Syllabus Structure 26-27

Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2636PC11]: Design and Analysis of Algorithms (DAA)

Semester	Credits	Teaching Scheme	Examination Scheme
6	02	L:02 Hrs. / Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Data Structure and algorithms, Advance Data structure, Discrete Mathematics

Course Objectives: The objective of this course is to provide students with

- To learn the analysis of different algorithms and solve recurrence relation for various problems.
- To understand Greedy and dynamic programming strategies for solving problems.
- To understand backtracking and branch & bound strategies for solving problems.
- To understand classification of algorithms and amortized analysis.

Course Outcomes: After completing this course, students will be able

CO1: To analyze time complexity for recursive and non-recursive algorithms and apply divide and conquer strategy for given problems.

CO2: To apply the Greedy and dynamic programming strategies for solving the given optimization problems.

CO3: To solve the given problem by using backtracking and branch and bound strategies.

CO4: To differentiate classification of algorithms and amortized analysis.

COURSE CONTENTS

Module-I	**Analysis of Algorithms and Complexity Theory**	**7 Hrs.**	**CO1**

Algorithm: The Role of Algorithms in Computing, Evolution of Algorithms, Iterative algorithm design issues, Correctness of Algorithm.
Analysis: Asymptotic growth, O, Ω, Θ notations.
Divide and Conquer Algorithm: Introduction, General Abstraction, analyzing of sorting and search algorithm, Recurrence and different methods to solve recurrence-substitution and master's theorem, Problem Solving using divide and conquer algorithm - Finding Max- Min, large integer Multiplication (Karatsuba's Algorithm) and Strassen's Matrix Multiplication

Module-II	**Greedy And Dynamic Programming algorithmic method**	**7 Hrs.**	**CO2**

Greedy strategy: Principle, Elements of Greedy Strategy, Activity selection problem **Graphs**: revision of graph traversal techniques (BFS, DFS), Shortest paths: Dijkstra's Algorithm, fractional knapsack Problem, Job Sequencing Problem.
Dynamic Programming: The Principle of Optimality, OBST, 0/1 knapsack, shortest path: Bellman Ford, multistage graph problem, Coin change problem, TSP.

Module-III	**Backtracking and Branch-n-Bound**	**6 Hrs.**	**CO3**

Tree representation, Revision on Tree Traversals,

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Backtracking: Principle, control abstraction, n-queen problem, graph coloring problem, sum of subsets problem. Generating combinations and Permutations. **Branch-n-Bound:** Principle, control abstraction, Bounding -FIFO branch and bound, LC branch and bound TSP, 0/1 knapsack problem.			
Module-IV	**Complexity Theory and Amortized Analysis**	**6 Hrs.**	**CO4**
Classification of Algorithms based on time complexity: polynomial and non-polynomial problems, deterministic and non-deterministic algorithms. NP Completeness: P- class problems, NP-class problems, Polynomial problem reduction, NP complete problems- vertex cover and 3-SAT and NP hard problem - Hamiltonian cycle **Amortized Analysis:** Aggregate Analysis, Accounting Method, Potential Function method, Amortized analysis-binary counter, stack Time-Space tradeoff, Introduction to Tractable and Non tractable Problems.			
Textbooks:			
T1: E. Horowitz and S. Sahni, *Fundamentals of Computer Algorithms*. New Delhi, India: Galgotia Publications, 2001, ISBN 81-7371-612-9.			
T2: G. Brassard and P. Bratley, *Fundamentals of Algorithms*. New Delhi, India: Pearson,1996, ISBN 978-81-317-1244-3.			
Reference Books:			
R1: T. H. Cormen and C. E. Leiserson, *Introduction to Algorithms*, 3rd ed.New Delhi, India: PHI Learning,2010, ISBN 9788120340077.			
R2: S. Sridhar, *Design and Analysis of Algorithms,* 2nd ed.New Delhi, India: Oxford University Press, 2023, ISBN 978-9354977886.			
R3: Jon Kleinberg and Éva Tardos,*Algorithm Design*, 2nd ed.Boston, MA, USA: Pearson,2022, ISBN 9780132131087.			
Relevant MOOCs Course (Course name and Weblink)			
1. NPTEL-Design and Analysis of Algorithms, IIT Madras, Prof. Madhavan Mukund https://nptel.ac.in/courses/106106131			
Relevant Topics for Self-study:			
1.Segment Tree / Binary Indexed Tree https://www.geeksforgeeks.org/dsa/fenwick-tree-vs-segment-tree/ 2. KMP Algorithm (Knuth-Morris-Pratt) https://www.geeksforgeeks.org/dsa/kmp-algorithm-for-pattern-searching/			

Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2636PC12]: Cloud Computing and Virtualization (CCV)

Semester	Credits	Teaching Scheme	Examination Scheme
6	02	L: 02 Hrs. / Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Computer Network and Technology

Course Objectives: The objective of this course is to provide students with

- A comprehensive understanding of the fundamental principles and concepts of cloud computing, including its service and deployment models.
- The ability to analyze and explain the core services of major cloud platforms (AWS, Google Cloud, Azure) and articulate the role of each service in a real-world application.
- The ability to assess the principles of virtualization and distinguish between Type-1 and Type-2 hypervisors, evaluating their performance, use cases, and the stages of a virtual machine's lifecycle.
- The ability to use advanced virtualization concepts and containerization that enable cloud infrastructure.

Course Outcomes: After completing this course, students will be able to

CO1: Explain core cloud concepts, service models, and deployment models and **differentiate** between various cloud service and deployment models and select the appropriate model for a given business scenario.

CO2: Evaluate the services of major cloud providers and open-source platforms to design a basic cloud-based application.

CO3: Analyze the core principles of virtualization and containerization and compare their performance and use cases.

CO4: Evaluate the architectural differences between virtual machines and containers to determine their respective applications in modern cloud environments using advanced virtualization and containerization technologies.

COURSE CONTENTS

Module-I	Introduction to Cloud Computing	08 Hrs.	CO1

Defining Cloud Computing: Characteristics (On-demand self-service, broad network access, resource pooling, rapid elasticity, measured service), benefits, and challenges. **Evolution of computing:** From mainframes to clusters and grid computing to cloud computing. **Cloud Service Models:** Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS). **Cloud Deployment Models:** Public, Private, Hybrid, and Community clouds. Discussion on the trade-offs of each model. **Key concepts:** Elasticity, scalability, Service-Level Agreements (SLAs), Cloud Interoperability and Portability, Multi-tenancy, Cloud Economics.
Case Study: Comparison between Cloud Service Models and Cloud Deployment Models.

Module-II	Cloud Services & Platforms	06 Hrs.	CO2

Compute Services (e.g., Amazon EC2, Google Compute Engine, Azure VMs), Storage Services (e.g., Amazon S3, Google Cloud Storage, Azure Storage), Database Services (e.g., Amazon RDS, Google Cloud SQL, Azure SQL Database), Application Services, Content Delivery Services (e.g., Amazon CloudFront), Analytics Services (e.g., Amazon Elastic MapReduce, Google BigQuery), Deployment & Management Services (e.g., Amazon Elastic Beanstalk), Identity & Access Management Services, Open Source Private Cloud Software (e.g., CloudStack, Eucalyptus, OpenStack).

Module-III	**Virtualization Technology**	**06 Hrs.**	**CO3**

Introduction to Virtualization: Defining virtualization and its role in cloud computing, History and evolution of virtualization. The virtualization layer and the concept of a hypervisor.
Hypervisor Architectures: Type-1 (bare-metal) hypervisors (e.g., KVM, VMware ESXi), Detailed explanation of Type-2 (hosted) hypervisors (e.g., VirtualBox),
Virtual Machine Lifecycle: The stages of a VM's life: creation, power on, suspension, and deletion, live migration of modern virtualization platforms.
Case Study: Comparison of Type-1 vs. Type-2 hypervisors, highlighting their performance and use cases.

Module-IV	**Advanced Virtualization and Containerization**	**06 Hrs.**	**CO4**

Virtual Networking and Storage: Virtual Private Clouds (VPCs), subnets, and routing. Load Balancing and Scaling, Virtual Storage**.** Resource Management and High Availability Introduction to Containerization: Containers Vs. Virtual Machines, Container runtime and its components.
Case Study: Docker as a key containerization platform, covering its architecture, images, registries, and the role of the Docker Daemon.

Text Books:

T1: A. Bahga and V. Madisetti, *Cloud Computing: A Hands-On Approach*, Upper Saddle River, NJ, USA: Pearson Education, 2014, ISBN: 978-1494435144.

T2: T. Erl, *et al., Cloud Computing: Concepts, Technology & Architecture*, Upper Saddle River, NJ, USA: Pearson Education, 2013, ISBN: 978-0-13-338752-4**.**

Reference Books:

R1: R. Buyya *et al.*, *Mastering Cloud Computing: Foundations and Applications Programming*, Waltham, MA, USA: Morgan Kaufmann (Elsevier), 2013, ISBN**:** 978-0124114548**.**

R2: G. Arbezzano *et al.*, *Proxmox VE: A Complete Guide to Open-Source Virtualization and Containerization*, Birmingham, UK: Packt Publishing, 2018, ISBN**:** 978-1-78839-760-6**.**

R3: J. A. Davis and S. R. Hall, *The VMware VCP6-DCV Official Cert Guide*, Upper Saddle River, NJ, USA: VMware Press, 2016, ISBN: 978-0-13-417511-3**.**

R4: N. Poulton, *Docker Deep Dive*. London, U.K.: Nigel Poulton Publishing, 2023. ISBN: 978-1-52182-280-7**.**

Relevant MOOCs Course (Course name and Weblink)

1. NPTEL: Cloud Computing by Prof. Soumya Kanti Ghosh, IIT Kharagpur, https://nptel.ac.in/courses/106105167.

Relevant Topics for Self-study:

1. **Infrastructure as Code (IaC):**
   - **Terraform:** https://www.terraform.io
   - **Ansible:** https://www.ansible.com
2. **Virtualization:**
   - **Proxmox VE:** https://www.proxmox.com
3. **Container Orchestration:**
   - **Kubernetes:** https://kubernetes.io

Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)
[2636PC31]: IT Laboratory - I (DAA & CCV)

Semester	Credits	Teaching Scheme	Examination Scheme
6	2	P: 4 Hrs. / Week	CIE (TW): 25 Marks ESE (PR): 25 Marks

Prerequisite: Students should have prior knowledge of

- Data Structure and Applications
- Computer Network and Technology

Course Objectives: The objective of this course is to provide students with

- To learn the various algorithmic design strategies.
- To apply efficiently in problem solving.
- The ability to apply cloud service models by interacting with IaaS, PaaS, and SaaS environments and to configure and manage core cloud services like compute, storage, and databases on a public cloud platform.
- The ability to implement virtualization technology by installing, configuring, and managing virtual machines on an open-source hypervisor, and to construct and deploy containerized applications using a modern platform like Docker.

Course Outcomes: After completing this course, students will be able to

CO1: Implement classic algorithms across paradigms and analyze their time behavior empirically and theoretically.

CO2: Select and evaluate suitable algorithmic strategies for optimization and graph problems.

CO3: Explain core cloud concepts and differentiate between cloud service and deployment models. Evaluate and configure the services of major cloud providers by provisioning and managing compute, storage, and database resources to design and deploy a basic cloud-based application.

CO4: Analyze the principles of virtualization, implement virtual machine lifecycle management, and determine the architectural differences between VMs and containers. Deploy and evaluate both technologies in a modern cloud environment.

COURSE CONTENTS

IT LAB-I (PART A: DAA)

Assign. No.	Description of Assignment	CO
	PART - A DAA	
1	Implement and do time analysis of matrix multiplication for square matrices.	CO1

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2	Implement Rotten orange problem simulation using BFS. A fruit storage company maintains oranges in a large cold storage, arranged in a rectangular grid. Each cell in the grid can contain: ● 0 → an empty cell (no orange), ● 1 → a fresh orange ● 2 → a rotten orange. Every minute, a rotten orange spreads rot to its adjacent oranges (up, down, left, right only). You are required to simulate this process and determine the minimum time required for all the oranges to become rotten. If it is impossible to rot all the oranges, the system should return -1.	CO1
3	Write a program to implement the fractional knapsack using the greedy algorithm and the 0/1 knapsack using dynamic programming. Show that Greedy strategy does not necessarily yield an optimal solution over a dynamic programming approach	CO1 and CO2
4	Write a recursive program to find the solution of placing n queens on the chessboard so that no two queens attack each other using Backtracking	CO1
5	Write a program to solve the travelling salesman problem and to print the path and the cost using LC Branch and Bound	CO2
6	Write a program to determine if a Hamiltonian cycle exists in the graph.	CO2
7	Write a program to color vertices of a graph using at most m colors so that no adjacent vertices share a color.	CO2

Textbooks:

T1: E. Horowitz and S. Sahni, *Fundamentals of Computer Algorithms*. New Delhi, India: Galgotia Publications, 2001,ISBN 81-7371-612-9.

T2: G. Brassard and P. Bratley, *Fundamentals of Algorithms*. New Delhi, India: Pearson,1996, ISBN 978-81-317-1244-3.

Reference Books:

R1: T. H. Cormen and C. E. Leiserson, *Introduction to Algorithms*, 3rd ed. New Delhi, India: PHI Learning,2010, ISBN 9788120340077.

R2: Michael T. Goodrich and Roberto Tamassia, *Algorithm Design: Foundations, Analysis and Internet Examples*.Hoboken, NJ, USA: Wiley, 2001, ISBN 978-81-265-0986-7.

R3: :Dan Gusfield, *Algorithms on Strings, Trees and Sequences*, Cambridge, U.K.: Cambridge University Press,1997, ISBN 0-521-67035-7.

Relevant MOOCs Course (Course name and Weblink):

1. NPTEL: Design and Analysis of Algorithms ,Prof. Madhavan Mukund, IIT Madras
https://nptel.ac.in/courses/106106131
2. Coursera: Design and Analysis of Algorithms , San Diego ,University of California,
https://www.coursera.org/specializations/data-structures-algorithms

Relevant Topics for Self-study

IT LAB-I (PART B: CCV)		
Assign. No.	**Description of Assignment**	**CO**
1.	**Exploring Cloud Service & Deployment Models:** To set up the necessary infrastructure to demonstrate the three core cloud service models: **IaaS**, **PaaS**, and **SaaS**. You should be able to access the static website (IaaS) and the dynamic application (PaaS) and showcase the interaction between your local machine and the cloud resources.	CO3
2.	Design and implement a solution using a public cloud platform where a compute instance reads data from a storage service and writes the output to a database service. **1. Compute Service:** Create a virtual machine instance. Use a cloud-based command-line interface or the web console to launch and connect to the instance. Install a simple data processing script (e.g., a Python script that reads a file and performs a simple calculation). **2. Storage Service:** Create a cloud storage bucket (e.g., Amazon S3, Google Cloud Storage, or Azure Blob Storage). Upload a sample dataset file to this bucket. Configure the permissions to allow public read access but private write access. **3. Database Service:** Set up a managed database service instance. Use a database client to connect to the instance and create a simple table to store customer information. Modify the data processing script from step 1 to read the dataset from the storage bucket and write the processed results back to the managed database.	CO3
3.	Virtualization and VM Lifecycle Management To set up VM environment (Type-2 hypervisor) and demonstrate the VM's lifecycle stages, including its creation, suspension, snapshotting, and restoration. Hypervisor **Installation:** Install a Type-2 hypervisor like Oracle VirtualBox or VMware Workstation on your local machine. Virtual **Machine Creation:** Create a new virtual machine and install a Linux-based operating system (e.g., Ubuntu Server). Pay attention to the configuration of virtual hardware like CPU cores, RAM, and storage. VM **Lifecycle Operations:** • **Suspension:** Suspend the running VM and then resume it. Note the time it takes to power back on. • **Snapshot:** Create a snapshot of the VM after installing some software (e.g., a web server). This represents a "known good state." • **Restoration:** Make a change to the VM's configuration or file system. Then, restore the VM from the snapshot you created to demonstrate its ability to revert to a previous state. • **Cloning:** Clone the VM to create an identical, but separate, copy. This demonstrates the ease of creating multiple instances from a single template. Resource **Monitoring:** Use tools within the hypervisor to monitor the resource usage of the VM (CPU and RAM).	CO4
4.	Containerization with Docker and Virtual Networking: To create a simple containerized application and set up the necessary networking. · Container **vs. VM Analysis:** Use a command like docker stats or docker top to monitor a running container's resource usage (CPU and memory). Compare	CO4

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	this to the resource usage of a VM running a similar application. Document your observations on resource efficiency. · **Dockerfile Creation:** Write a Dockerfile to create a Docker image for a simple web application (e.g., a "Hello, World!" application). Container **Networking:** • Run the container and map a port from the host machine to the container's port to allow access to the web application. • Create a virtual network for a multi-container application. • Launch two containers (e.g., a web server and a database) on the same virtual network and demonstrate that they can communicate with each other using the container names. Demonstrate **Portability:** Use the Docker commands to save the container image as a `.tar` file and then load it on a different machine, demonstrating the portability of containers.	
Text Books:		
T1:	A. Bahga and V. Madisetti, *Cloud Computing: A Hands-On Approach*, Upper Saddle River, NJ, USA: Pearson Education, 2014, ISBN: 978-1494435144.	
T2:	T. Erl, *et al., Cloud Computing: Concepts, Technology & Architecture*, Upper Saddle River, NJ, USA: Pearson Education, 2013, ISBN: 978-0-13-338752-4**.**	
Reference Books:		
R1:	R. Buyya *et al.*, *Mastering Cloud Computing: Foundations and Applications Programming*, Waltham, MA, USA: Morgan Kaufmann (Elsevier), 2013, ISBN**:** 978-0124114548**.**	
R2:	G. Arbezzano *et al.*, *Proxmox VE: A Complete Guide to Open-Source Virtualization and Containerization*, Birmingham, UK: Packt Publishing, 2018, ISBN**:** 978-1-78839-760-6**.**	
R3:	J. A. Davis and S. R. Hall, *The VMware VCP6-DCV Official Cert Guide*, Upper Saddle River, NJ, USA: VMware Press, 2016, ISBN: 978-0-13-417511-3**.**	
Relevant MOOCs Course (Course name and Weblink):		
NPTEL: Cloud Computing, Prof. Soumya Kanti Ghosh, IIT Kharagpur https://nptel.ac.in/courses/106105167.		
Relevant Topics for Self-study:		
1. **Infrastructure as Code (IaC):** • **Terraform:** https://www.terraform.io • **Ansible:** https://www.ansible.com 2.**Virtualization:** • **Proxmox VE:** https://www.proxmox.com 3.**Container Orchestration:** **Kubernetes:** https://kubernetes.io		

Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

Program Elective - II (PEC-II)

Semester	Credits	Teaching Scheme	Examination Scheme
6	03	T: 3 Hr. / Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2636PE11]: Wireless and Adhoc Network Technologies (WANT)

Semester	Credits	Teaching Scheme	Examination Scheme
6	3	L: 03 Hrs/ Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Computer Network Technology

Course Objectives: The objective of this course is to provide students with

- Understand the fundamentals of wireless communication technologies, standards, and their design challenges.
- Analyze wireless sensor networks (WSN), architectures, protocols, and applications.
- Explore mobile and ad hoc network (MANET) protocols, performance metrics, and design considerations.
- Examine applications, security issues, and emerging trends in adhoc and wireless networks.

Course Outcomes: After completing this course, students will be able to

CO1: Explain the principles, architectures, and standards of wireless technologies.
CO2: Analyze the architectures, routing protocols, and design issues in wireless sensor and mobile ad hoc networks.
CO3: Evaluate the performance of adhoc and sensor networks using QoS and performance metrics.
CO4: Design and propose secure and efficient wireless/adhoc networking solutions for real-world applications.

COURSE CONTENTS

Module-I	Fundamentals of Wireless Technologies	10 Hrs.	CO1

Introduction to Wireless Communication: Characteristics, spectrum allocation, challenges (fading, interference, mobility) , **Design Goals of Wireless Networks**: Scalability, energy efficiency, reliability, security, QoS support, low cost, **Issues in Wireless Networks**: Spectrum scarcity, interference, handoff management, power management, **Wireless Standards Overview**: IEEE 802.11 (WLAN), IEEE 802.15 (WPAN), IEEE 802.16 (WiMAX) basics, **Bluetooth Technology**: Architecture, Protocol Stack, Profiles Bluetooth Low Energy (BLE), Applications in IoT , **Wi-Fi Technology**: IEEE 802.11 Standards (a/b/g/n/ac/ax), Wi-Fi 6 and Wi-Fi 7 basics, WLAN Architecture & Protocols, WLAN Security (WEP, WPA, WPA2, WPA3)
Wi-Max Technology: Comparison of Bluetooth, Wi-Fi, WiMAX, v Applications of short-range wireless technologies

Module-II	Wireless Sensor Networks (WSN)	10 Hrs.	CO2

Wireless Sensor Nodes: Architecture, components (sensing, processing, communication, power)
Design Goals of WSN: Low power consumption, fault tolerance, scalability, adaptability, data accuracy, security
Issues in WSN: Limited energy, unreliable communication, node failures, synchronization, coverage, latency
WSN Architectures: Flat, Hierarchical, Cluster-based

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Routing in WSN: Flat Routing Protocols: Flooding, Gossiping, **Hierarchical Routing Protocols**: LEACH, PEGASIS, TEEN, APTEEN, **Location-based Protocols**: GAF, GEAR **Data-centric Protocols**: Directed Diffusion, Rumor Routing, **Energy Efficiency & Data Aggregation in WSN**, **QoS in WSN** (latency, reliability, throughput), **Applications**: Environmental monitoring, healthcare, smart cities, industrial IoT

Module-III	Adhoc & Mobile Adhoc Networks (MANETs)	10 Hrs.	CO3

Adhoc Networks: Characteristics, design goals, applications, **Design Goals of MANETs**: Dynamic topology handling, scalability, energy efficiency, distributed operation, security, QoS, **Issues in MANETs**: Routing overhead, limited bandwidth, frequent link failures, power constraints, security vulnerabilities, **Mobile Adhoc Networks (MANET)**: Features, challenges, applications, **Protocol Stack in MANET**:, **MAC Layer**: IEEE 802.11 MAC, Hidden & Exposed terminal problems, RTS/CTS, **Network Layer**: Proactive Routing (DSDV, OLSR), Reactive Routing (AODV, DSR), Hybrid Routing (ZRP) **Transport Layer**: TCP over MANET, congestion issues, performance improvements **Quality of Service (QoS) in MANETs**

Performance Metrics of MANETs (throughput, end-to-end delay, packet delivery ratio, routing overhead)

Module-IV	Applications and Security in Adhoc Networks	10 Hrs.	CO4

Applications of Adhoc and Sensor Networks: Disaster recovery, military operations, vehicular networks (VANETs), UAV/drone networks, IoT integration

Security in Adhoc Networks:Security goals (confidentiality, integrity, availability, authentication, non-repudiation), Security challenges (decentralization, mobility, resource constraints), Attacks: Blackhole, Wormhole, Sybil, DoS, Jamming, Eavesdropping, Routing attacks

Security Solutions: Intrusion detection systems (IDS), Key management in distributed environments, Trust and reputation-based security models

Emerging Trends: Cognitive Radio Networks and Security issues, Integration of Adhoc Networks with 5G/6G

, Software Defined Networking (SDN) for Adhoc Networks, Internet of Vehicles (IoV) and Vehicular Adhoc Networks (VANETs)

Case Studies & Research Challenges

Text Books:

T1: C. Siva Ram Murthy & B. S. Manoj, *Ad Hoc Wireless Networks: Architectures and Protocols*, Pearson Education (Prentice Hall PTR), Boston, 2004, 1st Edition, ISBN-10: 0132465744, ISBN-13: 978-0132465748.

T2: Holger Karl & Andreas Willig, *Protocols and Architectures for Wireless Sensor Networks,* Wiley, 2005, 1st Edition, ISBN-10: 0470519231, ISBN-13: 978-0470519233.

Reference Books:

R1: Ian F. Akyildiz & Mehmet Can Vuran, *Wireless Sensor Networks,* John Wiley & Sons, 2010, 1st Edition, ISBN-10: 047003601X, ISBN-13: 978-0470036013.

R2: Charles E. Perkins,*Ad Hoc Networking,* Addison-Wesley Professional, 2000, 1st Edition (illustrated), ISBN-10: 0201309769, ISBN-13: 978-0201309768.

R3: Jochen H. Schiller,*Mobile Communications*, Addison-Wesley, 2003, 2nd Edition (illustrated), ISBN-10: 1405890304, ISBN-13: 978-1405890304.

Relevant MOOCs Course (Course name and Weblink)

1. **NPTEL:** Wireless Ad Hoc and Sensor Networks, Prof. Sudip Misra- IIT Kharagpur https://nptel.ac.in/courses/106105160

Relevant Topics for Self-study:

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2636PE12]: Computer Vision (CV)

Semester	Credits	Teaching Scheme	Examination Scheme
6	03	L: 03 Hrs. / Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Vectors, Linear Algebra (i.e., Matrix Operations, Solution of Linear Equations).
- Programming Language (e.g., C/C++, Matlab, Python , OpenCV etc).

Course Objectives: The objective of this course is to provide students with

- Understanding of the fundamentals, origin, and applications of computer vision and digital image processing.
- The ability to apply feature extraction and shape representation techniques for analyzing images and visual data.
- Knowledge of 3D vision concepts, motion estimation, and both classical and deep learning–based object detection and tracking methods.
- Skills to design and implement computer vision solutions for real-world applications such as surveillance, photo collections, and intelligent vehicle systems.

Course Outcomes: After completing this course, students will be able to

CO1: Describe the origin, applications, and fundamental concepts of computer vision and digital image processing techniques.

CO2: Apply and compare feature extraction techniques (contour, region, interest points, Hough).

CO3: Evaluate 3D vision, motion estimation, object detection & tracking (classical + DL).

CO4: Design and demonstrate CV solutions for real-world applications (surveillance, photo, automotive).

COURSE CONTENTS

Module-I	Introduction to Computer Vision	11 Hrs.	CO1

Introduction, Origin, Applications and Examples of Computer Vision, Fundamental steps in Digital Image Processing, Components of Digital Image Processing, Elements of Visual Perception, Image Sensing and Acquisition, image enhancement (Spatial Domain, Frequency domain), image segmentation (Classification, region-based segmentation, thresholding, edge-based segmentation), morphological image processing

Case Study: Image enhancement application

Module-II	Image Feature Extraction	10 Hrs.	CO2

Taxonomy of shape representation techniques: contour-based shape representation (chain code, Bspline, Perimeter, shape signature, Fourier descriptor), Region based (area, Euler number, Shape Matrix, convex-hull, media axis), Corner and Interest Point Detection (Harris, FAST, SIFT, SURF, ORB), Line and circle detection using Hough Transform (HT)

Case Study: Corner and Interest Point Detection in Image Matching

Module-III	3D VISION AND MOTION	10Hrs.	CO3

The three-dimensional world, Methods for 3D vision, stereo vision, projection schemes for 3D vision, 3D Reconstruction, Structure from motion: triangulation, bundle adjustment, Dense motion estimation: translational alignment, parametric motion, spline-based motion, Optical flow layered motion

Object Detection and Tracking: classical object detection (Viola-Jones, HOG+SVM), Introduction to CNNs: fundamentals of convolution, pooling, feature maps, and classification, Deep Learning based object detection (YOLO, SSD), OBJECT TRACKING classical algorithm (Mean-Shift/CAM Shift, particle

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filters/Kalman Filter), Multi-object tracking (SORT, Deep SORT).			
Module-IV	**COMPUTER VISION APPLICATIONS**	**8Hrs.**	**CO4**
Application: Photo album – Object detection, Face detection, Pedestrian detection, Face recognition: Eigen faces, Active appearance and 3D shape models, Application: Personal Photo Collections, Category Recognition, Intelligent Photo Editing, Image Search, **Application: Surveillance** – The basic geometry, foreground-background separation, particle filters, Chamfer Matching, Tracking, and Occlusion, combining views from multiple cameras, License Plate Location, Occlusion Classification for Tracking, Human Gait Analysis, In-vehicle vision system: Locating the Roadway, Location of Road Markings, Location of Road Signs, Location of Vehicles, Locating Pedestrians			
Text Books:			
T1: S Jayaraman et.al, *Digital Image Processing*, 2nd Edition, Tata McGraw-Hill Education, ISBN :1259081435, 9781259081439,2009.			
T2: Simon J. D. Prince, *Computer Vision: Models, Learning, and Inference*, Cambridge University Press, 2012.			
Reference Books:			
R1: R. Davies, *Computer & Machine Vision*, 4th Edition, Academic Press, 2012.			
R2: R. Szeliski, *Computer Vision: Algorithms and Applications*, Springer 2011.			
R3: D. L. Baggio et al., *Mastering OpenCV with Practical Computer Vision Projects*, Packet Publishing, 2012.			
R4: Jan Erik Solem, *Programming Computer Vision with Python: Tools and algorithms for analyzing images*, O'Reilly Media, 2012.			
Relevant MOOCs Course (Course name and Weblink)			
1. NPTEL- Computer Vision, IIT, Kharagpur, Prof. J. Mukhopadhyay, https://nptel.ac.in/courses/106105216 2. NPTEL-Deep Learning for Computer Vision, IIT Hyderabad, Prof. Vineeth N Balasubramanian, **https://nptel.ac.in/courses/106106224**			
Relevant Topics for Self-study:			
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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2636PE13]: Modern Databases & Scalable Architecture (MDSA)

Semester	Credits	Teaching Scheme	Examination Scheme
6	03	L: 03 Hrs. / Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Database Management System, web development framework(WDF)

Course Objectives: The objective of this course is to provide students with

- To learn and understand Distributed database and web databases.
- To learn N Cloud database and Spatial data management.
- To learn Multimedia database and mobile transaction models.
- To Learn distributed systems, cloud computing technologies

Course Outcomes: After **completing this course, students will be able to**

CO1: To define Distributed databases and No SQL Databases using Cassandra

CO2: To design and evaluate advanced database solutions using spatial data, cloud-based architectures, graph models, and NewSQL technologies to meet real-world application requirements.

CO3: To express and apply Multimedia databases.

CO4: To design, build, and manage systems that can handle increasing workloads and user demands efficiently

COURSE CONTENTS

Module-I	NEXT GENERATION DATABASES	10 Hrs.	CO1

Distributed databases: Distributed DBMS architectures, storing data in a Distributed DBMS, Distributed catalog management, Distributed Query processing, Updating distributed data, Distributed transactions, Distributed Concurrency control and Recovery.

Web Databases: NoSQL Databases - **Cassandra** example, Embedded Databases - definition- Example - SQLite internal architecture and data types. Sharding and Replication-Consistency-The CAP Theorem. Key-Value data-stores – Examples- In-memory databases.

Module-II	CLOUD DATABASES	10 Hrs.	CO2

Advanced databases: Spatial Data Management: Types Of Spatial Data And Queries- Point And Region Data-Queries-Applications Involving Spatial Data -Spatial Indexes.

Cloud Databases- methods to run- virtual machine deployment, as a service- Column Stores-Examples-Cassandra, HBase-Aggregation and Join, - Case study- BigTable Google's distributed storage system for structured data-building blocks-GFS, Scheduler, Lock Service, MapReduce Graph databases- Comparison of Twitter's FlockDB and Neo4j- Overview of NewSQL- Case study -Google's Spanner.

Module-III	MULTIMEDIA DATABSES	10 Hrs.	CO3

Multimedia Databases-Multimedia Sources-Image Databases Compressed Representations-Similarity Based Retrieval. Mobile Databases- Mobile Database Systems – Transaction Execution in MDS- Mobile Transaction Models -Concurrency Control Mechanism-Transaction Commit Protocols Mobile database Recovery: Log management in mobile database systems - Mobile database recovery schemes.

Module-IV	Fundamentals of Scalable Architecture	9 Hrs.	CO4

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Introduction to Scalability: Scalability Principles, Performance Optimization, Scalability in Architectural Styles **Architectural Patterns for Scalability:** Microservices Architecture, Event-Driven Architecture, Service-Oriented Architecture (SOA), Cloud-Native Architecture, Serverless Architecture **Strategies for Scalability**: Horizontal Scaling, Vertical Scaling, Load Balancing, Caching ,Database Scaling
Text Books:
T1: AviSilberschatz et.al., *Database System Concepts, Sixth Edition*, ISBN-13: 978-93-3290-138-4, MCGraw Hill
T2: Raghu Ramakrishnan and Johannes Gehrke, *Database Management Systems*, McGraw Hill, Third Edition, 2004
Reference Books:
R1: DT Editorial Services, *Big Data Black Book.* Wiley-Dreamtech Press, ISBN-9789351197577, May 2015
R2: Mario Piattini, Oscar Diaz, *Advanced Database Technology and Design,* online book.
R3: Eben Hewitt, *Cassandra The Definitive Guide*, Publisher: Shroff Publishers & Distributors ISBN10: 9350231937 ISBN13: 9789350231937 ,Publication Year: 2010
R4: Elmasri R. and Navathe S.B., *Fundamentals of Database Systems*, Pearson Education/Addison Wesley, Fifth Edition, 2007.
Relevant MOOCs Course (Course name and Weblink)
1. **Big Data Computing,** Prof. Rajiv Misra- IIT Patna https://nptel.ac.in/courses/106104189 2. **Cloud Computing and** Distributed Systems, Prof. Rajiv Misra IIT Patna https://onlinecourses.nptel.ac.in/noc21_cs15/preview
Relevant Topics for Self-study:
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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2636PE14]: Cloud-Native Application Development (CAD)

Semester	Credits	Teaching Scheme	Examination Scheme
6	03	L: 03 Hrs. / Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Cloud Computing and Virtualization (3504141)

Course Objectives: The objective of this course is to provide students with

- A deep understanding of the fundamental principles of cloud-native software and the evolution of cloud platforms.
- The ability to design and implement applications based on microservices and various synchronous and asynchronous messaging patterns.
- Comprehensive knowledge of connectivity and composition patterns, including service mesh, service discovery, and orchestration.
- The ability to design and implement event-driven architectures using relevant patterns and technologies.

Course Outcomes: After completing this course, students will be able to

CO1: Explain and apply foundational cloud-native principles and platform design patterns to develop modern applications.

CO2: Design and implement applications using various synchronous and asynchronous messaging patterns for effective microservices communication.

CO3: Apply connectivity and composition patterns, such as service mesh and service orchestration, to build and manage robust cloud-native applications.

CO4: Develop and automate solutions based on event-driven architecture, including event delivery and state management patterns.

COURSE CONTENTS

Module-I	Foundational Principles & Platform	10 Hrs.	CO1

Introduction to Cloud Native: What is Cloud Native? Modern application requirements, introducing cloud-native software, Running cloud-native applications in production: The obstacles, The enablers. The platform for cloud-native software: The cloud (-native) platform evolution, Core tenets of cloud-native platform and capabilities. Methodology for Building Cloud Native Apps, Design Patterns for Building Cloud Native Apps, Reference Architecture for Cloud Native Apps.

Module-II	Communication Design Patterns	10 Hrs.	CO2

Synchronous Messaging Patterns: Request-Response Pattern, Remote Procedure Calls Pattern. Asynchronous Messaging Patterns: Single-Receiver Pattern, Multiple-Receiver Pattern, Asynchronous Request-Reply Pattern. Service Definition Patterns: Service Definitions in Synchronous Communication, Service Definition in Asynchronous Communication. Technologies to Implement Synchronous Messaging Patterns: RESTful Services, GraphQL, WebSocket, gRPC, Summary of Synchronous Messaging Technologies. Technologies to Implement Asynchronous Messaging Patterns: AMQP, Kafka, NATS.

Module-III	Connectivity and Composition Patterns	09 Hrs.	CO3

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Connectivity Patterns: Service Connectivity Pattern, Service Abstraction Pattern, Service Registry and Discovery Pattern, Resilient Connectivity Pattern, Sidecar Pattern, Service Mesh Pattern, Sidecarless Service Mesh Pattern, Technologies for Implementing Service Connectivity Patterns. **Service Composition Patterns:** Service Orchestration Pattern, Service Choreography Pattern, Saga Pattern, Technologies for Implementing Service Composition Patterns.

Module-IV	Event-Driven Architecture Patterns	10 Hrs.	CO4

Event-Driven Architecture, Exactly Once Processing, Message Broker Categories, CloudEvents, Event Schema. Event-Delivery Patterns: Producer-Consumer Pattern, Publisher-Subscriber Pattern, Fire and Forget Pattern, Store and Forward Pattern, Polling Pattern, Request Callback Pattern. State Management Patterns: Event Sourcing Pattern. Orchestration Patterns: Mediator Pattern, Pipe and Filter Pattern, Priority Queue Pattern, Summary of Orchestration Patterns. Technologies for Event-Driven Architecture: Apache ActiveMQ, RabbitMQ, Amazon SQS, Amazon SNS, Azure Event Grid, Azure Service Bus Queues, Google Cloud Pub/Sub.

Text Books:

T1: C. Davis, *Cloud Native Patterns: Designing Change-Tolerant Software*, Shelter Island, NY, USA: Manning Publications Co., 2019, ISBN: 978-1-61729-429-7.

T2: K. Indrasiri and S. Suhothayan, *Design Patterns for Cloud Native Applications*, Sebastopol, CA, USA: O'Reilly Media, 2021, ISBN: 978-1-098-10782-7.

Reference Books:

R1: J. Gilbert, *Cloud Native Development Patterns and Best Practices*, Birmingham, UK: Packt Publishing, 2018, ISBN: 978-1-78847-377-5.

R2: C. Fehling *et al.*, *Cloud Computing Patterns: Fundamentals to Design, Build, and Manage Cloud Applications*, Cham, Switzerland: Springer, 2014, ISBN: 978-3-319-05020-8.

Relevant MOOCs Course (Course name and Weblink)

1. NPTEL: Advanced Distributed Systems, Prof. Smruti Ranjan Sarangi, IIT Delhi
   https://nptel.ac.in/courses/106102237.

Relevant Topics for Self-study:

1. Docker: Docker Docs, "Reference," Docker, [Online]. Available: https://docs.docker.com/reference/.
2. Kubernetes (Container Orchestration): Kubernetes, "Reference," Kubernetes, [Online]. Available: https://kubernetes.io/docs/reference/.
3.gRPC (Microservices Architecture):gRPC, "gRPC," Cloud Native Computing Foundation, [Online]. Available: https://grpc.io/.
4. Apache Kafka (Messaging Systems): Apache Kafka, "Documentation," Apache Kafka, [Online]. Available: https://kafka.apache.org/documentation/.
5.RabbitMQ (Messaging Systems): RabbitMQ, "RabbitMQ: One broker to queue them all," VMware Tanzu, [Online]. Available: https://www.rabbitmq.com/.

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2636PE15]: Interface Design and Usability Engineering (IDUE)

Semester	Credits	Teaching Scheme	Examination Scheme
6	03	L: 03 Hrs. / Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Basic knowledge of programming and user interface design principles. Familiarity with prototyping tools such as Figma, Adobe XD, or Balsamiq

Course Objectives: The objective of this course is to provide students with

- Understand the fundamental concepts, principles, and evolution of Human-Computer Interaction (HCI) and Usability Engineering.
- Analyze human cognitive processes, perception, and mental models to improve user-centered design.
- Apply interaction design principles and usability heuristics to create intuitive user interfaces.
- Evaluate user interfaces through usability testing and heuristic evaluation.
- Design user-centered interfaces by integrating usability engineering methods, accessibility standards, and HCI best practices.
- Implement prototyping techniques and usability testing strategies to enhance system usability. .

Course Outcomes: After completing this course, students will be able to

CO1: Understand & Explain fundamental concepts of Human-Computer Interaction (HCI) and Usability Engineering, including user-centered design principles.

CO2: Analyze & Evaluate user interfaces using usability heuristics, cognitive models, and usability testing techniques to identify design issues.

CO3: Apply & Design interactive prototypes and user-friendly interfaces based on usability principles, accessibility guidelines, and industry best practices.

CO4: Conduct & Justify usability testing and user research, interpreting results to improve system usability and enhance user experience.

COURSE CONTENTS

Module-I	Human Factors and Interaction Design	10 Hrs.	CO1

Definition, scope, and importance of HCI, Evolution of HCI: From command-line interfaces to AI-based interactions, 5 Senses of information, human memory, Human error and its impact on design, Norman's design principles (e.g., visibility, feedback, constraints, affordances), User Centered Design, Measurable Human Factors, Case studies of good and bad interaction design.

Module-II	User Interface Design & Prototyping	10 Hrs.	CO2

UI design patterns and consistency, Wireframing and prototyping, Creating a low-fidelity wireframe, Usability testing methods: heuristic evaluation, cognitive walkthroughs, Metrics for measuring usability (efficiency, effectiveness, satisfaction), AI and HCI (voice interfaces, chatbots), AR/VR and multimodal interactions, Ethics and accessibility in HCI, Real-world case studies of successful and failed HCI implementations.

Module-III	Usability Engineering	10 Hrs.	CO3

Introduction to Usability Engineering (Definition, role, and importance of usability engineering, Difference between usability and UX), Ethnography and its types, User Research & Data Collection Methods: Interviews, surveys, observations, Personas, and scenarios), Usability Heuristics & Guidelines (Nielsen's 10 Usability Heuristics, ISO 9241 Usability Standards), Evaluating a real-world application using heuristics, Conducting a mini-user research study.

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Module-IV	Usability Testing & Metrics	9 Hrs.	CO4

Types of usability testing: remote, in-person, Key usability metrics: Time on task, error rate, satisfaction, conducting a small usability test, Web Content Accessibility Guidelines (WCAG), Assistive technologies and designing for diverse users, Evaluating a website's accessibility, Examples of usability successes and failures, Discussion on best practices and industry trends

Text Books:

T1: A. Dix et al., *Human-Computer Interaction*, 4th ed., Harlow, England: Pearson Education Ltd., 2004, ISBN: 978-0-13-046109-4.

T2: J. Nielsen, *Usability Engineering*, San Francisco, CA, USA: Morgan Kaufmann Publishers, 1993, ISBN: 0-12-518406-9.

Reference Books:

R1: D. Norman, *The Design of Everyday Things*, Revised and Expanded ed., New York, NY, USA: Basic Books, 2013, ISBN: 978-0-465-05065-9.

R2: J. Rubin and D. Chisnell, *Handbook of Usability Testing: How to Plan, Design, and Conduct Effective Tests*, 2nd ed., Indianapolis, IN, USA: Wiley Publishing, 2008, ISBN: 978-0-470-18548-3.

Relevant MOOCs Course (Course name and Weblink)

1. NPTEL: Human-Computer Interaction, Dr.Samit Bhattacharya, Pradeep P Yammiyavar , IIT Guwahati https://nptel.ac.in/courses/106103115
2. NPTEL: Human Computer Interactions, Prof. K. Ponnurangam,IIIT Hyderabad , https://nptel.ac.in/courses/106106177

Relevant Topics for Self-study:

https://www.coursera.org/specializations/human-factors-and-usability-engineering-designing-for-humans

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

Program Elective - III (PEC-III)

Semester	Credits	Teaching Scheme	Examination Scheme
6	03	T: 3 Hr. / Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2636PE21]: Blockchain Technology (BT)

Semester	Credits	Teaching Scheme	Examination Scheme
6	03	L: 03 / Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Information security and Cryptography.

Course Objectives: The objective of this course is to provide students with

- Develop a strong foundation of blockchain concepts, cryptographic principles, and decentralized systems.
- Apply consensus mechanisms and blockchain architectures (Bitcoin, Ethereum, Hyperledger) to solve real-world problems.
- Analyze the design and implementation of smart contracts, decentralized applications, and tokenization in enterprise contexts.
- Evaluate blockchain applications, security risks, legal/ethical issues, and propose innovative solutions using emerging trends such as Web3, DeFi, and Metaverse integration.

Course Outcomes: After completing this course, students will be able to

CO1: Explain blockchain foundations, cryptography, and decentralization principles.
CO2: Implement consensus mechanisms and differentiate between blockchain architectures such as Bitcoin, Ethereum, and Hyperledger.
CO3: Design and deploy smart contracts and decentralized applications (DApps) for real-world use cases.
CO4: Evaluate blockchain applications, security challenges, and propose solutions considering risks, regulations, and future trends.

COURSE CONTENTS

Module-I	Blockchain Foundations & Cryptography	10 Hrs.	CO1

History & evolution of blockchain, Characteristics: transparency, immutability, decentralization, Centralized vs decentralized systems, Distributed Ledger Technology (DLT), Structure of a block: hash, nonce, timestamp, Merkle tree
Cryptographic fundamentals: Hash functions (SHA-256, Keccak), Symmetric & asymmetric cryptography, Public/private key infrastructure, Digital signatures & certificates, Security, integrity, and privacy in decentralized systems, Transactions and Blocks, Security Principles in Blockchain

Module-II	Consensus & Blockchain Architectures	10 Hrs.	CO2

Consensus mechanisms: Proof of Work (PoW), Proof of Stake (PoS), Delegated PoS, PBFT, Energy efficiency & scalability issues, Blockchain architectures: Bitcoin blockchain (UTXO model), Ethereum blockchain (account-based model), Permissioned vs permissionless blockchains, Layer-1 vs Layer-2 solutions
Tokenization: concepts, enterprise use, challenges

Module-III	Blockchain Platforms & Smart Contracts	10 Hrs.	CO3

Ethereum blockchain: Features, components, ecosystem, Ethereum programming languages & bytecode, How smart contracts work
Hyperledger Fabric: Architecture & working principles, Features & benefits, Differences between Ethereum & Hyperledger, Smart contracts design & deployment, Introduction to Decentralized Applications (DApps)

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Module-IV	Security, Applications & Future Directions	10 Hrs.	CO4

Security challenges: Double-spending, Sybil attacks, 51% attacks, Privacy mechanisms (zk-SNARKs, ring signatures, mixers), Scalability & sharding
Applications of blockchain: Cryptocurrency & digital payments, DeFi, supply chain, healthcare, e-voting, digital identity, Case studies (supply chain financing, health insurance)
Regulatory & ethical issues: Risks (privacy, ransomware, money laundering), Legal aspects & data privacy (GDPR), Blockchain governance & DAOs
Future trends: Web3, Metaverse, Blockchain + AI/IoT

Text Books:

T1: Imran Bashir, *Mastering Blockchain: Unlocking the Power of Cryptocurrencies, Smart Contracts, and Decentralized Applications,* 4th Edition, Packt Publishing, 2023, ISBN-13: 978-1804612569 | ISBN-10: 1804612565.

T2: Arvind Narayanan et.al, *Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction,* Princeton University Press, 2016 ISBN-13: 978-0691171692 | eBook ISBN: 978-1400884155

Reference Books:

R1: Kumar Saurabh and Ashutosh Saxena, *Blockchain Technology: Concepts and Applications,* 1st Edition, Wiley India, 2019, ISBN-13: 978-8126557660

R2: Melanie Swan, *Blockchain: Blueprint for a New Economy,* 1st Edition, O'Reilly Media, 2015, ISBN-13: 978-1491920497 | eBook ISBN: 978-1491920459

R3: Roger Wattenhofer, *The Science of the Blockchain,* 1st Edition, CreateSpace Independent Publishing, 2016, ISBN-13: 978-1522751830

R4: Andreas M. Antonopoulos, *Mastering Bitcoin: Programming the Open Blockchain*, 2nd Edition, O'Reilly Media, 2017, ISBN-13: 978-1491954386 | ISBN-10: 1491954388

Relevant MOOCs Course (Course name and Weblink)

1. NPTEL: Blockchain and its Applications, Prof. Sandip Chakraborty Prof. Shamik Sural, IIT Kharagpur
   https://nptel.ac.in/courses/106105235
2. NPTEL: Blockchain Architecture Design and Use Cases, Prof. Sandip Chakraborty & Prof. Praveen Jayachandran, IIT Madras
   https://nptel.ac.in/courses/106105184

Relevant Topics for Self-study:

1. Introduction to Blockchain Technology and Application

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2636PE22]: Scalable AI (SAI)

Semester	Credits	Teaching Scheme	Examination Scheme
6	03	L: 03 Hrs. / Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite:

- Computer Science Fundamentals, Programming, AI and Machine Learning Concepts and System Design and Architecture

Course Objectives: The objective of this course is to provide students with

- **Identify** and solve scalability challenges in machine learning and deep learning. This includes managing large datasets, optimizing model training, and ensuring models perform well in production environments.
- **Apply** distributed computing principles to AI. Students will understand how to use distributed frameworks and parallel processing to train models on large-scale datasets more effectively.
- **Design** and implement scalable AI architectures. This involves choosing the right cloud infrastructure, hardware, and software tools to support the growth of an AI application.
- **Optimize** AI models for performance. The course covers techniques such as model compression, quantization, and algorithmic improvements to reduce computational costs without losing accuracy.

Course Outcomes: After completing this course, students will be able to

CO1: **Solving** problems related to distributed AI systems by simulating & analyzing their scalability, fault tolerance, & consistency.
CO2: **Develop** big data pipelines & implement scalable feature engineering techniques for large datasets.
CO3: **Implement** & optimize distributed & trained machine learning models for efficient inference at scale.
CO4: **Assess**, automate, & manage scalable AI/ML deployment pipelines using best practices in MLOps.

COURSE CONTENTS

Module-I	Foundations of Scalable AI and Distributed Systems	10 Hrs.	CO1

Introduction to Scalable AI: The need for scalability in modern AI, and scalability metrics. **Explainable AI overview:** Understanding Black-box Models, Techniques for Model Interpretability, Practical Applications of XAI, Future Trends and Challenges in XAI, Comparing AI and XAI.
Distributed Computing Fundamentals: Introduction to distributed systems: key concepts and challenges (e.g., fault tolerance, consistency), Data Parallelism vs. Model Parallelism
Introduction to distributed frameworks: Apache Spark and Dask, **Cloud Infrastructure for AI: Overview of major cloud platforms:** AWS, Google Cloud, and Azure. **Cloud computing services for AI**: virtual machines, containerization (Docker, Kubernetes), and serverless computing.
Case Study: Setting up a basic distributed computing environment on a cloud platform.

Module-II	Scalable Data Engineering and Feature Extraction	10 Hrs.	CO2

Big Data Processing with Apache Spark: Introduction to the Spark architecture: RDDs, DataFrames, and the Spark ecosystem, Data ingestion and transformation using PySpark, **Scalable Data Engineering**

Algorithms Scalable Feature Engineering: Challenges of feature engineering with big data, Feature selection and dimensionality reduction at scale: PCA, hashing trick, Handling categorical and unstructured data (text, images) in a distributed setting. **Scalable Feature Extraction Algorithms Stream Processing and Real-Time Data:** Introduction to stream processing: concepts and use cases, Frameworks for streaming data: Apache Kafka and Spark Streaming.

Case Study: Building a simple real-time data pipeline for an AI application.

Module-III	Scalable Model Training and Optimization	10 Hrs.	CO3

Distributed Machine Learning: Introduction to parallel and distributed training of ML models, Parameter server architecture and collective communication libraries (e.g., NCCL). **Optimizing Deep Learning Models :** Techniques for improving training efficiency: mixed-precision training and gradient accumulation, Model compression techniques: pruning, quantization, and knowledge distillation, **Distributed Optimization Algorithms, Scaling Inference and Deployment:** Challenges of serving models in production at scale, Frameworks for high-performance inference: TensorFlow Serving, TorchServe, Triton Inference Server. **Scalable Model-Specific Algorithms**

Case Study: Deploy a model as a scalable API endpoint using a modern serving framework.

Module-IV	MLOps and Security in SAI	09 Hrs.	CO4

The MLOps Lifecycle: Introduction to MLOps: defining the MLOps lifecycle from experimentation to production, Tools for MLOps: MLflow for experiment tracking, DVC for data versioning,Threat Modeling for Scalable AI Systems , **Data Security & Privacy in AI Pipelines, Model Security & Trustworthiness, Secure Deployment & Infrastructure (Cloud, Edge), Monitoring and Governance: Monitoring AI systems in production:** data drift, model drift, and performance degradation, Alerting and logging for AI systems, Governance and ethics in scalable AI: fairness, transparency, and accountability , Incident Response in Scalable AI .

Case Study: Design a scalable AI system architecture ensuring edge & cloud components with security controls; produce design document, threat model, governance plan.

Text Books:

T1: Abhishek Mishra, *Scalable AI and Design Patterns: Design, Develop, and Deploy Scalable AI Solutions.* New York: Apress, 2024.

T2: Munn, M., & Pitman, D., *Explainable AI for Practitioners: Designing and Implementing Explainable ML Solutions,* O'Reilly Media,2022.

Reference Books:

R1: Kleppmann, M. *Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems,* O'Reilly Media, 2017.

R2: Leskovec, J., Rajaraman, A., & Ullman, J. D., *Mining of Massive Datasets* ,3rd ed., Cambridge University Press, 2020.

R3: Chambers, B., & Zaharia, M., Spark*: The Definitive Guide: Big Data Processing Made Simple*, O'Reilly Media, 2018.

Relevant MOOCs Course (Course name and Weblink)

1. NPTEL- Scalable Data Science, IIT Kharagpur, Prof. Anirban Dasgupta and Prof. S. Bhattacharya https://nptel.ac.in/courses/106105186
2. NPTEL- Applied Accelerated Artificial Intelligence, IIT Palakkad, Prof. Satyajit Das, Prof. S.Chickerur, Prof. Bharatkumar Sharma, Prof. Adesuyi Tosin, Prof.Ashrut Ambastha, https://nptel.ac.in/courses/106106238.

Relevant Topics for Self-study:

1. Big Data Fundamentals & Distributed Computing
2. Model Optimization & Scalability

Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2636PE23]: Web Development Framework (WDF)

Semester	Credits	Teaching Scheme	Examination Scheme
6	03	L: 03 Hrs. / Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- HTML, CSS

Course Objectives: The objective of this course is to provide students with

- Ability to Understand, connect, and manage fully functional web-based database systems.
- The knowledge and skills to learn front-end technologies for website development.
- Ability to develop the skills to build, manage, and deploy scalable server-side web applications using Node.js and MongoDB.
- Skills to learn REST API-based enterprise website development using React, Node.js, and Spring Boot with various database technologies.

Course Outcomes: After completing this course, students will be able to

CO1: Understand database types, interfacing methods, challenges, and connect databases to web applications.
CO2: Build a web application using server-side technologies.
CO3: Develop an interactive web application using client-side technologies.
CO4: Develop, secure, test, and deploy Spring Boot applications with database connectivity, messaging, and API testing.

COURSE CONTENTS

Module-I	Databases in Web Development	10 Hrs.	CO1

What is a Database? Role of Database in Web Application, types of Databases in Web Application: Relational, Object Oriented, Graph Database. Interfacing Databases with the Web: Server-Side Approaches, Client-Side Approaches, Middle-Layer Approaches. issues in developing fully fledged Web-based database applications. steps to connect databases with your Web Applications.

Module-II	Server-Side Technologies	10 Hrs.	CO2

MongoDB: NoSQL and MongoDB Basics, MongoDB-Node.JS Communication, CRUD Operations using Node.JS, MongoDB–Node.js Communication, Mongoose ODM for Middleware, Advanced MongoDB, Deployment & DevOps with MongoDB.
Node.JS: Introduction to Node.JS, Environment Setup, Node.JS Events, Node.JS Functions, Node.JS Built-in Modules, File System, NPM, Install External Modules, Handling Data I/O in Node.JS, Create HTTP Server, Create Socket Server, Microservices- PM2.

Module-III	Client-Side Technologies	10 Hrs.	CO3

Front-End Frameworks: What is a web framework? Why Web Framework? Web Framework Types.
MVC: What is MVC, MVC Architecture, MVC in Practical, MVC in Web Frameworks.
AngularVersion 10+: Angular CLI, Angular Architecture, Angular Project Structure, Angular Lifecycle, Angular Modules, Angular Components, Angular Data Binding, Directives and Pipes, Angular Services and Dependency Injections (DI), Angular Routers, Angular Forms.
ReactJS: Introduction to ReactJS, React Components, Inter Components Communication, Components Styling, Routing, Redux- Architecture, Hooks- Basic hooks, useState() hook, useEffect() hook useContext() hook.

Module-IV	Full stack development Technologies	9 Hrs.	CO4

Spring Framework, Spring Boot Framework, Installing Spring Boot, Build Tool Maven/Gradle/Ant, Core

Features, Spring Security, Web Applications, JPA for database connectivity, working with SQL and NoSQL, Messaging, Testing, Deploying Spring Boot Applications, Monitoring and Testing. POSTMAN Tool for API testing.
Text Books:
T1: A. Kumar, *Web Technology: Theory and Practice*. Boca Raton, FL, USA: CRC Press, Taylor & Francis Group, 2019. ISBN: 978-1-13855-0438-8.
T2: P. Scobey and P. Lingras, *Web Programming and Internet Technologies: An E-Commerce Approach*, 2nd ed. Burlington, MA, USA: Jones & Bartlett Learning, 2016. ISBN: 978-1284070699.
Reference Books:
R1: Marty Hall, Larry Brown,*Core Web Programming*, Second Edition, Pearson Education, 2001, ISBN 9780130897930.
R2: M. Casciaro and L. Mammino, *Node.js Design Patterns – Third Edition: Design and Implement Production-Grade Node.js Applications Using Proven Patterns and Techniques.* Birmingham, UK: Packt Publishing, 2020.
R3: S. Almog, *Java Basics: A Practical Introduction to Full-Stack Java,* Independently Published, 2023, ISBN: 979-8851595783
Relevant MOOCs Course (Course name and Weblink)
1. NPTEL- Introduction to Modern Application Development,IIT Madras, Prof. Gaurav Raina, Tanmai Gopal : https://nptel.ac.in/courses/106106156 2. NPTEL- Internet Technology, IIT Kharagpur, Prof. Indranil Sengupta https://nptel.ac.in/courses/106105084
Relevant Topics for Self-study:
1. HTMl,HTML-5,CSS,Javascript,Typescript

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2636PE24]: Serverless Computing and Application Development (SCAD)

Semester	Credits	Teaching Scheme	Examination Scheme
6	03	L: 03 Hrs. / Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Cloud Computing and Virtualization (3504141)
- Cloud-Native Application Development (3604142)

Course Objectives: The objective of this course is to provide students with

- A comprehensive understanding of the fundamental principles and concepts of cloud computing, including its service and deployment models.
- The skills to design, develop, and deploy applications using serverless platforms like AWS Lambda, Azure Functions, or Google Cloud Functions.
- The ability to introduce students to common serverless application design patterns and best practices for building scalable, event-driven architectures.
- The ability to address security concerns and optimize the performance of serverless applications.

Course Outcomes: After completing this course, students will be able to

CO1: Explain the core concepts of serverless computing and differentiate it from traditional and containerized architectures.

CO2: Develop and deploy a basic serverless application using a major cloud provider's platform.

CO3: To design and implement event-driven architectures by utilizing serverless functions and other managed services.

CO4: Analyze and apply security best practices and performance optimization techniques for serverless applications.

COURSE CONTENTS

Module-I	Foundations of Serverless Architectures	10 Hrs.	CO1

The Serverless Paradigm: Beyond "no servers," understanding abstraction and resource management. Monolithic vs. Serverless: A detailed comparison of architecture. Benefits and limits of Serverless Computing, Core Concepts: Function-as-a-Service (FaaS): FaaS Infrastructure, API Gateways with FaaS, Backend-as-a-Service (BaaS), and Serverless Containers. The Serverless Ecosystem: Major platforms and open-source alternatives. Microservice Architecture and Event Driven Architecture: Structure and Components.

Module-II	Serverless Application Design	10 Hrs.	CO2

Architectural Patterns: Designing for scalability, resilience, and cost-efficiency. Event-Driven Architectures: The role of events, triggers, and message queues. Data Management: Strategies for handling data in a serverless environment, including databases and data streams. API Gateways: Designing and securing APIs for serverless backends.

Module-III	Building Serverless Applications	09 Hrs.	CO3

FaaS Development: Best practices for writing efficient, stateless functions. Application Implementation: Building complete applications using serverless functions and integrated services. Orchestrating Workflows: Using state machines and queues to manage complex, multi-step business logic. Hands-on: Building a complete application with a FaaS, a database, and an API Gateway.

Module-IV	Deployment, Operations, and Best Practices	10 Hrs.	CO4

The Serverless Lifecycle: Using automated tools and frameworks for deployment. Testing: Unit, integration, and end-to-end testing for serverless applications. Observability: Implementing logging, metrics, and tracing to monitor application health. Security & Cost Optimization: Best practices for securing functions and managing costs.

Text Books:

T1: S. M. Safeer, *Architecting Cloud-Native Serverless Solutions*, Birmingham, UK: Packt Publishing, 2021, ISBN: 978-1-80107-834-9.

T2: P. Sbarski, *Serverless Architectures on AWS*, Shelter Island, NY, USA: Manning Publications Co., 2017, ISBN: 978-1-61729-382-5.

Reference Books:

R1: S. Newman, *Building Microservices: Designing Fine-Grained Systems*, Sebastopol, CA, USA: O'Reilly Media, 2015, ISBN: 978-1-49195-035-7**.**

R2: P. Johnston and A. P., *Serverless Computing*, Dallas, TX, USA: Pragmatic Bookshelf, 2019, ISBN: 978-1-68050-240-9**.**

R3: A. Banks and E. Porcello, *The Serverless Framework: A Guide for Beginners*, Sebastopol, CA, USA: O'Reilly Media, 2018, ISBN: 978-1-49196-491-0**.**

Relevant MOOCs Course:

1. NPTEL-Advanced Distributed Systems, IIT Delhi, Prof. Smruti Ranjan Sarangi
   https://nptel.ac.in/courses/106102237.

Relevant Topics for Self-study:

1. **Serverless Architectures and FaaS**:
   - G. Arbezzano, M. Cavalli, S. M. N. E. G. H. I. N. I. et al., Proxmox VE: A Complete Guide to Open Source Virtualization and Containerization. Packt Publishing, 2018.
2. **Infrastructure as Code (IaC)**:
   - G. Arbezzano, M. Cavalli, S. M. N. E. G. H. I. N. I. et al., Proxmox VE: A Complete Guide to Open Source Virtualization and Containerization. Packt Publishing, 2018.
3. **Microservices Architecture**:
   - S. Newman, Building Microservices: Designing Fine-Grained Systems. O'Reilly Media, 2015.

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2636PE25]: Virtual and Augmented Reality (VAR)

Semester	Semester	Teaching Scheme	Examination Scheme
6	VI	L: 3 Hrs. /Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Computer Graphics

Course Objectives: The objective of this course is to provide students with

- To understand the fundamentals of XR technologies and VR system architecture.
- To explore VR design principles, interaction models, and performance considerations.
- To study AR system pipeline, tracking mechanisms, and AR frameworks.
- To examine Mixed Reality concepts, WebXR systems, and future trends in immersive technologies.

Course Outcomes: After completing this course, students will be able to

CO1: Explain the architecture and working principles of Virtual Reality systems and interaction models.
CO2: Analyze VR design principles, user interfaces, and performance requirements for immersive environments.
CO3: Describe Augmented Reality system pipeline, tracking techniques, and AR frameworks.
CO4: Compare VR, AR, and MR technologies and evaluate their applications, limitations, and future trends.

COURSE CONTENTS

Module-I	Module-I	10 Hrs.	CO1

Introduction to Extended Reality (XR)-VR, AR, MR – Definitions and differences, Milestones in immersive technologies, Applications of XR in industry, **VR System Architecture-** Components of VR system [Head Mounted Display (HMD), Sensors (Gyroscope, Accelerometer, Magnetometer), Controllers & haptic devices], Tracking systems[Inside-out tracking, Outside-in tracking], Rendering pipeline for VR, Stereoscopic vision and depth perception, **Human Perception in VR-** Binocular disparity, Motion parallax, Field of View (FOV), Latency and simulator sickness, **VR Interaction Models-** Gaze-based interaction, Controller-based interaction, Hand tracking, Locomotion techniques (Teleportation, Smooth movement)

Module-II	Module-II	10 Hrs.	CO2

VR Environment Design Principles- Presence and immersion, Spatial layout design, Scale and proportion in virtual spaces, **VR User Interface (UI/UX)-** 2D vs 3D UI, Diegetic vs Non-diegetic interfaces, Comfort and accessibility in VR, **Spatial Audio in VR-** 3D audio concepts, Sound localization, Role of audio in immersion, **Performance and Optimization in VR-** Frame rate requirements (90 FPS concept), Reducing latency, Minimizing motion sickness, **Case Studies of VR Applications-** Healthcare simulation, Training and education, Industrial design, Gaming environments.

Module-III	Module-III	10 Hrs.	CO3

Fundamentals of Augmented Reality- AR system components, AR working pipeline, Types of AR:

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Marker-based, Markerless, Projection-based, Location-based AR, **Tracking and Registration Techniques-** Feature detection basics, Image recognition, SLAM (Simultaneous Localization and Mapping) – concept, Depth sensing, **AR Display Technologies-** Optical see-through, Video see-through, Head-up displays, Mobile AR systems, **AR SDKs & Frameworks (Conceptual Study)-** Vuforia, ARKit, ARCore, **Applications of AR-** Retail & marketing, Maintenance & remote assistance, Education, Navigation systems.

Module-IV	Module-IV	9 Hrs.	CO4

Mixed Reality (MR)- MR vs AR vs VR, Spatial mapping, Holographic rendering, Example Device: Microsoft HoloLens, **WebXR & Browser-Based Immersive Systems-** Concept of WebXR, Advantages and limitations, Cloud-based XR, **Ethical, Privacy & Social Issues in XR-** Data privacy, Biometric tracking concerns, Psychological effects, Digital addiction, **Emerging Trends in XR-** Metaverse concepts, AR Cloud, AI in XR, Haptic feedback evolution, **Research Directions in XR-** Lightweight headsets, Full-body tracking, Brain–computer interfaces (introductory)

Text Books:

T1: D. Schmalstieg and T. Hollerer, *Augmented Reality: Principles and Practice*, 1st ed. Boston, MA, USA: Addison-Wesley, 2016.

T2: T. Parisi, *Learning Virtual Reality: Developing Immersive Experiences and Applications*, 1st ed. Sebastopol, CA, USA: O'Reilly Media, 2015.

Reference Books:

R1: S. Aukstakalnis, *Practical Augmented Reality: A Guide to the Technologies, Applications, and Human Factors for AR and VR*, 1st ed. Boston, MA, USA: Addison-Wesley, 2017.

R2: S. Lavalle, *Virtual Reality*, 1st ed. Cambridge, UK: Cambridge University Press, 2017.

R3: A. Thorn, *Mastering Unity Game Development*, 2nd ed. Birmingham, UK: Packt Publishing, 2018.

Relevant MOOCs Course (Course name and Weblink)

1. **C# Programming for Unity Game Development Specialization**
   Learn to Program Unity Games with C#. Learn C# programming and how to use C# in Unity games Dr. Tim "Dr. T" Chamillard https://www.coursera.org/specializations/programming-unity-game-development
2. **User Experience & Interaction Design for AR/VR/MR/XR — Coursera (University of Michigan)**
   Focuses on interaction design and UX for immersive technologies, with rapid prototyping and design thinking for XR. **https://www.mooc-list.com/tags/virtual-reality**

Relevant Topics for Self-study:

1. Cloud gaming and multiplayer game architecture basics
2. Advanced C# concepts for Unity (object-oriented patterns, events, coroutines)
3. Storyboarding & prototyping for AR/VR experiences

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

Program Elective – II Laboratory (PECL-II)

Semester	Credits	Teaching Scheme	Examination Scheme
6	01	P: 2 Hr. / Week	CIE(TW): 25 Marks OR : 25 Marks

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)
[2636PE31]: Wireless and Adhoc Network Technologies Laboratory (WANTL)

Semester	Credits	Teaching Scheme	Examination Scheme
6	01	P: 02 Hrs. / Week	CIE (TW): 25 Marks ESE (OR): 25Marks

Prerequisite: Students should have prior knowledge of

- Computer Network Technology

Course Objectives: The objective of this course is to provide students with the ability to

- **Understand** the configuration, standards, and security aspects of Wireless LANs (WLANs).
- **Explore** the applications of Bluetooth and Wi-Fi in short-range communication and IoT scenarios.
- **Demonstrate** the coexistence and performance analysis of heterogeneous wireless technologies such as WiMAX and Wi-Fi.
- **Analyse** routing strategies and energy efficiency issues in Wireless Sensor Networks (WSN).
- **Evaluate** proactive, reactive, and hybrid routing protocols in Mobile Adhoc Networks (MANETs).
- **Design and propose** solutions to mitigate security threats and attacks in Adhoc and Wireless Networks.

Course Outcomes: After completing this course, students will be able to

CO1: **Configure and secure** Wireless LANs (WLANs) using IEEE 802.11 standards and authentication protocols.
CO2: **Compare and justify** the applicability of Bluetooth and Wi-Fi technologies in IoT environments.
CO3: **Evaluate** the coexistence of WiMAX and Wi-Fi networks based on performance metrics such as throughput and latency.
CO4: **Implement and analyse** hierarchical routing protocols (e.g., LEACH, PEGASIS) for energy-efficient Wireless Sensor Networks.
CO5: **Assess and compare** proactive, reactive, and hybrid routing protocols in Mobile Adhoc Networks under varying mobility conditions.
CO6: **Simulate and propose** effective countermeasures against security attacks (e.g., Blackhole, Wormhole) in Adhoc Networks.

COURSE CONTENTS

Assign. No.	Description of Assignment	CO
1	**WLAN Setup and Security Configuration -** Configure a Wi-Fi network using IEEE 802.11 standards in Packet Tracer. • Set up access points and wireless clients. • Configure WLAN security protocols (WEP, WPA2, WPA3). • Test connectivity and secure communication.	CO-1
2	**Bluetooth and IoT Integration Simulation -**Demonstrate short-range wireless technology in IoT use cases using cisco packet tracer. • Configure Bluetooth connections between IoT devices and a gateway. • Show BLE application for smart-home automation. • Compare Wi-Fi and Bluetooth setups in terms of coverage and throughput.	CO-2

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3	**WiMAX and Wi-Fi Coexistence -** Simulate long-range (WiMAX) and short-range (Wi-Fi) wireless access using cisco packet tracer. • Create a WiMAX backbone and connect multiple Wi-Fi hotspots. • Test client handoff between Wi-Fi and WiMAX networks. • Analyse performance metrics (latency, throughput).	CO-3
4	**Wireless Sensor Network (WSN) Routing Protocols -** : Using Network Simulation 3, Compare energy efficiency of WSN routing protocols. • Simulate LEACH vs PEGASIS protocols. • Measure metrics: energy consumption, packet delivery ratio. • Discuss scalability of routing approaches.	CO-4
5	**MANET Routing Performance Analysis -** Using Network Simulation 3, Analyse proactive and reactive routing in MANETs. • Simulate AODV, DSR, and OLSR protocols. • Compare throughput, end-to-end delay, and packet delivery ratio. • Demonstrate impact of mobility on performance.	CO-5
6	**Security Attack Simulation in Adhoc Networks –** using Network Simulator 3, Demonstrate security vulnerabilities and countermeasures in MANETs. • Simulate a Blackhole or Wormhole attack in a MANET. • Evaluate network performance with and without attack. • Apply a simple security mechanism (e.g., trust-based routing) and measure improvement.	CO-6

Text Books:

T1:	C. Siva Ram Murthy & B. S. Manoj, *Ad Hoc Wireless Networks: Architectures and Protocols*, Pearson Education (Prentice Hall PTR), 1st Edition, 2004, ISBN: 978-0132465748.
T2:	Holger Karl & Andreas Willig, *Protocols and Architectures for Wireless Sensor Networks*, John Wiley & Sons, 1st Edition, 2005, ISBN: 978-0470519233.

Reference Books:

R1:	Ian F. Akyildiz & Mehmet Can Vuran, *Wireless Sensor Networks*, John Wiley & Sons, 1st Edition, 2010, ISBN: 978-0470036013.
R2:	Charles E. Perkins, *Ad Hoc Networking*, Addison-Wesley Professional, 1st Edition, 2000, ISBN: 978-0201309768.
R3:	Theodore S. Rappaport, *Wireless Communications: Principles and Practice*, Prentice Hall, 2nd Edition, 2002, ISBN: 978-0130422323
R4:	Jochen H. Schiller, *Mobile Communications*, Addison-Wesley, 2nd Edition, 2003, ISBN: 978-1405890304.

Relevant MOOCs Course (Course name and Weblink):

1. NPTEL: Wireless Ad Hoc and Sensor Networks, IIT Kharagpur, Prof. Sudip Misra- https://nptel.ac.in/courses/106105160
2. NPTEL: Introduction to Wireless and Cellular Communications , IIT Madras, Prof. David Koilpillai https://nptel.ac.in/courses/106106167

Relevant Topics for Self-study:

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)
[2636PE32]: Computer Vision Laboratory (CVL)

Semester	Credits	Teaching Scheme	Examination Scheme
6	01	P: 02 Hrs. / Week	CIE (TW): 25 Marks ESE (OR): 25 Marks

Prerequisite: Students should have prior knowledge of

- Fundamentals of Digital Image Processing
- Fundamentals of Python Programming
- Familiarity with NumPy and OpenCV

Course Objectives: The objective of this course is to provide students with

- Equip students with the knowledge and skills to process and enhance digital images using filtering and edge detection techniques.
- Enable students to analyze shapes and structures in binary images through morphological operations and region-based processing.
- Develop the ability to extract, describe, and match features across multiple images for real-world applications.
- Facilitate hands-on experience in estimating depth and motion using stereo vision and optical flow.
- Train students to implement real-time object and face detection using pretrained computer vision models.

Course Outcomes: After completing this course, students will be able to

CO1: Apply image filtering, enhancement, and edge detection techniques to process real-world visual data.
CO2: Analyze binary images and extract shape-related features using morphological and region-based techniques.
CO3: Detect key points and perform feature matching across images for applications like tracking and registration.
CO4: Implement shape and object detection algorithms like Hough Transform for lines and circles in static images.
CO5: Estimate depth and detect motion using stereo vision and optical flow methods in dynamic environments.
CO6: Utilize pretrained models for real-time face, object, and pedestrian detection in images and video streams.

COURSE CONTENTS

Assign. No.	Description of Assignment	CO
1.	**Image Preprocessing and Edge Detection** Implement fundamental image preprocessing techniques including filtering, histogram equalization, and edge detection to enhance grayscale images for further analysis.	CO1
2.	**Binary Image Processing and Shape Analysis** Convert grayscale images to binary, perform object labeling, apply morphological operations, and compute basic shape features such as area, centroid, and aspect ratio.	CO2
3.	**Keypoint Detection and Feature Matching** Implement an approach (ML/DL) for keypoint detection and feature matching between two related images using pre-trained or custom-trained models.	CO3
4.	**Line and Circle Detection** Implement a learned models/ neural network-based pipelines to detect and analyze geometric shapes (lines and circles) in images	CO4
5.	**Depth Estimation and Motion Detection** Implement depth estimation and motion detection using **deep neural networks**	CO5
6.	**Object and Face Detection using Pretrained Models** Implement object and face detection using modern deep learning models such as YOLO, SSD, RetinaNet, or MTCNN. Detect	CO6

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	faces, pedestrians, or vehicles in static images or real-time video, and draw bounding boxes around the detected objects.	
Text Books:		
T1:	Simon J. D. Prince, *Computer Vision: Models*, Learning, and Inference", Cambridge University Press, 2012.	
T2:	Rafael C. Gonzalez, Richard E. Woods, *Digital Image Processing*, 3rd Edition, Pearson, ISBN: 978-81-317-2695-2, 2007.	
Reference Books:		
R1:	Richard Szeliski, *Computer Vision: Algorithms and Applications*, Springer, 2011.	
R2:	Jan Erik Solem, *Programming Computer Vision with Python*, O'Reilly, 2012.	
R3:	Adrian Kaehler and Gary Bradski, *Learning OpenCV 4*, O'Reilly, 2019.	
Relevant MOOCs Course (Course name and Weblink):		
1. OpenCV Python Docs. https://docs.opencv.org/ 2. GitHub – OpenCV Python Tutorials. https://github.com/opencv/opencv-python 3. PyImageSearch Blog. https://pyimagesearch.com/ (Computer Vision Problems) 4. PyTorch. https://pytorch.org/		
Relevant Topics for Self-study:		
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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2636PE33]: Modern Databases and Scalable Architecture Laboratory (MDSAL)

Semester	Credits	Teaching Scheme	Examination Scheme
6	01	P: 02 Hrs. / Week	CIE (TW): 25 Marks ESE (PR): 25 Marks

Prerequisite: Students should have prior knowledge of

- Database Management System, web development framework (WDF)

Course Objectives: The objective of this course is to provide students with

- To learn the various algorithmic design strategies.
- To apply efficiently in problem solving.
- To provide hands-on experience with modern DevOps tools and practices such as Git, Jenkins, Docker, Kubernetes, Prometheus, and Grafana.
- To automate software development, deployment, and monitoring processes in real-world scenarios.

Course Outcomes: After completing this course, students will be able to

CO1: Implement the various algorithmic design strategies and use them to solve real-time problems/ applications
CO2: Apply Divide & Conquer as well as Greedy approach to design algorithms.
CO3: Understand and analyze optimization problems using dynamic programming.
CO4: Implement version control, build automation, containerization, and deployment using industry-standard DevOps tools.
CO5: Design and execute CI/CD pipelines with automated testing and monitoring to ensure reliable software delivery.

COURSE CONTENTS

Assign. No.	Description of Assignment	CO
1.	Assignment 1: Case Study on Distributed Databases (Google Spanner, Amazon Aurora). Discuss in terms of its architecture, query processing and also discuss real-world use cases	CO1
2.	Study of Cassandra and SQLite. Also, implement CRUD Operations on	CO1, CO2
3.	Design a database structure for storing student data using Cassandra • Create a table of students with fields: • student_id (PRIMARY KEY) • name (TEXT) • department (TEXT) • year (INT) also implement the role of partition keys and clustering keys in Cassandra.	CO1, CO2, CO3
4.	To perform Create, Read, Update, and Delete operations using Cassandra Query Language (CQL). 1. Insert a minimum of **10 records** into the student's table. 2. Retrieve: . All student records. a. Students from a specific department using filtering. 2. Update the department of a specific student. 3. Delete a student record based on student_id.	CO2,CO4

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5.	To implement batch operations and indexing to improve query performance.	CO3
6.	To perform CRUD operations on Cassandra using a programming interface (Python or Java).	CO4, CO5

Text Books:	
T1:	R. Ramakrishnan and J. Gehrke, *Database Management Systems*, 3rd ed., New Delhi, India: McGraw-Hill, 2003.
T2:	C. J. Date, *An Introduction to Database Systems*, 8th ed., Boston, MA, USA: Pearson, 2003.
Reference Books:	
R1:	M. Kleppmann, *Designing Data-Intensive Applications*, 1st ed., Sebastopol, CA, USA: O'Reilly Media, 201
R2:	A. Beaulieu, *Learning SQL*, 3rd ed., Sebastopol, CA, USA: O'Reilly Media, 2020.
R3:	K. Morris, *Infrastructure as Code: Managing Servers in the Cloud*, 2nd ed., Sebastopol, CA, USA: O'Reilly Media, 2020.
R4:	A. Petrov, *Database Internals: A Deep Dive into How Distributed Data Systems Work*, 1st ed., Sebastopol, CA, USA: O'Reilly Media, 2019.

Relevant MOOCs Course (Course name and Weblink):

1. NPTEL: Big Data Computing, Prof. Rajiv Misra- IIT Patna
   https://nptel.ac.in/courses/106104189
2. NPTEL: Cloud Computing and Distributed Systems, Prof. Rajiv Misra IIT Patna
   https://onlinecourses.nptel.ac.in/noc21_cs15/preview

Relevant Topics for Self-study:

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2636PE34]: Cloud-Native Application Development Laboratory (CADL)

Semester	Credits	Teaching Scheme	Examination Scheme
6	01	P: 02 Hrs. / Week	CIE (TW): 25 Marks ESE (OR): 25 Marks

Prerequisite: Students should have prior knowledge of

- Cloud Computing and Virtualization

Course Objectives: The objective of this course is to provide students with

- The ability to set up and configure a cloud-native development environment, including containerization with Docker and a language runtime.
- The skills to design and implement applications using both synchronous and asynchronous messaging patterns for effective microservices communication.
- Comprehensive knowledge of connectivity and composition patterns, such as service discovery and service orchestration, to build and manage robust cloud-native applications.
- The ability to design and implement event-driven architectures using relevant patterns and technologies for handling event delivery and state management.

Course Outcomes: After completing this course, students will be able to

CO1: Set up a cloud-native development environment and implement a simple microservice using synchronous communication patterns like REST or gRPC.

CO2: Design and implement multi-service applications using both synchronous (Request-Response) and asynchronous (Producer-Consumer) messaging patterns for effective communication.

CO3: Deploy and manage containerized microservices on a Kubernetes cluster, applying service registry, discovery, and orchestration patterns to manage the lifecycle of applications.

CO4: Develop event-driven applications using the Publisher-Subscriber pattern and demonstrate how to handle event delivery and state management.

COURSE CONTENTS

Assign. No.	Description of Assignment	CO
1.	**Set up the development environment**: Install Docker and a language runtime (e.g., Python, Node.js). Implement a simple microservice using either a RESTful API or gRPC to demonstrate synchronous communication, which is a core principle of cloud-native development.	CO1
2.	**Synchronous Messaging Patterns:** Implement a multi-service application to demonstrate the Request-Response pattern. Develop at least two microservices using RESTful APIs and establish communication between them	CO2
3.	**Asynchronous Messaging Patterns:** Implement the Producer-Consumer pattern using a message broker like Apache Kafka or RabbitMQ. Design one service to produce messages to a queue and to consume and process them.	CO2
4.	**Service Connectivity and Discovery:** Deploy a containerized microservice to a local Kubernetes cluster. Implement a service registry and discovery pattern to ensure other services can locate and communicate with it.	CO3
5.	**Service Orchestration and Management:** Use a tool like Kubernetes to orchestrate the deployment and scaling of a multi-container application. Apply service composition	CO3

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	patterns to manage the lifecycle and dependencies of the services.	
6.	**Event-Driven Architecture:** Implement the Publisher-Subscriber pattern using a public cloud-based messaging service (e.g., Google Cloud Pub/Sub) to build an event-driven application. This will demonstrate how to handle event delivery and state management.	CO4

Text Books:	
T1:	K. Indrasiri and S. Suhothayan, *Design Patterns for Cloud Native Applications*. Sebastopol, CA: O'Reilly Media, 2021.
T2:	C. Fehling, F. Leymann, R. Retter, W. Schupeck, and P. Arbitter, Cloud Computing Patterns: Fundamentals to Design, Build, and Manage Cloud Applications. Springer, 2014.
Reference Books:	
R1:	J. Gilbert, *Cloud Native Development Patterns and Best Practices*, Birmingham, UK: Packt Publishing, 2018.
R2:	C. Fehling et.al, *Cloud Computing Patterns: Fundamentals to Design, Build, and Manage Cloud Applications*, Cham, Switzerland: Springer, 2014.
R3:	K. Indrasiri and S. Suhothayan, *Design Patterns for Cloud Native Applications: Patterns in Practice Using APIs, Data, Events, and Streams,* O'Reilly Media, 2021.

Relevant MOOCs Course (Course name and Weblink):

1. NPTEL: Advanced Distributed Systems, IIT Delhi, Prof. Smruti Ranjan Sarangi - https://nptel.ac.in/courses/106102237.
2. G. K. N. I., "Docker Tutorial for Beginners," https://www.youtube.com/watch?v=x28SihmTjWw.

Relevant Topics for Self-study:

Docker: Docker Docs, "Reference," Docker, [Online]. Available: https://docs.docker.com/reference/.
Kubernetes (Container Orchestration): Kubernetes, "Reference," Kubernetes, [Online]. Available: https://kubernetes.io/docs/reference/.
gRPC (Microservices Architecture):gRPC, "gRPC," Cloud Native Computing Foundation, [Online]. Available: https://grpc.io/.
Apache Kafka (Messaging Systems): Apache Kafka, "Documentation," Apache Kafka, [Online]. Available: https://kafka.apache.org/documentation/.
RabbitMQ (Messaging Systems): RabbitMQ, "RabbitMQ: One broker to queue them all," VMware Tanzu, [Online]. Available: https://www.rabbitmq.com/.

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2636PE35]: Interface Design and Usability Engineering Laboratory (IDUEL)

Semester	Credits	Teaching Scheme	Examination Scheme
6	01	P: 02 Hrs. / Week	CIE (TW): 25 Marks ESE (OR): 25 Marks

Prerequisite:

- Basic knowledge of programming and user interface design principles. Familiarity with prototyping tools such as Figma, Adobe XD, or Balsamiq.

Course Objectives: The objective of this course is to provide students with

- To introduce the principles of human factors and their role in interface design.
- To enable students to design wireframes and prototypes for interactive systems.
- To develop skills in applying usability heuristics for evaluating interfaces.
- To provide hands-on experience in conducting usability testing and gathering user feedback.
- To create awareness of accessibility standards and evaluation tools.
- To apply methods for structuring information and designing effective navigation systems.

Course Outcomes: After completing this course, students will be able to

CO1: Explain the role of human perception, memory, and error in HCI design.
CO2: Develop low-fidelity and high-fidelity prototypes using industry tools.
CO3: Critically assess applications using Nielsen's usability heuristics.
CO4: Conduct user research and usability testing to gather actionable insights.
CO5: Perform accessibility evaluation using automated and manual tools aligned with WCAG standards.
CO6: Design information architecture and navigation systems for improved usability.

COURSE CONTENTS

Assign. No.	Description of Assignment	CO
1.	**Exploring Human Factors in HCI** *Objective:* Understand the role of human perception, memory, and error in interface design. *Tasks:* 1. Identify and analyze at least two examples of good and bad interaction designs (apps/websites). 2. Map Norman's design principles (visibility, feedback, constraints, affordances) to your examples. 3. Write a short reflection on how human error impacts usability.	CO1
2.	**Wireframing & Prototyping** *Objective:* Learn how to create low-fidelity and high-fidelity prototypes. *Tasks:* 1. Choose a simple application (e.g., food ordering, bus booking, college portal). 2. Create low-fidelity wireframes for at least 3 screens. 3. Convert one of the wireframes into a high-fidelity prototype.	CO2
3.	Usability Heuristic Evaluation Objective: Apply usability heuristics to real-world applications. Tasks: 1. Select a website or mobile app.	CO3

	2. Evaluate it against Nielsen's 10 Usability Heuristics. 3. Rate each heuristic (scale 1–5) and note violations. 4. Suggest redesign recommendations.	
4.	User Research & Usability Testing Objective: Practice user research methods and conduct usability testing. Tasks: 1. Design a short user survey/interview (5–7 questions). 2. Recruit 2–3 users and observe them performing tasks on an application. 3. Record usability metrics: time on task, error rate, satisfaction. 4. Summarize findings and suggest improvements.	CO4
5.	Accessibility Evaluation Objective: Evaluate accessibility of digital content. Tasks: 1. Pick a public website. 2. Run an accessibility audit (color contrast, alt text, navigation, ARIA labels, keyboard accessibility). 3. Compare findings against WCAG 2.1 guidelines. 4. Suggest improvements to make the site accessible to diverse users.	CO5
6.	Information Architecture & Navigation Design Objective: Explore how content organization and navigation impact usability. Tasks: 1. Choose a content-heavy site (e.g., e-commerce, news, university portal). 2. Conduct a card sorting exercise (with 2–3 peers/users) to group and label content categories. 3. Create a site map (hierarchical structure) based on results. 4. Evaluate if the navigation supports findability and efficiency. 5. Suggest redesign improvements.	CO6

Text Books:	
T1:	A. Dix, et al., *Human-Computer Interaction*, 3rd ed., Harlow, England: Pearson Education Ltd., 2003, ISBN: 978-0-13-046109-4.
T2:	D. Norman, *The Design of Everyday Things*, Revised and Expanded ed., New York, NY, USA: Basic Books, 2013, ISBN: 978-0-465-05065-9.
Reference Books:	
R1.	B. Shneiderman et.al., *Designing the User Interface: Strategies for Effective Human-Computer Interaction*, 6th ed., Boston, MA, USA: Pearson, 2016, ISBN: 978-0-13-438038-4.
R2.	Y. Rogers et.al,, *Interaction Design: Beyond Human-Computer Interaction*, 5th ed., Indianapolis, IN, USA: Wiley, 2019, ISBN: 978-1-119-54725-1.
R3.	J. Sauro and J. R. Lewis, *Quantifying the User Experience: Practical Statistics for User Research*, 2nd ed., Cambridge, MA, USA: Morgan Kaufmann, 2016, ISBN: 978-0-12-802308-2.

Relevant MOOCs Course (Course name and Weblink):

1. NPTEL-Human-Computer Interaction – Prof. Rajiv Ratn Shah, IIIT Delhi
https://nptel.ac.in/courses/106106575
2. NPTEL-User Interface Design - Prof. Saptarshi Kolay , IIT Roorkee
https://nptel.ac.in/courses/124107008

Relevant Topics for Self-study:

1. Eye-tracking and gaze-based interaction.
2. Emotional design and affective computing.
3. Usability testing in Virtual and Augmented Reality.
4. Accessibility for elderly and differently-abled users.

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

Program Elective – III Laboratory (PECL-III)

Semester	Credits	Teaching Scheme	Examination Scheme
6	01	P: 2 Hr. / Week	CIE(TW): 25 Marks OR : 25 Marks

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)
[2636PE41]: Blockchain Technology Laboratory (BTL)

Semester	Credits	Teaching Scheme	Examination Scheme
6	01	P: 02 / Week	CIE (TW): 25 Marks ESE (OR): 25 Marks

Prerequisite: Students should have prior knowledge of

- **Information Security and Cryptography**
-

Course Objectives: The objective of this course is to provide students with

1. **Understand** the working of different crypto wallet types and their role in blockchain ecosystems.
2. **Apply** blockchain development tools to set up local test environments and deploy smart contracts.
3. **Analyze** the differences between Bitcoin and Ethereum development frameworks with respect to system setup and contract programming.
4. **Evaluate** security, usability, and performance trade-offs of blockchain tools, wallets, and smart contracts.
5. **Create** decentralized applications (DApps) such as voting systems on Ethereum testnets by integrating frontend and smart contracts.

Course Outcomes: After completing this course, students will be able to

CO1: **Demonstrate** the ability to set up and operate hosted, self-custody, and hardware crypto wallets, including buying, sending, and receiving cryptocurrencies.
CO2: **Implement** Ethereum smart contracts using Solidity and interact with them via testnets and frontend tools.
CO3: **Analyze** Bitcoin Core operations and apply basic Solidity concepts like arrays, mappings, and functions in smart contracts.
CO4: **Deploy** smart contracts on Ethereum testnets and document the deployment process including wallet configuration, gas fee estimation, and transactions.
CO5: **Design and Develop** a blockchain-based voting/polling system with essential features such as candidate registration, double-vote prevention, and decentralized participation.

COURSE CONTENTS

Assign. No.	Description of Assignment	CO
1.	**Crypto Wallets** 1. **Setup and Evaluate Different Wallet Types** • **Hosted Wallets** (e.g., exchanges like Binance, Coinbase) • **Self-Custody Wallets** (e.g., Metamask, Trust Wallet) • **Hardware Wallets** (e.g., Ledger, Trezor) *– optional but recommended for exploration* • Compare and evaluate each wallet type based on: • Security • Ease of use • Control over private keys • Accessibility and recovery options 2. **Basic Wallet Operations (using Bitcoin as an example):** • Buying cryptocurrency	CO1

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	• Selling cryptocurrency • Sending cryptocurrency • Receiving cryptocurrency • Exchanging one cryptocurrency for another • Understanding the role of **mining** in transaction verification	
0.	**Ethereum Local Development & Smart Contracts** 1. **Set up a Local Ethereum Development Network** • Use **Hardhat** (or Ganache/Truffle) to create a local blockchain. • Write and deploy a simple smart contract that allows users to send a 👋 (wave). • Keep track of the **total number of waves**. 2. **Deploy and Interact via Testnet** • Connect an Ethereum wallet (e.g., **Metamask**) to your project. • Deploy the contract on an Ethereum **testnet** (e.g., Goerli, Sepolia). • Build a **web frontend** to interact with the deployed contract: • Connect wallet • Send wave messages • Store messages in an **array of structs** for retrieval/display	CO2
0.	**Bitcoin Core & Solidity Basics** 1. **Build System Setup** • Prepare a development environment for **Bitcoin Core** (compilation & basic usage). 2. **Solidity Basics** • Write a **Hello World smart contract** in Solidity. • Implement a **Solidity program** demonstrating: • Arrays (static & dynamic) • Functions (e.g., add, retrieve, update)	CO3
0.	**Ethereum Smart Contract Deployment** • Deploy a **simple smart contract** (e.g., a storage contract or counter contract) to the **Ethereum blockchain**. • Document the **deployment process**, including: • Network used (testnet or mainnet) • Wallet setup • Gas fees and transactions	CO4
0.	**Blockchain-based Voting System** • Design and implement a **polling/voting smart contract** using Solidity. • Features to include: • Candidate/option registration • Voting by Ethereum addresses • Prevention of double voting	CO5

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	• Use of a **mapping (hashmap)** data structure for efficiency (optional) • Deploy the voting system on Ethereum testnet and interact through a web interface (Metamask + frontend).	

Text Books:	
T1:	Arvind Narayanan et.al, *Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction*, Princeton University Press, 1st Edition, July 2016, ISBN: 978-0691171692.
T2:	Chris Dannen, *Introducing Ethereum and Solidity: Foundations of Cryptocurrency and Blockchain Programming for Beginners*, Apress, 1st Edition, March 2017, ISBN: 978-1484225349.
Reference Books:	
R1:	Andreas M. Antonopoulos & Gavin Wood – *Mastering Ethereum: Building Smart Contracts and DApps*, O'Reilly Media, 1st Edition, December 2018, ISBN: 978-1491971949.
R2:	Imran Bashir, *Mastering Blockchain: Unlocking the Power of Cryptocurrencies, Smart Contracts, and Decentralized Applications*, Packt Publishing, 4th Edition, May 2023, ISBN: 978-1803241067.
R3:	Syed M. Haque – *Solidity Programming Essentials: A Beginner's Guide to Build Smart Contracts for Ethereum and Blockchain*, Packt Publishing, 2nd Edition, April 2023, ISBN: 978-1804611159
R4:	Andreas M. Antonopoulos – *Mastering Bitcoin: Unlocking Digital Cryptocurrencies*, O'Reilly Media, 2nd Edition, July 2017, ISBN: 978-1491954386.

Relevant MOOCs Course (Course name and Weblink):

1. NPTEL: Prof. Sandip Chakraborty Prof. Shamik Sural, IIT Kharagpur https://nptel.ac.in/courses/106105235
2. NPTEL:Prof. Sandip Chakraborty & Prof. Praveen Jayachandran ,IIT Madras https://nptel.ac.in/courses/106105184

Relevant Topics for Self-study:

1. https://buildspace.so/p/build-solidity-web3-app/lessons/welcome
2. https://www.theinsaneapp.com/2022/05/best-web3-projects.html
3. https://www.coinbase.com/learn/tips-and-tutorials/how-to-set-up-a-crypto-wallet

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2636PE42]: Scalable AI Laboratory (SAIL)

Semester	Credits	Teaching Scheme	Examination Scheme
6	01	P: 02 Hrs. / Week	CIE (TW): 25 Marks ESE (OR): 25 Marks

Prerequisite: Students should have prior knowledge of

- Python Programming, Machine Learning Fundamentals, Linux Basics, Basic Distributed Systems and Cloud Concepts

Course Objectives: The objective of this course is to provide students with

- **Apply** hands-on techniques to design and operate scalable AI systems effectively.
- **Analyze** and utilize distributed computation frameworks, scalable feature engineering methods, and model optimization and deployment strategies.
- **Design** and implement MLOps workflows and tools to manage large-scale data processing and AI model lifecycle.
- **Evaluate** the performance and scalability of AI systems in real – world scenarios to ensure robustness and efficiency.

Course Outcomes: After completing this course, students will be able to

CO1: **Solving** problems related to distributed AI systems by simulating & analyzing their scalability, fault tolerance, & consistency.
CO2: **Develop** big data pipelines & implement scalable feature engineering techniques for large datasets.
CO3: **Implement** & optimize distributed & trained machine learning models for efficient inference at scale.
CO4: **Assess**, automate, & manage scalable AI/ML deployment pipelines using best practices in MLOps.

COURSE CONTENTS

Assign No.	Description of Assignment	CO
1.	Develop a PySpark application that reads JSON data from a file, performs data cleansing, and calculates summary statistics.	CO2
2.	Optimize a PySpark job by using broadcast variables and Spark's built-in functions. Measure performance improvement.	CO2
3.	a.Demonstrate setting up a single-node, single-broker Kafka cluster and show basic operations such as creating topics and producing/consuming messages. b. Extend the cluster to multiple brokers on a single node.	CO2
4.	Write a script to create a topic with specific partition and replication factor settings.	CO2
5.	Implement operations such as listing topics, modifying configurations, and deleting topics.	CO2
6.	. Implement a simple word count stream processing application using Kafka Stream. a. Implement Kafka integration with the Hadoop ecosystem.	CO2

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7.	Implement operations such as listing topics, modifying configurations, and deleting topics.	CO2
8.	Analyze website performance using Google Analytics and identify areas for optimization and improvement	CO3
9.	Build a recommendation engine that uses customer browsing and purchase data to suggest relevant products.	CO4

Text Books:	
T1:	Abhishek Mishra, *Scalable AI and Design Patterns: Design, Develop, and Deploy Scalable AI Solutions.* New York: Apress, 2024
T2:	Munn, M., & Pitman, D., *Explainable AI for Practitioners: Designing and Implementing Explainable ML Solutions,* O'Reilly Media,2022.
Reference Books:	
R1:	Kleppmann, M. *Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems,* O'Reilly Media, 2017
R2:	Leskovec, J., Rajaraman, A., & Ullman, J. D., *Mining of Massive Datasets* ,3rd ed., Cambridge University Press, 2020.
R3:	Chambers, B., & Zaharia, M., Spark*: The Definitive Guide: Big Data Processing Made Simple*, O'Reilly Media, 2018.

Relevant MOOCs Course (Course name and Weblink):

1. NPTEL- Scalable Data Science, IIT Kharagpur, Prof. Anirban Dasgupta and Prof. S. Bhattacharya https://nptel.ac.in/courses/106105186
2. NPTEL- Applied Accelerated Artificial Intelligence, IIT Palakkad, Prof. Satyajit Das, Prof. S. Chickerur, Prof. Bharatkumar Sharma, Prof. Adesuyi Tosin, Prof. Ashrut Ambastha. https://nptel.ac.in/courses/106106238

Relevant Topics for Self-study:

- https://learn.microsoft.com/en-us/azure/aks/best-practices-ml-ops

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)
[2636PE43]: Web Development Framework Laboratory (WDFL)

Semester	Credits	Teaching Scheme	Examination Scheme
6	01	P: 02 Hrs. / Week	CIE (TW): 25 Marks ESE (OR): 25 Marks

Prerequisite: Students should have prior knowledge of

- Python Programming, Machine Learning Fundamentals, Linux Basics, Basic Distributed Systems and Cloud Concepts

Course Objectives: The objective of this course is to provide students with

- Designing and developing responsive and interactive web pages using HTML5, CSS3, and JavaScript.
- Building backend applications using Node.js and Express.js with MongoDB for data management.
- Creation of RESTful APIs for full-stack web application development.
- Development of dynamic frontend applications using modern frameworks such as React and Angular integrated with backend services.

Course Outcomes: After completing this course, students will be able to

CO1: To create responsive and interactive web interfaces using HTML5, CSS3, and JavaScript.
CO2: To develop backend applications using Node.js and Express.js with MongoDB integration.
CO3: To design, implement, and consume RESTful APIs for web application functionality.
CO4: To build dynamic, data-driven frontend applications using React and Angular connected to backend services.

COURSE CONTENTS

Assign No.	Description of Assignment	CO
1.	**HTML5, CSS3, JavaScript, and API:** Create a responsive personal portfolio website showcasing skills, education, and contact details using HTML5 and CSS3. Add a JavaScript-powered registration form with validation, password toggle, and dynamic styling. Integrate a GitHub User Finder tool using the Fetch API to retrieve and display user profiles with loading indicators and error handling.	CO1
2.	**Node.js & Express.js:** • Create a Node.js script that allows users to: 1. Add a student's name and marks to a MongoDB students collection via command-line arguments. 2. View all stored student records when the user enters the view command. • Create an Express.js REST API connected to MongoDB with the following features: 1. GET /tasks → Fetch all tasks. 2. POST /tasks → Add a new task (title, completed status). 3. DELETE /tasks/:id → Delete a task by its ID.	CO2

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3.	**REST API Development:** Build a 'Product Management API' with routes for adding, viewing, updating, and deleting products	CO3
4.	**React Application Using Components, Props, and Hooks:** Build a React app with a To-Do List for adding/removing tasks and a Feedback Form with validation. Use functional components, props, useState, and useEffect to manage data, lifecycle events, and interactivity.	CO4
5.	**Interactive Expense Management System in Angular with Database Integration:** Build an Angular app to manage daily expenses with features to add, edit, delete, and filter records by date or category. Display spending summaries using Angular Chart.js, and store data in a database through API integration to ensure persistent and synchronized expense records.	CO2
6.	**Web Application Development Using Spring Boot Framework:** Build a Spring Boot web application with REST APIs, database connectivity using MongoDB, and authentication. Implement messaging for asynchronous communication, test APIs with POSTMAN, write unit/integration tests, deploy the application.	CO2, CO4
7.	**Mini Project:** For the mini-project, students must develop a full-stack web application using Angular or React for the frontend, Spring Boot for the backend, and MongoDB for database storage. The application should expose secure RESTful APIs with authentication and role-based access control implemented via Spring Security. All database operations should be handled using Spring Data for MongoDB. APIs must be tested and documented using Postman, and the frontend should consume these APIs to display and manage data in a responsive, user-friendly interface. The completed application should be deployed on a server or cloud platform, with proper documentation including system architecture, API endpoints, screenshots, and a GitHub repository containing the full source code.	CO1, CO2, CO3, CO4

Text Books:	
T1:	E. Brown, *Web Development with Node and Express: Leveraging the JavaScript Stack*, 2nd ed. Sebastopol, CA, USA: O'Reilly Media, 2019. ISBN: 978-1-492-05351-4.
T2:	B. Dayley and C. Dayley, *Node.js, MongoDB, and Angular Web Development*, 2nd ed. Indianapolis, IN, USA: Addison-Wesley Professional, 2017. ISBN: 978-0-13-465553-0.
Reference Books:	
R1:	D. Flanagan, *JavaScript: The Definitive Guide*, 7th ed. Sebastopol, CA, USA: O'Reilly Media, 2020. ISBN: 978-1-491-95202-3.
R2:	A. Banks and E. Porcello, *Learning React: Modern Patterns for Developing React Apps*, 2nd ed. Sebastopol, CA, USA: O'Reilly Media, 2020. ISBN: 978-1-492-05172-5.
R3:	S. Seshadri, *Angular: Up and Running: Learning Angular, Step by Step*, 2nd ed. Sebastopol, CA, USA: O'Reilly Media, 2018. ISBN: 978-1-491-99439-9.

Relevant MOOCs Course (Course name and Weblink):

1. NPTEL: Introduction to Modern Application Development, IIT Madras- Prof. Gaurav Raina, Tanmai Gopal
https://nptel.ac.in/courses/106106156

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2. nptel: Modern Application Development, Persistent Computing Systems & CMI- Prof. Aamod Sane, Prof. Abhijat Vichare, Prof. Madhavan Mukund, https://nptel.ac.in/courses/106106222
Relevant Topics for Self-study:
• Basic HTML structure, TypeScript

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)

[2636PE44]: Serverless Computing & Application Development Laboratory (SCADL)

Semester	Credits	Teaching Scheme	Examination Scheme
6	01	P: 02 Hrs. / Week	CIE (TW): 25 Marks ESE (OR): 25 Marks

Prerequisite: **Students should have prior knowledge of**

Course Objectives: **The objective of this course is to provide students with**

- a comprehensive understanding of the fundamental principles and concepts of cloud computing, including its service and deployment models.
- the skills to design, develop, and deploy applications using serverless platforms like AWS Lambda, Azure Functions, or Google Cloud Functions.
- the ability to introduce students to common serverless application design patterns and best practices for building scalable, event-driven architectures.
- the ability to address security concerns and optimize the performance of serverless applications.

Course Outcomes: **After completing this course, students will be able to**

CO1: Explain the core concepts of serverless computing and differentiate it from traditional and containerized architecture.

CO2: Develop and deploy a basic serverless application using a major cloud provider's platform.

CO3: Design and implement event-driven architectures by utilizing serverless functions and other managed services.

CO4: Analyze and apply security best practices and performance optimization techniques for serverless applications.

COURSE CONTENTS

Assign. No.	Description of Assignment	CO
1.	**Foundations of Serverless Computing:** Explore the Serverless Paradigm by setting up a basic serverless development environment. Create and deploy a simple "Hello, World!" function using a free-tier FaaS platform (e.g., AWS Lambda, Azure Functions, Google Cloud Functions, or an open-source alternative like OpenFaaS)	CO1
2.	**Building with API Gateway:** Develop and deploy a serverless application that utilizes an API Gateway to expose the function created in Assignment 1. The focus is on understanding the integration between FaaS and API Gateway.	CO2
3.	**Event-Driven Architectures:** Design and implement an event-driven architecture where a serverless function is triggered by an event from a cloud storage service (e.g., an object upload). The function should process the object (e.g., resize an image, convert a file format) and store the output in another location.	CO2
4.	**Full-Stack Serverless Application:** Build a complete application using a FaaS, a NoSQL database (e.g., DynamoDB, Cosmos DB, Firestore), and an API Gateway. The application should demonstrate how to manage data in a serverless environment.	CO3
5.	**Orchestrating Workflows:** Implement a serverless workflow using a	CO3

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	state machine service (e.g., AWS Step Functions) or queues to manage a multi-step business process like a simple order processing system or data pipeline.	
6.	**Security and Cost Optimization:** Apply security best practices to a serverless application by using managed Identity and Access Management (IAM) roles. Additionally, implement monitoring and logging to analyze and optimize the application's performance and cost. This can be done using free-tier monitoring tools provided by the cloud vendor.	CO4

Text Books:	
T1:	S. M. Safeer, *Architecting Cloud-Native Serverless Solutions*. Packt Publishing, 2021.
T2:	Newman, *Building Microservices: Designing Fine-Grained Systems*. O'Reilly Media, 2015.
Reference Books:	
R1:	Sbarski, *Serverless Architectures on AWS*. Manning Publications, 2017.
R2:	Johnston and A. P., *Serverless Computing*. Pragmatic Bookshelf, 2019.
R3:	A. Banks and E. Porcello, *The Serverless Framework: A Guide for Beginners*. O'Reilly Media, 2018.

Relevant MOOCs Course (Course name and Weblink):

1. NPTEL: Cloud Computing by S. K. Ghosh , IIT Delhi https://nptel.ac.in/courses/106102237.

Relevant Topics for Self-study:

- IBM, "Application Development using Microservices and Serverless." [Online Course]. Coursera. Available: https://www.coursera.org/learn/applications-development-microservices-serverless-openshift.
- Packt, "Learn Azure Serverless Functions in a Weekend." [Online Course]. Coursera. Available: https://www.coursera.org/learn/packt-learn-azure-serverless-functions-in-a-weekend-13g34.
- Amazon Web Services (AWS), "Python for Serverless Applications and Automation on AWS." [Online Course]. Coursera. Available: https://www.coursera.org/learn/aws-python-serverless.
- Amazon Web Services (AWS), "Building serverless applications on AWS." [Online Course]. edX. Available: https://www.edx.org/school/aws.
- The Linux Foundation, "Introduction to Serverless on Kubernetes." [Online Course]. edX. Available: https://www.edx.org/school/linuxfoundationx.
- Microsoft, "Azure Synapse Serverless SQL: Build data analytics solutions." [Online Course]. edX. Available: https://www.edx.org/school/microsoft.
- J. Gomes, "Serverless Application Development with AWS & Typescript." [Online Course]. Udemy. Available: https://www.udemy.com/topic/serverless-computing/.
- M. Villalba, "Free AWS Lambda Tutorial - Practical beginners course to Serverless Applications." [Online Course]. Udemy. Available: https://www.udemy.com/course/introduction-to-serverless-aws/.
- A. Rodrigues, "Serverless computing in AWS." [Online Course]. Udemy. Available: https://www.udemy.com/course/serverless-computing-in-aws/.

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)
[2636PE45]: Virtual and Augmented Reality (VAR) Lab (VARL)

Semester	Credits	Teaching Scheme	Examination Scheme
6	1	P: 2 Hrs. / Week	CIE (TW): 25 Marks ESE (OR): 25 Marks

Prerequisite: Students should have prior knowledge of

- Students should have prior knowledge of Computer Graphics and basic Programming in C#/OOPs.

Course Objectives: The objective of this course is to provide students with

- To design simple immersive VR and AR applications.
- To implement interaction and tracking techniques.
- To analyze usability and performance in XR environments.
- To develop a small-scale immersive prototype solving a real-world problem.

Course Outcomes: After completing this course, students will be able to

- CO1: Design and implement interactive VR environments.
- CO2: Apply locomotion and UI principles in immersive systems.
- CO3: Develop marker-based and markerless AR applications.
- CO4: Build a mini-project addressing a real-world XR problem.

COURSE CONTENTS

Assign. No.	Description of Assignment	CO
1.	XR Development Environment Setup & Verification	CO1
2.	VR Interactive Environment – Virtual Training Room	CO1
3.	VR User Interface & Spatial Audio System	CO2
4.	Marker-Based AR – Product Visualization System	CO3
5.	Markerless AR – Surface Detection & Placement	CO4
6.	Capstone Mini Project – Real-World XR Solution	CO4

Textbooks:

T1: D. Schmalstieg and T. Hollerer, *Augmented Reality: Principles and Practice*, 1st ed. Boston, MA, USA: Addison-Wesley, 2016.

T2: T. Parisi, *Learning Virtual Reality: Developing Immersive Experiences and Applications*, 1st ed. Sebastopol, CA, USA: O'Reilly Media, 2015.

Reference Books:

R1: S. Aukstakalnis, *Practical Augmented Reality: A Guide to the Technologies, Applications, and Human Factors for AR and VR*, 1st ed. Boston, MA, USA: Addison-Wesley, 2017.

R2: J. Jerald, *The VR Book: Human-Centered Design for Virtual Reality*, 1st ed. New York, NY, USA: Morgan & Claypool, 2015.

R3: A. Thorn, *Mastering Unity Game Development*, 2nd ed. Birmingham, UK: Packt Publishing, 2018.

Relevant MOOCs Course (Course name and Weblink):

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1. **C# Programming for Unity Game Development Specialization,Coursera** Dr. Tim "Dr. T" Chamillard https://www.coursera.org/specializations/programming-unity-game-development
2. **User Experience & Interaction Design for AR/VR/MR/XR — Coursera (University of Michigan)** Focuses on interaction design and UX for immersive technologies, with rapid prototyping and design thinking for XR. **https://www.mooc-list.com/tags/virtual-reality**

Relevant Topics for Self-study:

1. Cloud-based game streaming and VR content delivery.
2. AI in AR/VR for adaptive user experiences.

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Third Year B. Tech (T. Y B. Tech) AY (2026-27) Information Technology (IT) [2636MD1X]: Multidisciplinary Minor (MDM-4)			
Semester	**Credits**	**Teaching Scheme**	**Examination Scheme**
6	**02**	**L: 02 Hr. / Week**	**CIE: 20 Marks** **ISE: 20 Marks** **ESE: 60 Marks**
Refer Annexure-I			

Third Year B. Tech (T. Y B. Tech) AY (2026-27) Information Technology (IT) [2636MDXX]: Multidisciplinary Minor Laboratory (MDM-4#)			
Semester	**Credits**	**Teaching Scheme**	**Examination Scheme**
6	**01**	**P: 02 Hr. / Week**	**CIE (TW): 25 Marks** **ISE: NA** **ESE: NA**
Refer Annexure-I			

Third Year B. Tech (T. Y B. Tech) AY (2026-27) Information Technology (IT) [2636OE5X]: Open Elective-IV (OE-IV)			
Semester	**Credits**	**Teaching Scheme**	**Examination Scheme**
VI	**02**	**T: 2 Hr. / Week**	**CIE: NA** **ISE: NA** **ESE: 50 Marks**
Refer Annexure-II			

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Third Year B. Tech (T. Y B. Tech) AY (2026-27)
Information Technology (IT)
[2636VE51]: Seminar (SEM)

Semester	Credits	Teaching Scheme	Examination Scheme
6	01	T: 01 Hr. / Week	CIE (TW): 25 Marks ESE (OR): NA ESE (PR): NA

Prerequisite: Students should have prior knowledge of

- Project Based Learning
- Software Engineering

Course Objectives: The objective of this course is to provide students with

- ability to gather the literature of specific area in a focused manner.
- skills summarize the literature to find state-of-the-art in proposed area.
- capability to recognize and define the scope for future research or development.
- competence to effectively present the case for the intended work to be done as project.
- knowledge to report literature review and proposed work in scientific way.

Course Outcomes: After completing this course, students will be able to

CO1: Interpret and summarize technical literature.
CO2: Demonstrate the techniques used in the paper.
CO3: Distinguish the various techniques required to accomplish the task.
CO4: Identify intended future work based on the technical review.
CO5: Prepare and present the content through various presentation tools and techniques in effective manner.

COURSE CONTENTS

Assign. No.	Description of Assignment/ Tasks
1.	Discussion to understand what is technical paper, how to search, where to search?
2.	Download technical papers (minimum four), getting approved from Guide and Prepare abstract summary of all papers downloaded.
3.	Read and understand in detail the decided research papers about the problem statement, techniques used, experimental details and results with conclusion from identified papers
4.	Review of the studied papers by Guide / Panel.
5.	Search / Find equivalent techniques (other than the one proposed in technical paper) so performance / complexities can be improved (by amortized analysis, not actual implementation).
6.	Prepare presentation with outline as the topic, its significance, The research problem, studied solutions (through research papers) with strengths and weaknesses of each solution, comparison of the solutions to research problem, future directions of work, probable problem statement of project, tentative plan of project work
7.	Write Seminar report.
8.	Deliver Presentation to Guide/ Panel.
9.	Verification of Seminar report and Submission.

Guidelines for Seminar Selection and Presentation:

1. Student shall identify the area or topics in Information Technology referring to recent trends and developments in consultation with industry (for their requirement) and institute guide.

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2. Student must review sufficient literature (reference books, journal articles, conference papers, white papers, magazines, web resources etc.) in relevant area on their topic as decided.
3. Seminar topics should be based on recent trends and developments. Guide should approve the topic by thoughtfully observing different techniques, comparative analysis of the earlier algorithms used or specific tools used by various researchers in the domain.
4. Research articles could be referred from IEEE, ACM, Science direct, Springer, Elsevier, IETE, CSI or from freely available digital libraries like Digital Library of India (dli.ernet.in), National Science Digital Library, JRD Tata Memorial Library, citeseerx.ist.psu.edu, getcited.org, arizona.openrepository.com, Open J-Gate, Research Gate, worldwidescience.org etc.
5. Student shall present the study as individual seminars in 20 – 25 minutes in English which is followed by Question Answer session
6. Guide should ensure that students are doing literature survey and review in proper manner.
7. Guide should give appropriate instructions for effective presentation.
8. Attendance of all other students in the class for presentation is mandatory.

Guidelines for Seminar Report:

1. Each student shall submit two copies of the seminar report in appropriate text editing tool/software as per prescribed format duly signed by the guide and Head of the department/Principal.
2. Broad contents of review report (20-25 pages) shall be
    - Title Page with Title of the topic, Name of the candidate with Exam Seat Number /Roll Number, Name of the Guide, Name of the Department, Institution, Year & University.
    - Seminar Approval Sheet/Certificate.
    - Abstract and Keywords.
    - Acknowledgments.
    - Table of Contents, List of Figures, List of Tables and Nomenclature.
    - Chapters need to cover topic of discussion.
        - Introduction with section including organization of the report
        - Literature Survey
        - Motivation, purpose and scope and objective of seminar
        - Details of design/technology/Analytical and/or experimental work, if any/
        - Discussions and Conclusions
        - Bibliography/References (in IEEE Format
        - Plagiarism Check report
3. Students are expected to use open-source tools for writing seminar report, citing the references and plagiarism detection.

Text Books:

T1: Sharon J. Gerson and Steven M. Gerson, *Technical Writing: Process and Product*, Pearson Education Asia, ISBN: 130981745, 4th Edition, 2002.

Reference Books:

R1: Andrea J. Rutherford, *Basic Communication Skills for Technology*, Pearson Education Asia,2ndEdition, 2001.

R2: Lesikar, *Lesikar's Basic Business Communication,* Tata McGraw, ISBN: 256083274, 1st Edition, 1999.

Relevant MOOCs Course (Course name and Weblink):

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Relevant Topics for Self-study:

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Annexures

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Annexure-I: Structure of Multi-Disciplinary Minor Courses

The structure for the multidisciplinary Minor courses is as follows.

			Teaching Scheme (Hours/ Week)			Credits			Examination Scheme and Marks				
Sem	**Course code**	**Name of Course**	**L**	**P/ T**	**Total**	**L**	**P/ T**	**Total credits**	**Theory**			**Practical**	**Semester**
									CIE	**ISE**	**ESE**	**CIE/ ESE**	**Total**
									[20]	**[20]**	**[60]**	**TW/ P/ OR**	**550**
3	**2603MD1X**	MDM-1	2	--	**2**	2	--	**2**	20	20	60	--	**100**
3	**2603MDXX**	MDM-1 #	-	2	**2**	-	1	**1**	-	-	-	25	**25**
4	**2604MD1X**	MDM-2	2	--	**2**	2	--	**2**	20	20	60	--	**100**
4	**2604MDXX**	MDM-2 #	-	2	**2**	-	1	**1**	-	-	-	25	**25**
5	**2605MD1X**	MDM-3	2	--	**2**	2	--	**2**	20	20	60	--	**100**
5	**2605MDXX**	MDM-3 #	-	2	**2**	-	1	**1**	-	-	-	25	**25**
6	**2606MD1X**	MDM-4	2	--	**2**	2	--	**2**	20	20	60	-	**100**
6	**2606MDXX**	MDM-4 #	-	2	**2**	-	1	**1**	-	-	-	25	**25**
7	**2607MD5X**	MDM-5	-	2	**2**	-	2	**2**	-	-	-	50	**50**
		Total	**8**	**10**	**18**	**8**	**6**	**14**	**80**	**80**	**240**	**150**	**550**

Note: In course code X is course serial number in that semester, XX: first X is laboratory or tutorial value 3 or 5 respectively and second X course serial number in that semester.
#: is laboratory or tutorial as per course requirements.

1. Students are expected to choose one of the eligible domains of MDM at the beginning of the Semester III.
2. Students will complete the chosen set of all multidisciplinary minor courses mentioned under the chosen MDM domain.
3. Students are not permitted to change from one domain to another.
4. Refer to the last column of following table for eligibility to choose a particular MDM domain.

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Domains of Multidisciplinary Minor Course

Label	Multi-Disciplinary Minor Domains	SY		TY		
		MD1-1	MD2-2	MD3-3	MD4-4	
		Sem-III	Sem-IV	Sem-V	Sem-VI	
MD1	Smart and Sustainable Systems (SSS)	Fundamentals of Smart and Sustainable Systems & Tutorial (FSSST)	IoT for Smart and Sustainable Systems & Lab (ISSSL)	Data Analytics for Smart and Sustainable Systems Lab (DASSSL)	Security for Smart and Sustainable Systems Lab (SSDL)	Sr Sy La
MD2	Finance and Management (F&M)	Fundamentals of Financial Management & Tutorial (FFMT)	Fundamentals of Financial Engineering Tutorial (FFET)	Fundamentals of Stock Market Tutorial (FSMT)	Fintech: Foundations and Applications Tutorial (FFAT)	Fi Ri Tu
MD3	3D- Printing (3DP)	3D Modeling and Design & Lab (3DMD)	Fundamentals of Additive Manufacturing & Lab (FAM)	Materials and Processes in Additive Manufacturing (MPAM)	Industry 4.0 and Digital Manufacturing (IDM)	Ap Pr (A Pr
MD4	Electric Vehicles (EV)	EV Foundation – Principles and Concepts & Lab (EVPCL)	Advanced Motor Technologies & Power Electronics for EV & Lab (AMTL)	EV Powertrain Dynamics and Control System & Lab (PDCL)	Intelligent EV Systems:AI Iot and Automation Lab (IEVL)	Ca El (C
MD5	Applied Mathematics for Engineering (AME)	Linear Algebra with Python & Lab (LAPL)	Statistical Techniques and Numerical Methods with R & Lab (STNMRL)	Graph Neural Networks with Python & Lab (GNNPL)	Mathematical Optimization Techniques using R & Lab (MOTRL)	Fi La
MD6	Software Development (SD)	Data Structures and Algorithms & Lab (DSAL)	Object Oriented Programming & Lab (OOPL)	Database and Management Systems & Lab (DBMSL)	Web Development & Lab (WDL)	Sy Op (S
MD7	Autonomous and Intelligent Systems (AIS)	Digital Systems and Organization & Lab. (DSOL)	Smart System Engineering & Lab. (SSEL)	Embedded IoT Systems & Lab (EISL)	Autonomous Systems & Lab (ASL)	Ca
MD8	Embedded Systems (ES)	Fundamentals of Microcontrollers & Lab (FML)	Embedded Processors & Lab (EPL)	Microcontrollers and IoT & Lab (MIL)	Embedded System Design and RTOS & Lab (ESD-RTOSL)	Ca M (C
MD9	AI & Machine Learning (AI-ML)	Statistical Data Analysis & Lab (SDAL)	Machine Learning & Lab (MLL)	Artificial Intelligence & Lab (AIL)	Next Gen AI & Lab (NGAIL)	De
MD10	Quantum Computing (QC)	Foundations of Quantum Technologies & Lab	Foundations of Quantum Computing & Lab	Basic Programming for Scientific Computing & Lab	Introduction to Quantum Computation & Lab	M La

Link: Detailed Syllabus

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Annexure -II: Guidelines for Open Elective Courses

i. Open Elective – I will be offered in third semester as foreign language as prescribed in the structure.

ii. Open Electives – II, III, IV will be offered through SWAYAM/NPTEL MOOCs of Equivalent Credits.

iii. Departments shall prepare the baskets of open elective courses from discipline/faculty other than respective major programs. Students may choose any course from the basket without adhering to any one stream.

iv. Credits & Grade will be awarded based on the Marks Obtained through the certification including assignments and proctored examination as per the MOOCs Policy.

			Teaching Scheme (Hours/Week)				Credits				Examination/ Evaluation Scheme and Marks						
											Theory			Practical			
Sem	Course code	Name of the Course	L	P	T	Total	L	P	T	Total	CIE [20]	ISE [20]	ESE [60]	CIE TW	ESE P	ESE OR	Total
3	2603OE5X	Foreign Language Studies (FLS)	-	-	2	2	-	-	2	2	-	-	-	50	-	-	50
4	2604OE5X	MOOCs	-	-	2	2	-	-	2	2			50	-	-	-	50
5	2605OE5X	MOOCs	-	-	2	2	-	-	2	2	-	-	50	-	-	-	50
6	2606OE5X	MOOCs	-	-	2	2	-	-	2	2	-	-	50	-	-	-	50

Guidelines for MOOCs

1. The department shall release a list of approved SWAYAM-NPTEL courses before the commencement of every semester.
2. Students shall register for the approved Courses as per the schedule announced by SWAYAM-NPTEL.
3. A student shall undergo the courses only from the list notified by the department through SWAYAM/NPTEL platform and complete all the assignments and examination requirements as specified by SWAYAM/NPTEL.
4. SWAYAM-NPTEL Courses are considered for transfer of credits only if the student concerned has successfully completed and obtained the SWAYAM-NPTEL Certificate.
5. The credit equivalence for SWAYAM-NPTEL Courses: 12 weeks – 3credits; 8 weeks – 2 credits; 4 weeks – 1 credit.
6. Equivalent marks will be considered for awarding the grades as specified in examination rules and regulations. The weightage for assignments is 40%, while the weightage for the proctored examination will be 60% for award calculating SGPA/CGPA. Students must score

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a minimum of 40% of the total marks by combining both assignments and proctored examinations

7. A student must submit the original SWAYAM-NPTEL Course Certificates to the Head of the Department concerned, with a written request for the transfer of the equivalent credits. On verification of the SWAYAM-NPTEL Course Certificates and approval by the head of the department, credits will be awarded.
8. The Institute shall not reimburse any fees/expenses a student may incur for the SWAYAM-NPTEL Courses.
9. If the SWAYAM/NPTEL course calendar does not align with the institute's calendar, the department shall facilitate and conduct examination of the relevant course of equivalent credits in physical/virtual mode and award the credits accordingly.



Annexure -III: Relevant Topics for Self-study

These topics are optional and suggested for students for self-study to get additional exposure. However, these topics will not be considered for any evaluation and credits.

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Annexure -IV: Honors Course (HC)

Guidelines for B. Tech Honors Program (As per NEP 2020)

1. Objective

The Honors program is designed to provide **advanced knowledge and deeper specialization in the student's major discipline**. It aims to encourage **academic excellence, research orientation, and advanced technical competency**, enabling students to gain expertise beyond the standard B. Tech curriculum.

2. Eligibility

- Students enrolled in the **B. Tech program** are eligible to opt for the Honors program.
- The option for Honors shall normally be available from **Semester V to Semester VIII**.
- Students should have **successfully completed all courses up to Semester IV without backlog**, or as per institute regulations.
- A **minimum CGPA (for example 7.5 or as decided by the institute)** may be required to enroll in the Honors program.
- The number of students allowed for Honors may be **limited based on academic performance and departmental capacity**.

3. Credit Requirement

- Students must complete **20 additional credits** in advanced courses of their **major discipline**.
- These credits shall be **over and above the 160 credits required for the regular B. Tech program**.
- The courses may include **advanced electives, research-oriented subjects, laboratory work, or project-based learning**.

4. Structure of Honors Program

- The Honors program shall normally consist of **4–5 advanced courses totaling 20 credits**.
- These courses shall be offered during **Semester V to Semester VIII**.
- A typical credit distribution may be:

Semester	Course Type	Credits
Semester V	Honors Course – I	5
Semester VI	Honors Course – II	5
Semester VII	Honors Course – III	6
Semester VIII	Honors Course – IV	4

***Institutions may modify the structure depending on curriculum requirements.

5. Course Selection Rules

- Honors courses must belong to the **student's major discipline**.
- The list of **approved Honors courses** shall be notified by the department.
- Honors courses should be **advanced or specialized subjects beyond the core curriculum**.
- Students must register for Honors courses as **additional courses along with their regular semester workload**.

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6. Registration Procedure

- Interested students must **apply for the Honors program at the beginning of Semester V**.
- The registration shall be approved by the **Head of the Department (HoD)** of the parent department.
- Course registration shall follow the **standard academic registration procedures of the institute**.

7. Evaluation and Assessment

- Evaluation of Honors courses shall follow the **same assessment and examination pattern as regular B. Tech courses**.
- Students must secure the **minimum passing grade as prescribed by the institute**.
- If a student fails in an Honors course, they must **clear the course as per institute examination rules**.

8. Award of Honors Degree

- Students who successfully complete the **20 additional credits of Honors courses**, along with the **160 credits required for the B. Tech program**, will be awarded the degree as **"B. Tech with Honors in <Discipline Name>."**
- The Honors specialization will be **mentioned on the degree certificate and academic transcript**.

9. Exit Option

If a student **does not complete all the required Honors credits**, the additional credits earned may be **recorded in the academic transcript**, but the **Honors distinction will not be awarded**.

Available courses are as below

Sr. No	Course Name	Mentor Dept.	Remark
1.	VLSI Design Technology (VLSI-DT)	E&TCE	For all E&TCE, E&CE
2.	Block Chain Technology (BCT)	CE	For all CE, AIDS
3.	Internet of Things (IoT)	IT	For all IT

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Structure of Honors Course

Honors Courses			Teaching Scheme (Hours/ Week)			Credits			Examination Scheme and Marks				
									Theory			Practical	Semester
Sem	Course Code	Name of Course	L	P/T	Total	L	P/T	Total	CIE	ISE	ESE	CIE/ESE	Total
									[20]	[20]	[60]	TW/P/OR	500
5	**2605HC1X**	HC Theory-1	4	-	4	4	-	4	20	20	60	-	**100**
5	**2605HCXX**	Tutorial/Laboratory-1	-	2	2	-	1	1	-	-	-	25	**25**
6	**2606HC1X**	HC Theory-2	4	-	4	4	-	4	20	20	60	-	**100**
6	**2606HCXX**	Tutorial/Laboratory-2	-	2	2	-	1	1	-	-	-	25	**25**
7	**2607HC1X**	HC Theory-3	4	-	4	4	-	4	20	20	60	-	**100**
7	**2607HCXX**	Tutorial/Laboratory-3	-	4	4	-	2	2	-	-	-	50	**50**
8	**2608HCXX**	Capstone Project	-	8	8	-	4	4	-	-	-	100	**100**
			12	**16**	**28**	**12**	**8**	**20**	**60**	**60**	**180**	**200**	**500**

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